DATASHEET

AXS15260D-S1

In-Cell IC Integrates 810-channel 8-bit Source Driver and GIP Gate Driver and Touch Panel Controller Into a Single Chip with TP Controller Supports Real Multi-Touch Capability

Version V1.2
2024-03-06

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REVISION HISTORY

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2023/01/05	V0.1	Preliminary version	179	Sand
2023/03/07	V0.2	improve all diagram	173	Sand
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2023/07/24	V0.6	Update the Power On/Off Sequence;Delete the 2 power mode	161	Sand
2023/08/26	V0.7	Update Note: VGH1:When the external on-load current is below 1.25mA, the maximum voltage can reach 16V, and the register can be adjusted normally; VGL1:When the external on-load current is below 1.35mA, the minimum voltage can reach -13V, and the register can be adjusted normally. Update the AC characteristic; Update the power Consumption. Add power block diagram Add power Supply Peripheral Component List	164	Sand
2023/10/12	V0.8	Update Block Diagram	168	AXS
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GENERAL DESCRIPTION

The AXS15260D-S1 highly integrates a-Si/LTPS TFT LCD driver and Super in-cell Touch controller, is a 16,777,216-color System-on-Chip (SOC) driver LSI designed for small and medium size TFT LCD display, and is capable of supporting up to 540RGBx1600(Dual gate) pixels in resolution. The 810-channel source driver can provide true 8-bit resolution and generate 256 Gamma-corrected values with an internal D/A converter.

The AXS15260D-S1 is able to operate with low IO interface power supply. Incorporating with several charge pumps, the AXS15260D-S1 can generate various voltage levels by an on-chip power management system for gate and source driver. Moreover, PWM for LED backlight, wake up-button enabling, respiratory lights and other functions, to provide customers with better experience.

In addition, the external Flash of AXS15260D-S1 can store not only the firmware used for Touch controller, but also the Initial code of LCD driver. After loading the initial code through the external Flash, the HOST only needs to send out "Sleep out" and "Display on" to turn on the LCD.

The built-in timing controller in the AXS15260D-S1 can support several functions to meet a wide variety of requirements for portable display applications. It provides several system interfaces, including MIPI/QSPI/DSPI/SPI/RGB/MCU, which can be used to configure the system. Furthermore, it can also archive high speed display data transmission by using the MIPI interface.

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1 Features

1.1 Display

- One-chip solution for color amorphous a-Si TFT-LCD with incell
  Display Resolution
  - Dual gate: 540RGB(up to 540,2 Columns/step) x 1600(up to 2048, 2 lines/step)
  - Single gate:270RGB(up to 270,2 Columns/step) x 1600(up to 2048,2 lines/step)
- Frame rate 60Hz/90Hz
- Support LTPS TFT-LCD:60Hz/90Hz,540RGB*1350
- Support IGZO TFT-LCD
- Display Data Memory: external PSRAM
- Support in-chip OTP
  - 2K*8 bit
  - OTP can stores trimming、ID1，ID2（factory ID，3bit）data
- System Interfaces
  - MIPI DSI (1/2/3/4 data lane,1.25Gbps/lane):
  - MIPI DSI (DSI v1.01.00, D-PHY v1.00.00 and DCS v1.01).
  - SPI/DSPI
  - QSPI
  - Parallel 8080-series MCU Interface (8-bit, 9-bit, 16-bit, 18bit)
  - 16bit/18bit/24bit RGB(60Hz) interface
- Display Features
  - Colour depth 888，666，565
  - Analog gamma，256 gray level
  - Support 8-color and 2-color mode
- On Chip Function
  - RAM-less driver with MIPI video mode
  - Column Inversion & Dot Inversion
  - Support loading FLASH to driver registers & TP registers
  - Support 0 Flash
  - Support SPR communication protocol
  - Support 810 source channel
  - Built-in 17 GOUT signals(CGOUTL1-L17,CGOUTR1-R17) on each side of the chip,providing directly the driver signals to the GIP circuit on the LCD panel
  - Built-in internal oscillator

1.2 Touch

- 8-bit embedded MCU
- 480 SX channels
- Support five-point detection
- Point reporting rate 60Hz(frame rate 60Hz)/Point reporting rate 90Hz(frame rate 90Hz)/Point reporting rate 120Hz(frame rate 60Hz)

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- Super self-capacitance detection technology
- Support wake up gesture function
- Anti-RF interference
- Automatic frequency hopping
- Water proof
- I2C/SPI data communication interface
- Internal ESD detection
- Power saving mode
- VDDI_TP and VDDI supplied independently
- Center area ≤1.0mm, edge ≤1.5mm@5mm copper column

1.3 Power

- Supply Voltage Range

3-power mode

- VSP：4.5V~6.5V，typ ：5.5V
- VSN：-6.5V~-4.5V，typ：-5.5V
- VDDI：1.65V ~ 1.95V typ：1.8V or 2.8V~3.6V typ：3.3V

- Output Voltage Range

- Gamma Positive Voltage Range: VGSP~VGMP（VGMP≤VSP-0.3）

VGMP：2.55~5.7V
VGSP：0.05~2.5V

- Gamma Negative Voltage Range: VGMN~VGSN（VGMN≥VSN+0.3）

VGMN：-5.7~-2.55V
VGSN：-2.15~1V

- Source Output Range: VGSP~VGMP
- Positive Gate Driver Output Voltage Level(VGH):

IGZO：VGH1 12~16V
LTPS：VGH2 6.5~10.7V

- Negative Gate Driver Output Voltage Level(VGL):

IGZO：VGL1 -13~-7.6V
LTPS：VGL2 -10~-6V

- Common Electrode Output Voltage Level(VCOM): -2.5V~0V (10mv/step)

Note:
VGH1:When the external on-load current is below 1.25mA, the maximum voltage can reach 16V, and the register can be adjusted normally;
VGL1:When the external on-load current is below 1.35mA, the minimum voltage can reach -13V, and the register can be adjusted normally.

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1.4 Others

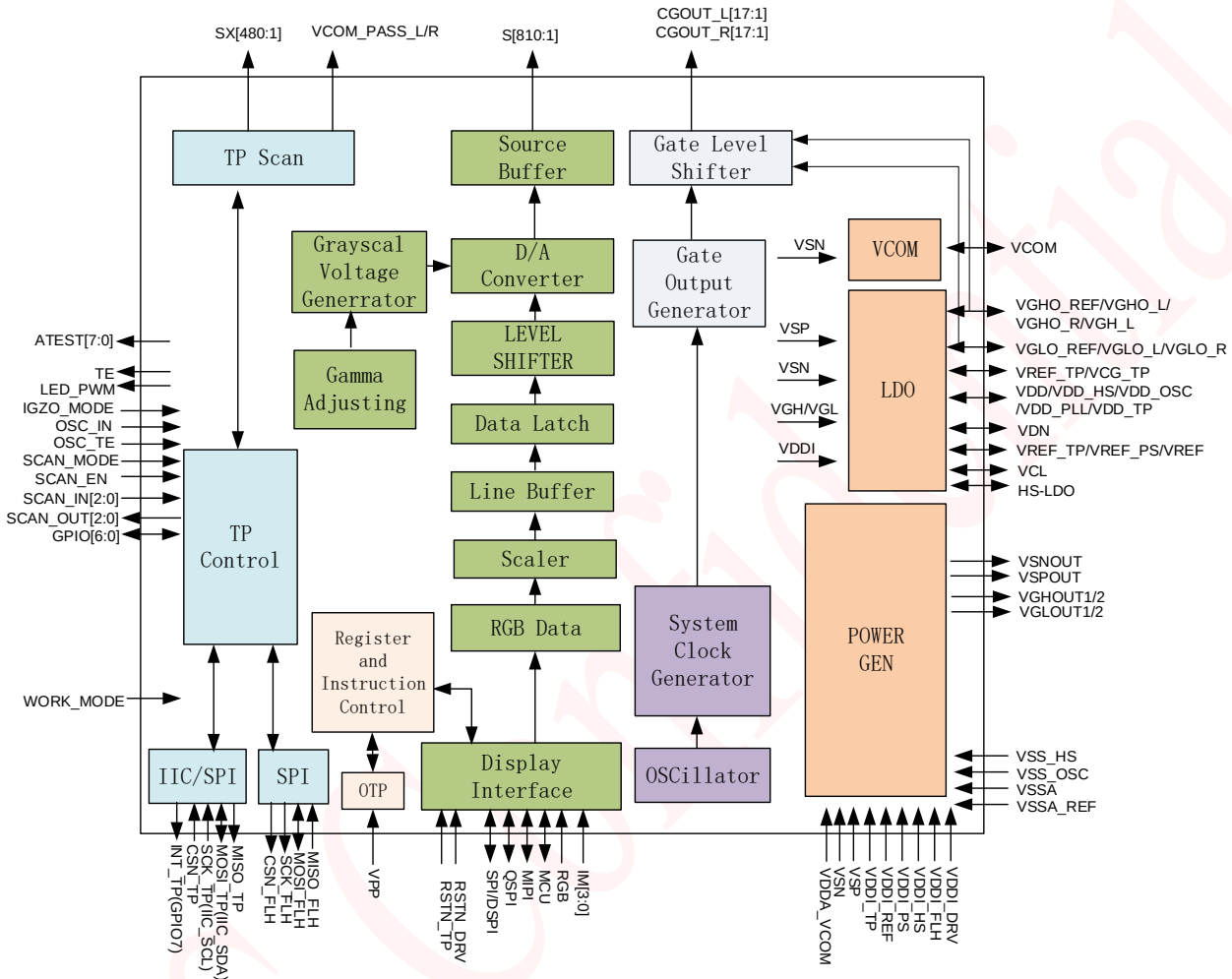
- ESD
  - HBM≥2000V
  - MM≥200V
  - Latch up≥±100mA
- COG Package

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2 Block Diagram

2.1 Block Function



2.1.1 Touch Function

The touch part of AXS15260D-S1 mainly consists of the following components:

- AFE Controller

AFE controller completes the scanning of the sensors in the touch panel, and sends the data of touch sensors after scanning to the MCU for data processing.

- Embedded MCU

MCU and SOC subsystems complete the control, data processing, LCD operation and coordination, HOST communication and other functions of the whole touch systems.

- I2C/SPI serial interface

The Slave end of I2C/SPI in AXS15260D-S1 is the interface for touch communication with HOST. The control interface consists of two signals INT (is a GPIO, specified in the firmware) and RSTN

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(only one rstn-pin, shared by driver and touch). Whenever there is effective touch sensed on the touch screen, Touch controller will send data transfer request to the HOST via INT port, and complete the point report to the HOST. HOST can communicate with AXS15260D-S1 via I2C or SPI. HOST can also reset Touch controller through RSTN port.

- External Flash

External Flash, used to store the Firmware, and LCD initialization code, can be added into the Touch controller.

- Watchdog

Watchdog is used to ensure the stability of the chip when in operation

- Internal voltage regulator

Internal voltage regulator generates power supply, which is to provide power to logic circuit.

2.1.2 Touch Operation Mode

Touch controller has the following three operation modes:

- Normal operating mode

In this mode, Touch controller scans the screen, and detects the touch actions.

- Monitor Mode

In this mode, Touch controller scans the screen intermittently to save power.

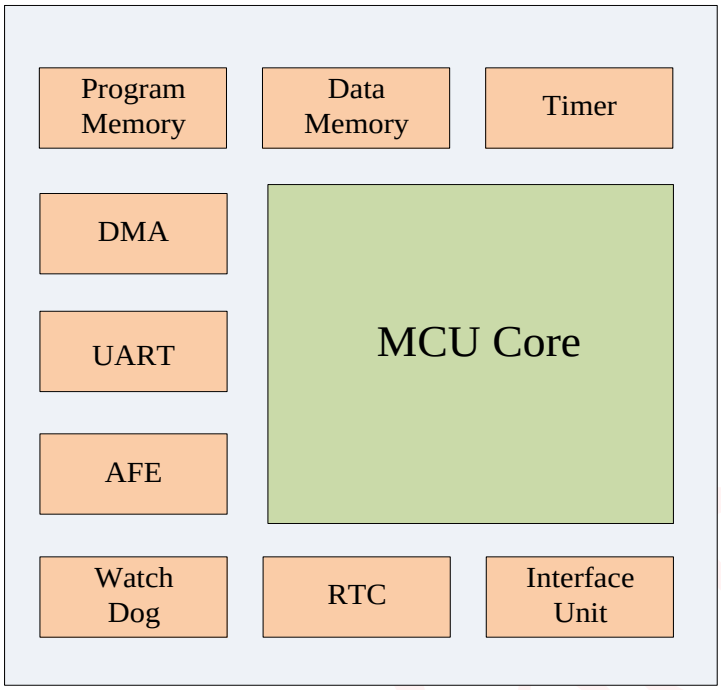
- Sleep mode

In this mode, Touch controller enters ultra low-power standby mode, HOST can only wake-up Touch controller via external signal to enter the normal operating mode. The power consumption in this mode is extremely small, and can greatly extend the standby time of mobile portable devices.

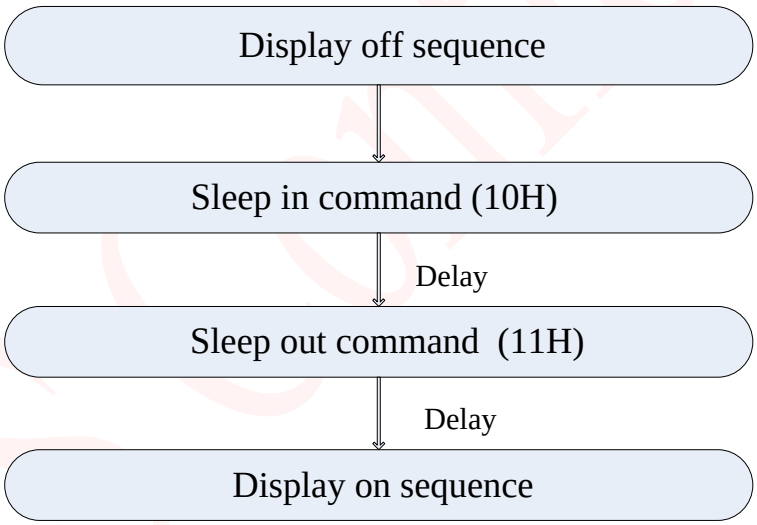
2.1.3 MCU

This section describes some critical features and operations supported by the MCU.The figure below shows the overall structure of the MCU block. In addition to the MCU core, we have added the following circuits.

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2.1.4 Sleep In/Out Sequence



2.1.5 System interface

The AXS15260D-S1 supports the video data transmitted through the high-speed system interface, MIPI (Mobile Industry Processor Interface).

2.1.6 Grayscale voltage generating circuit

AXS15260D-S1 has true 8-bit resolution D/A converter, digital gamma cooperates with OP-AMP structure to enhance display quality. The grayscale voltage can be adjusted by grayscale data set in the γ-correction register and RGB can be adjusted separately.

2.1.7 Timing controller

AXS15260D-S1 has a timing controller, which can generate a timing signal for internal circuit

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operation such as gate output timing, image data accessing timing, etc.

2.1.8 Oscillator (OSC)

The AXS15260D-S1 has an internal oscillator to generate system clock.

2.1.9 Source driver circuit

AXS15260D-S1 consists of a 810-output source driver circuit.

2.1.10 Gate driver circuit

AXS15260D-S1 consists of a gate driver control circuit. The gate driver circuit outputs gate driver signals level at either VGH or VGL level.

2.1.11 LCD driving power supply circuit

The LCD driving power supply circuit generates the voltage levels VDD, VSP, VSN, VGH, VGL, VCOM for driving a LCD. All these voltages can be adjusted by register setting.

3 Pin Description

3.1 Power Supply and Regulator pins

PIN	I/O	Default value	Description
VDDI	I	Digital Power	External input voltage.1.8V power supply for digital interface
VDDI_DRV	I	Digital Power	External input voltage.1.8V power supply for digital interface
VDDI_FLH	I	Digital Power	External input voltage for external spi-flash
VDDI_HS	I	Digital Power	External input voltage for mipi-ldo
VDDI_PS	I/O	Digital Power	Psram IO power supply,connect VDDI when VDDI=1.8V, connect 1uF cap to gnd when VDDI=3.3V
VDDI_REF	I	Analog Power	Bandgap power supply
VDDI_TP	I	Digital Power	Power supply for TP digital interface.
VSN	I	Analog Power	-5.5V voltage power supply
VSNOUT	O	Analog Power	Charge pump output
VSP	I	Analog Power	5.5V voltage power supply
VSPOUT	O	Analog Power	Charge pump output,Connect VSP
VREF	O	Internal LDO	Output reference voltage.
VREF_PS	O	Internal LDO	Output reference voltage, need connect a stabilizing capacitor to VDDI_PS.
VREF_TP	O	Internal LDO	Regulator output voltage.
VGH	I	GIP power suuply	GIP power supply connect VGHOUT1 or VGHOUT2
VGH_L	O	GIP power suuply	Connect VGH
VGHO_L/R	O	GIP power suuply	GIP power supply,VGHO_L connect to VGHO_R

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VGHO_REF	O	Charge pump	Connect a capacitor to VGHO_R
VGHOUT1	O	Charge pump	Charge pump output
VGHOUT2	O	Charge pump	Charge pump output
VGL	I	GIP power suuply	Power supply need connect a stabilizing capacitor to ground.
VGLO_L/R	O	GIP power suuply	GIP power supply,VGLO_L connect to VGLO_R
VGLO_REF	O	GIP power suuply	Connect a capacitor to VGLO_L
VGLOUT1	O	Charge pump	Charge pump output
VGLOUT2	O	Charge pump	Charge pump output
VDD	O	Internal LDO	Digital power supply, need connect a stabilizing capacitor to ground.
VDD_HS	O	Internal LDO	MIPI lower power circuit power supply, connect to VDD on FPC.
VDD_OSC	O	osc power supply	Osc power supply,conect to VDD
VDD_PLL	O	Internal LDO	Regulator output voltage for PLL.
VDD_TP	O	Internal LDO	Regulator output voltage. FPC Connect together.Connect a capacitor for stabilization.
VDN	O	Digital Power	Internal LDO output
HS_LDO	O	Internal LDO	MIPI high speed circuit power supply, need connect a stabilizing capacitor to ground.
VCG_TP	O	Internal LDO	Regulator output voltage. FPC Connect together.Connect a capacitor for stabilization.
VCL	O	Internal LDO	LDO negative output.
VCOM	O	Analog	VCOM signal output for panel.Need connect a stabilizing capacitor to ground.
VPP	I	Digital Power	Power supply.
VDDA_VCOM	I	analog power	VCOM power supply,conect VDN.
VSS_HS	I	Ground	MIPI ground, connect to VSSD on FPC
VSS_OSC	I	Ground	OSC ground
VSSA	I	Ground	Analog ground
VSSA_REF	I	Ground	Bandgap ground
VSSD	I	Ground	Digital ground

3.2 Drive interface

PIN	I/O	Default value	Description
DB<23:0>	I/O	Digital (VDDI)	DB[23:0] input data of rgb interface.
HSYNC	I	Digital (VDDI)	Line synchronizing signal of rgb interface.
VSYNC	I	Digital (VDDI)	Frame synchronizing signal of rgb interface.
PCLK	I	Digital (VDDI)	pixel clock signal of rgb interface.

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DE	I	Digital (VDDI)	Data enable signal of rgb interface;
DIN_SDA_DUAL	I/O	Digital (VDDI)	The second data input pin in spi dual data lane of spi slave.
DIN_SDA	I	Digital (VDDI)	The bidirectional data pin of SPI slave
SCL	I	Digital (VDDI)	Synchronous clock signal in SPI slave.
RS	I	Digital (VDDI)	command or parameter selection in spi 4wire 8bits.
CSX	O	Digital (VDDI)	Chip select input pin (“Low” enable) in SPI slave
SWIRE	O	Digital (VDDI)	Swire protocol setting pin (Note: “H” = VDDI level, “L” = VSSI level.)Output load 50pf.
TE	O	Digital (VDDI)	Output tearing effect signal from IC to phone,Default output hiz.
RSTN_DRV	I	Digital (VDDI)	Global reset, low active.

3.3 TP interface

PIN	I/O	Default value	Description
SX[480:1]	I/O	Analog	Separate COM Electrode(TP sensor PAD)
VCOM_PASS_L/R	I/O	Analog	Pass line for VCOM from ILB to OLB
RSTN_TP	I	Digital (VDDI_TP)	Reset, low active，weak pull up
WORK_MODE	I	Digital (VDDI_TP)	Selection of work interface;1:I2C , 0:spi；pull down
CSN_TP	I	Digital (VDDI_TP)	When WORK_MODE =1 ，floating；when WORK_MODE =0，connect to phone spi interface chip selection signal, low active.
SCK_TP	I	Digital (VDDI_TP)	When WORK_MODE =1 ，connect to phone i2c slave interface clock signal , need open drain and extern pull up. when WORK_MODE =0，connect to phone spi interface clock signal.

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MISO_TP	O	Digital (VDDI_TP)	Connect to phone spi interface data output signal, from SCK negative edge to MISO output need 6ns delay in digital internal, detail as followed spi timing. (phone input ,chip output, chip is spi_slv)
MOSI_TP	I/O	Digital (VDDI_TP)	Connect to phone I2C slave interface data input/output，need open drain and extern pull up. (chip is i2c_mst). connect to phone spi interface data input signal.(phone output ,chip input, chip is spi_slv)
CSN_FLH	O	Digital (VDDI_FLH)	Chip select of flash, low active，individual power for spi flash interface.Master output.
SCK_FLH	O	Digital (VDDI_FLH)	Clock output to flash，individual power for spi flash interface.
MISO_FLH	I	Digital (VDDI_FLH)	Data input from flash; need weak pull up to avoid floating, flash data output need 7ns delay from clock negative edge，individual power for spi flash interface.
MOSI_FLH	O	Digital (VDDI_FLH)	Data output to flash，individual power for spi flash interface.

3.4 MIPI Interface

PIN	I/O	Default value	Description
HS_CN HS_CP	I	MIPI	MIPI-DSI clock Lane positive-end/ negative-end input pin.These pins are MIPI-DSI CLK+/- differential clock signals if MIPI interface is used.HS_CP/N are differential small amplitude signals. Ensure the trace length is shortest ,so that the COG resistance is less than 10 ohm.If not used, please connect these pins to VSSAM.
HS_D0N HS_D0P	I/O	MIPI	MIPI-DSI data Lane 0 positive-end/ negative-end pin.These pins are MIPI-DSI D0+/- differential data signals if MIPI interface is used.HS_D0P/N are differential small amplitude signals. Ensure the trace length is shortest so that the COG resistance is less than 10 ohm. If not used, please connect these pins to VSSAM.
HS_D1N HS_D1P	I	MIPI	MIPI-DSI data Lane 1 positive-end/ negative-end input pin.These pins are MIPI-DSI D1+/- differential data signals if MIPI interface is used.HS_D1P/N are differential small amplitude signals. Ensure the trace length is shortest so that the COG resistance is less than 10 ohm. If not used, please connect these pins to VSSAM.
HS_D2N HS_D2P	I	MIPI	MIPI-DSI data Lane 2 positive-end input pin.These pins are MIPI-DSI D2+/- differential data signals if MIPI interface is used.HS_D2P/N are differential small amplitude signals. Ensure the trace length is shortest so that the COG resistance is less than 10 ohm. If not used, please connect these pins to VSSAM.
HS_D3N	I	MIPI	MIPI-DSI data Lane 3 positive-end input pin.These pins are MIPI-DSI D3+/- differential data signals if MIPI interface is used.HS_D3P/N are differential small amplitude signals. Ensure

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HS_D3P			the trace length is shortest so that the COG resistance is less than 10 ohm. If not used, please connect these pins to VSSAM.

3.5 PSRAM interface

PIN	I/O	Default value	Description
PS_ADQ<7:0>	I/O	Digital (VDDI_PS)	Address/DQ bus [7:0]
PS_CEN	O	Digital (VDDI_PS)	Chip select,active low,when CE=1 chip is standby state
PS_CLKN/P	O	Digital (VDDI_PS)	Clock signals,they must always be complementary,even in standby
PS_DM	O	Digital (VDDI_PS)	Data mask,active high.DM=1 mens “do not write”
PS_DQS	I/O	Digital (VDDI_PS)	PSRAM DQ Strobe clock

3.6 Test/Dummy Signal/ Other

PIN	I/O	Default value	Description
ATEST0_NV	O	Analog (VSP/VSN)	Test pin
ATEST1_PV			
ATEST2_LV			
ATEST3_LV			
ATEST4_NV			
ATEST5_PV			
ATEST6_PV			
ATEST7_NV			
SWIRE	O	Digital (VDDI)	Swire protocol setting pin (Note: “H” = VDDI level, “L” = VSSI level.)Output load 50pf.
TE	O	Digital (VDDI)	Output tearing effect signal from IC to phone,Default output hiz.
TS_SEL	I	Digital (VDDI)	test pad

3.7 Panel driver Signals

PIN	I/O	Default value	Description
S[810:1]	O	Analog	Output source driver signals. The D/A converted 256-gray-scale analog voltage output.
CGOUT_L<17:1>	O	Analog (VGHO/VGLO)	Gate control signals for panel in left side of IC.

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CGOUT1_R<17:1>	O	Analog (VGHO/VGLO)	Gate control signals for panel in right side of IC.

3.8 Logic Function Control

PIN	I/O	Default value	Description
OTP_EXT_EN	I	Digital	Conect VSP or floating.
IGZO_MOD	I	Digital (VDDI)	0:IGZO panel 1:normal panel
PMOD_SEL	I	Digital (VDDI)	power voltage set pin: 0:when VDDI=3.3v 1(floating):when VDDI=1.8v
PSW_ENN	I	Digital (VDDI)	1:Power switching function enable 0:Some I/O power supplies are fixed to VDDI
PSW_PRIO_SEL	I	Digital (VDDI)	Select PIN for some IO power supplies 0:VDDI_TP is preferred 1:VDDI_DRV is preferred
LED_PWM	O	Digital (VDDI)	Used to adjust the brightness of panel backlight.
IM<3:0>	I	Digital (VDDI)	External interface select, the default /floating is 1.
BS<3:0>	I	Digital (VDDI)	Polarity and order of MIPI selection, default 4'bXXX.

外部连接 BS[0]	外部连接 BS[3:1]	HS_D3P	HS_D3N	HS_D0P	HS_D0N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D2P	HS_D2N
1	000	HS_D2P	HS_D2N	HS_D3P	HS_D3N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D0P	HS_D0N
	001	HS_D0P	HS_D0N	HS_D3P	HS_D3N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D2P	HS_D2N
	010	HS_D3P	HS_D3N	HS_D2P	HS_D2N	HS_CP	HS_CN	HS_D0P	HS_D0N	HS_D1P	HS_D1N
	011	HS_D3P	HS_D3N	HS_D0P	HS_D0N	HS_CP	HS_CN	HS_D2P	HS_D2N	HS_D1P	HS_D1N
	100	HS_D2P	HS_D2N	HS_D3P	HS_D3N	HS_CP	HS_CN	HS_D0P	HS_D0N	HS_D1P	HS_D1N
	101	HS_D0P	HS_D0N	HS_D3P	HS_D3N	HS_CP	HS_CN	HS_D2P	HS_D2N	HS_D1P	HS_D1N
	110	HS_D3P	HS_D3N	HS_D2P	HS_D2N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D0P	HS_D0N
	111	HS_D3P	HS_D3N	HS_D0P	HS_D0N	HS_CP	HS_CN	HS_D1P	HS_D1N	HS_D2P	HS_D2N
0	000	HS_D2N	HS_D2P	HS_D3N	HS_D3P	HS_CN	HS_CP	HS_D1N	HS_D1P	HS_D0N	HS_D0P
	001	HS_D0N	HS_D0P	HS_D3N	HS_D3P	HS_CN	HS_CP	HS_D1N	HS_D1P	HS_D2N	HS_D2P

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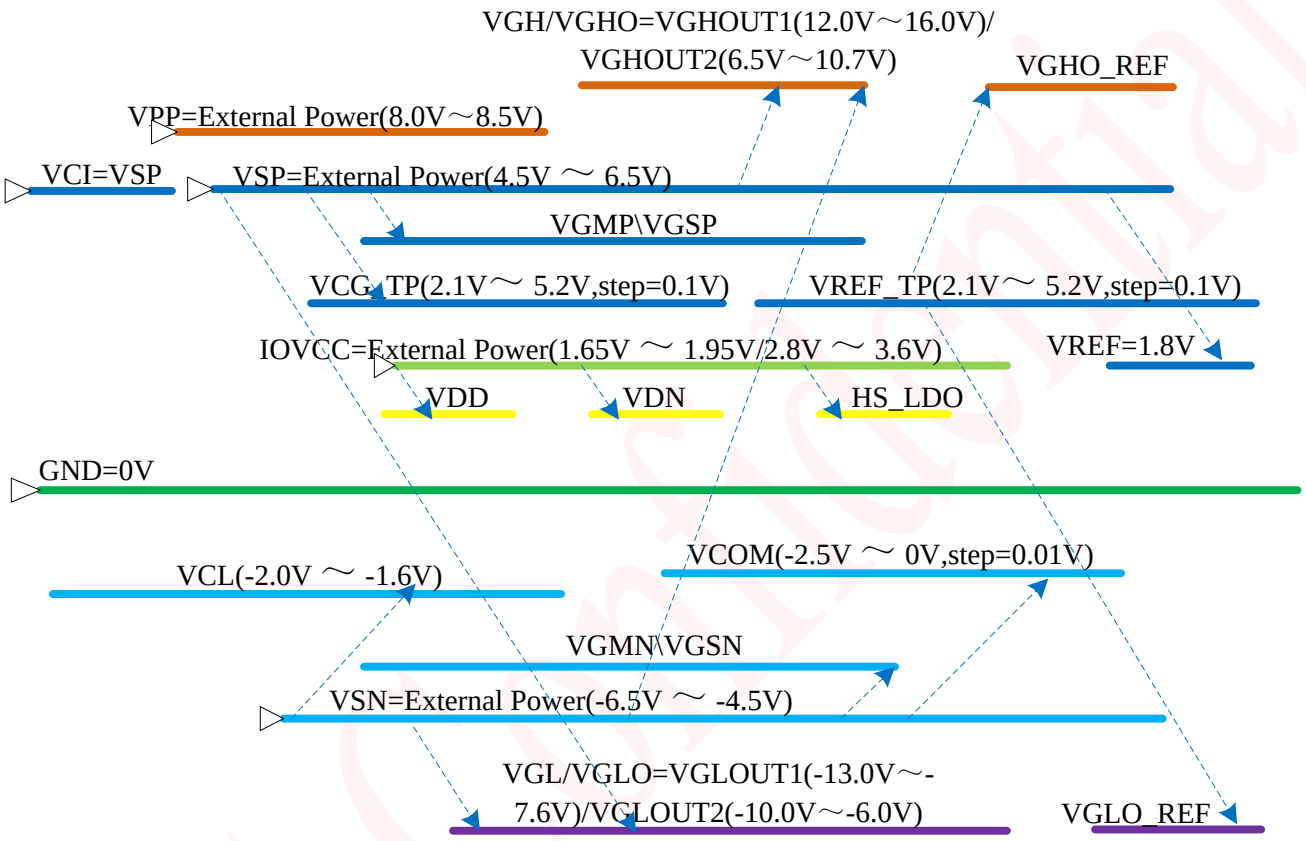
	010	HS_D 3N	HS_D 3P	HS_D 2N	HS_D 2P	HS_C N	HS_C P	HS_D 0N	HS_D 0P	HS_D 1N	HS_D 1P
	011	HS_D 3N	HS_D 3P	HS_D 0N	HS_D 0P	HS_C N	HS_C P	HS_D 2N	HS_D 2P	HS_D 1N	HS_D 1P
	100	HS_D 2N	HS_D 2P	HS_D 3N	HS_D 3P	HS_C N	HS_C P	HS_D 0N	HS_D 0P	HS_D 1N	HS_D 1P
	101	HS_D 0N	HS_D 0P	HS_D 3N	HS_D 3P	HS_C N	HS_C P	HS_D 2N	HS_D 2P	HS_D 1N	HS_D 1P
	110	HS_D 3N	HS_D 3P	HS_D 2N	HS_D 2P	HS_C N	HS_C P	HS_D 1N	HS_D 1P	HS_D 0N	HS_D 0P
	111	HS_D 3N	HS_D 3P	HS_D 0N	HS_D 0P	HS_C N	HS_C P	HS_D 1N	HS_D 1P	HS_D 2N	HS_D 2P

IM[3:0]	Interface	Connect Pin
1111	MIPI	HS_DN,HS_DP,HS_CN,HS_CP,RSTN
1110	RGB 16/18/24-bit (Can be used with SPI 4wire)	DB[15/17/23:0],VSYNC_QSPI_DIN2,HSYNC,DE_QSPI_DIN3,PCLK,RSTN (DIN_SDA(MISO),DIN_SDA_DUAL(MOSI),RS(D/CX),SCL,CSX) ——The SPI interface can only support sending register instructions, not stream data

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3.9 Power Block Diagram

The following is the power supply generation scheme, AXS1560A-S1 consists of VSP, VSN and IOVCC as the input power supply, and other power supply levels are generated in the chip.



Note：

1.VDD include：VDD VDD_TP VDD_HS VDD_OSC

2.VDDI include：VDDI VDDI_DRV VDDI_FLH VDDI_HS VDDI_REF VDDI_TP VDDI_PS

3.VSS_HS=VSS_OSC=VSSA=VSSA_REF=VSSD=GND=0V

3.10 Power Supply Peripheral Component List

No.	NAME	Signal name	Values	Max ability	Note
1	C1	VCOM	4.7uF	10V	stabilizing capacitor
2	C2	VCG_TP	4.7uF	10V	stabilizing capacitor
3	C3	VDD	1uF	10V	stabilizing capacitor

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4	C4	HS_LDO	1uF	10V	stabilizing capacitor
5	C5	VSP	4.7uF	10V	stabilizing capacitor
6	C6	VDDI	1uF	10V	stabilizing capacitor
7	C7	VSN	4.7uF	10V	stabilizing capacitor
8	C8	VGL	1.0uF	25V	stabilizing capacitor
9	C9	VGH	1.0uF	25V	stabilizing capacitor
10	C10	VGHO_REF/VGHO	1uF	25V	stabilizing capacitor
11	C11	VGLO_REF/VGLO	1uF	25V	stabilizing capacitor
12	D1	VGL/VSN	-	-	
13	R1	IGZO_MOD/GND	0R	-	
14	R2	RSTN_DRV/RSTN_TP	0R	-	

4 INSTRUCTIONS

4.1 Outline

The AXS15260D-S1 supports high speed serial interface, MIPI, to configure the system via accessing command registers. While accessing the command registers, the information that indicates which register would be accessed should be sent first. After that, the new value can be updated via system interface. The MIPI-DSI is compliant with MIPI Alliance Standard for Display Serial Interface (DSI), Version 1.01.00 and D-PHY Version 1.00.00. Updating command instructions can also be accomplished by using all supporting system interfaces.

The AXS15260D-S1 has the following major categories of instructions:

(1). User Command List and Description.

(2). Manufacturer Command List and Description.

Since updating these instructions is asynchronous to the internal clock of the AXS15260D-S1, the updating procedure will require no waiting cycles. Furthermore, the updating procedure will not interfere with the processing of the host controller, this makes instructions can be handled smoothly and efficiently.

The following contents of this chapter will describe the supported instructions in detail.

4.2 User Command List and Description

4.2.1 Introduction

After the H/W reset by RESX pin or S/W reset by SWRESET command, each internal register will return to the default state. The commands 10h, 20h, 21h, 28h, 29h, 36h will be updated only during V-sync periods while the module is in the “Sleep Out” mode to avoid abnormal visual effects, and will be updated immediately in the “Sleep In” mode.

4.2.2 System Command List

Name	Hex	Write/Read Command	Description	Parameter Number	Transmission
NOP	00h	C	No operation	0	MIPI/QSPI/DSPI/SPI/RGB/MCU

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SWRESET	01h	C	Software reset	0	MIPI/QSPI/DSPI/SPI/RGB/MCU
RDDID	04h	R	Read display	4	MIPI/QSPI/DSPI/SPI/RGB/MCU
RDNUMED	05h	R	Read Number of the Errors on DSI	1	MIPI/QSPI/DSPI/SPI/RGB/MCU
RDDST	09h	R	Read display status	4	MIPI/QSPI/DSPI/SPI/RGB/MCU
RDDPM	0Ah	R	Read display power	1	MIPI/QSPI/DSPI/SPI/RGB/MCU
RDDMADCTL	0Bh	R	Read memory data access control	1	MIPI/QSPI/DSPI/SPI/RGB/MCU
RDDIPF	0Ch	R	Read Interface Pixel Format	1	MIPI/QSPI/DSPI/SPI/RGB/MCU
RDDIM	0Dh	R	Read display image	1	MIPI/QSPI/DSPI/SPI/RGB/MCU
RDDSM	0Eh	R	Read display signal	1	MIPI/QSPI/DSPI/SPI/RGB/MCU
RDDSDR	0Fh	R	Read display self-diagnostic result	1	MIPI/QSPI/DSPI/SPI/RGB/MCU
SLPIN	10h	C	Sleep in	0	MIPI/QSPI/DSPI/SPI/RGB/MCU
SLPOUT	11h	C	Sleep out	0	MIPI/QSPI/DSPI/SPI/RGB/MCU
PTLON	12h	C	Partial mode on	0	MIPI/QSPI/DSPI/SPI/RGB/MCU
NORON	13h	C	Partial mode off(Normal)	0	MIPI/QSPI/DSPI/SPI/RGB/MCU
INVOFF	20h	C	Display inversion off	0	MIPI/QSPI/DSPI/SPI/RGB/MCU
INVON	21h	C	Display inversion on	0	MIPI/QSPI/DSPI/SPI/RGB/MCU
DISPOFF	28h	C	Display off	0	MIPI/QSPI/DSPI/SPI/RGB/MCU
DISPON	29h	C	Display on	0	MIPI/QSPI/DSPI/SPI/RGB/MCU
CASET	2Ah	W	Column address set	4	MIPI/QSPI/DSPI/SPI/MCU
RASET	2Bh	W	Row address set	4	MIPI/QSPI/DSPI/SPI/MCU
PTLAR	30h	W	Partial start/end address set	4	MIPI/QSPI/DSPI/SPI/RGB/MCU
PTLAR	31h	W	set_partial_columns	4	MIPI/QSPI/DSPI/SPI/RGB/MCU
VSCRDEF	33h	W	Vertical scrolling definition	6	MIPI/QSPI/DSPI/SPI/MCU
TEOFF	34h	C	Tearing effect line off	0	MIPI/QSPI/DSPI/SPI/RGB/MCU
TEON	35h	W	Tearing effect line on	1	MIPI/QSPI/DSPI/SPI/RGB/MCU
MADCTL	36h	W/R	Memory data access control	1	MIPI/QSPI/DSPI/SPI/MCU
VSCRSADD	37h	W	Vertical scrolling start address	2	MIPI/QSPI/DSPI/SPI/MCU
IDMOFF	38h	C	Idle mode off	0	MIPI/QSPI/DSPI/SPI/RGB/MCU
IDMON	39h	C	Idle mode on	0	MIPI/QSPI/DSPI/SPI/RGB/MCU
IPF	3Ah	W	Interface pixel format	1	MIPI/QSPI/DSPI/SPI/RGB/MCU
TESCAN	44h	W	Set tear scanline	4	MIPI/QSPI/DSPI/SPI/RGB/MCU

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RDTESCAN	45h	R	Get tear scanline	4	MIPI/QSPI/DSPI/SPI/RGB/MCU
WRDISBV	51h	W	Write display brightness value	1	MIPI/QSPI/DSPI/SPI/RGB/MCU
RDDISBV	51h/52h	R	Read display brightness value	1	MIPI/QSPI/DSPI/SPI/RGB/MCU
WRCTRLD	53h	W	Write CTRL display	1	MIPI/QSPI/DSPI/SPI/RGB/MCU
RDCTRLD	54h	R	Read CTRL dsiplay	1	MIPI/QSPI/DSPI/SPI/RGB/MCU
RDFCHKSUM	Aah	R	Read First Checksum	2	MIPI/QSPI/DSPI/SPI/RGB/MCU
RDCCHKSUM	Afh	R	Read Continue Checksum	2	MIPI/QSPI/DSPI/SPI/RGB/MCU

4.2.2.1 User Command Description

4.2.2.1.1 SWRESET (01H): Software Reset

8'H01	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Description			Without parameters; Soft reset of the drv. If cr_soft_rstn_tp_opt=0, TP soft reset resets drv. If cr_soft_rstn_tp_opt=1, TP soft reset does not reset drv. When the TP register cr_soft_rstn_drv_opt=0, DRV soft reset can reset TP. When the TP register cr_soft_rstn_drv_opt=1, DRV soft reset non-resettable TP.								

4.2.2.1.2 ICI (04H): IC Information

8'HC0	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	-	↑	cr_manufacture_id								8'h00
Par1	-	↑	cr_driver_version_id								8'h00
Par2	-	↑	cr_driver_id								8'h00
Par3	-	↑	cr_id_dummy								8'h00
Description			cr_manufacture_id：the manufacture_id of the IC. cr_driver_version_id：the version_id of the IC. cr_driver_id：the IC id. cr_id_dummy：is cr_id_dummy_rev0,the dummy id0 that reserved. See the C0 command for details on the above registers.								

4.2.2.1.3 RDNUMED (05H): Read Number of the Errors on DSI

8'H05	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	-	↑	dsi_err_overflow	dsi_err_num							8'h00

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Description	The first parameter is telling a number of the errors on DSI. When register cr_ecc_en=1, enable the mipi ecc function. When register cr_mipi_crc_en=1, the mipi crc function is enabled. dsi_err_num：ecc error number +crc error num dsi_err_overflow：dsi err num Indicates the overflow flag bit.
	Readable error status (ECC and CRC)

4.2.2.1.4 RDDST (09H): Read Display Status

8'H09	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	-	↑	booster_en	cr_my	cr_mx			cr_bgr	cr_ca	cr_gs	8'h40
Par1	-	↑		rgb_format_new			idle_mode	cr_partial_en	pwr_on	normal_dsp_en	8'h70
Par2	-	↑	cr_vscroll_en		src_inv			dsp_on	tear_on		8'h00
Par3	-	↑			cr_tear_mode						8'h00
Description			Read back command(21/20/36/10/11/28/29/3a) rgb_format_new: indicates the actual pixel format booster_en: charge pump Enables the booster function (vgh/vgl/vsp/vsn). idle_mode: 38/39h command flag. This bit is 1 when the 39h command is executed and 0 when the 38h command is executed. cr_partial_en: Part of the enable flag is displayed. The bit is 1 when the 12h command is executed and 0 when the 13h command is executed. normal_dsp_en: the opposite of cr_partial_en; pwr_on: indicates that the bit is 1 when the 11 command is executed, and the bit is 0 when the 10 command is executed. cr_vscroll_en: 33/37h Indicates that the running lamp function is enabled. 1 indicates that the running lamp function is enabled. src_inv: indicates the reverse flag bit of 0 and 1. If the flag bit is 1, the 21h command is executed. If the flag bit is 0, the 20 command is executed. dsp_on: indicates that the bit is 1 when the command is executed 29 and 0 when the command is executed 28. tear_on: This bit is set to 1 for issuing 35 and 0 for issuing 34. cr_tear_mode: 1 indicates that TE is H-blanking+V-blanking; 0 indicates that TE is V-blanking.								

4.2.2.1.5 RDDMADCTL (0AH): Read Display MADCTL

8'H0A	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	-	↑	booster_en	idle_mode	cr_partial_en	pwr_on	normal_dsp_en	dsp_on	-	-	8'h00
Description			booster_en: Enables the charge pump function. idle_mode: 38/39h command flag. This bit is 1 when the 39h command is executed and 0 when the 38h command is executed. pwr_on: indicates that the bit is 1 when the 11 command is executed, and the bit is 0 when the 10 command is executed. cr_partial_en: Part of the enable flag is displayed. The bit is 1 when the 12h command is executed and 0 when the 13h command is executed. normal_dsp_en: the opposite of cr_partial_en; dsp_on: indicates that the bit is 1 when the command is executed 29 and 0 when the command is executed 28.								

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4.2.2.1.6 RDDMADCTL (0BH): Read Display MADCTL

8'H0B	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	-	↑	cr_my	cr_mx			cr_bgr	cr_mh			8'h00
Description			Read back the parameters of command word 36h.								

4.2.2.1.7 RDDCOLMOD(0CH): Read Display Pixel Format

8'H0C	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	-	↑	0	dpi_format			0	dbi_format			8'h77
Description			Read back the parameters of command word 3Ah. dpi_format: video format rgb/mipi pixel format. dbi_format: command format pixel format of dbi, spi, and qspi interfaces								

4.2.2.1.8 RDDMADCTL (0DH): Read Display MADCTL

8'H0D	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	-	↑	cr_vscroll_en		src_inv						8'h00
Description			Read back 21/20h cr_vscroll_en: 33/37h Indicates that the running lamp function is enabled. 1 indicates that the running lamp function is enabled. src_inv: indicates the reverse flag bit of 0 and 1. If the flag bit is 1, the 21h command is executed. If the flag bit is 0, the 20 command is executed.								

4.2.2.1.9 RDDSM (0Eh): Read Display Signal Mode

8'H0E	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	-	↑	tear_on	cr_tear_mode	cr_hs_flag	cr_vs_flag	cr_pclk_flag	cr_de_flag	0	dsi_err_en	8'h00
Description			This command indicates the current status of the display as described in the table below:								

Bit	Description	Value
TEON	Tearing effect line on/off	'1' = ON, '0' = OFF,
TEM	Tearing effect line mode	'1' = mode2, '0' = mode1,
HS	Horizontal Sync (RGB interface)	'1' = ON, '0' = OFF,
VS	Vertical Sync (RGB interface)	'1' = ON, '0' = OFF,
PixelClk	Pixel Clock (DOTCLK, RGB interface)	'1' = ON, '0' = OFF,
DataEn	Data Enable (DE, RGB interface)	'1' = ON, '0' = OFF,
ErrorDSI	Error On DSI (MIPI Interface)	'1' = Error, '0' = No Error

4.2.2.1.10 RDDSDR (0Fh): Read Display Self-Diagnostic Result

8'H0F	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par1	-	↑	otp_crc_done	ctn_crc_done	0					otp_crc_ok	8'h00

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Description	This command indicates the current status of the display self-diagnostic result after sleep out command as described (test function) below: otp_crc_done: The otp is loaded and the crc check is complete ctn_crc_done: The host reconfigures the register and the crc check ends otp_crc_ok:otp Checksums Comparison, '0' = Checksums are same, '1' = Checksums are not same See sections: "Read First Checksum (Aah)" and "Read Continue Checksum (Afh)"

4.2.2.1.11 POFF (10H): Power Off Command

8'H10	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Description			power off command(10) with no parameter is used to turn off power								

4.2.2.1.12 PON (11H): Power On Command

8'H11	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Description			power on command(11) with no parameter is used to turn on power								

4.2.2.1.13 PTLON (12H): Partial Display Mode On

8'H12	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Description			Without parameters; Enable the partial display function, that is, cr_partial_en =1. The upper and lower boundary positions of the local display area are determined by the 30H command word. The left and right boundary of the local display area is determined by 31H command word. Local display area (within the boundary position) shows normal picture content; The contents of the non-local display area (outside the boundary position) are configured by registers (cr_partial_value_r, cr_partial_value_g, cr_partial_value_b).								

4.2.2.1.14 NORON (13H): Normal Display Mode On（Partial mode off）

8'H13	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Description			Without parameters; Disable the partial display function and switch to the normal display mode, that is, cr_partial_en =0.								

4.2.2.1.15 NSI (20H): No Src_Inv Command

8'H20	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Description			no src_inv command(20) with no parameter is used to exit the invertion of black and white picture								

4.2.2.1.16 SI (21H): Src_Inv Command

8'H21	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Description			src_inv command(21) with no parameter is used to enter the invertion of black and white picture								

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4.2.2.1.17 DOFF (28H): Display Off Command

8'H28	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Description			display off command(28) with no parameter is used to turn off display								

4.2.2.1.18 DON (29H): Display On Command

8'H29	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Description			display on command(29) with no parameter is used to turn on display								

4.2.2.1.19 CWA (2AH): Column Windows Address Command

8'H2a	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	↑	-						cr_2a_sc_rx[10:8]			8'h00
Par1	↑	-	cr_2a_sc_rx[7:0]								8'h00
Par2	↑	-						cr_2a_ec_rx[10:8]			8'h00
Par3	↑	-	cr_2a_ec_rx[7:0]								8'h00
Description			When cr_win_en =1 of command table2, windowing functions 2a and 2b are effective; cr_2a_sc_rx: represent row/column start/stop address in command mode display cr_2a_ec_rx: represent row/column start/stop address in command mode display;								

4.2.2.1.20 RWA (2BH): Row Windows Address Command

8'H2b	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	↑	-					cr_2b_sr_rx[11:8]				8'h00
Par1	↑	-	cr_2b_sr_rx[7:0]								8'h00
Par2	↑	-					cr_2b_er_rx[11:8]				8'h00
Par3	↑	-	cr_2b_er_rx[7:0]								8'h00
Description			cr_2b_sr_rx: represent row/column start/stop address in command mode display; cr_2b_er_rx: represent row/column start/stop address in command mode display;								

4.2.2.1.21 PTLAR (30H):set_partial_rows

8'H30	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	↑	-					cr_partial_strrow[11:8]				8'h00
Par1	↑	-	cr_partial_strrow[7:0]								8'h00
Par2	↑	-					cr_partial_endrow[11:8]				8'h00
Par3	↑	-	cr_partial_endrow[7:0]								8'h00
Description			cr_partial_strrow [11:0]: Locally displays the start row. cr_partial_endrow [11:0]: locally displays the endrow.								

4.2.2.1.22 PTLAR (31H):set_partial_columns

8'H31	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	↑	-						cr_partial_strcol[10:8]			8'h00

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Par1	↑	-	cr_partial_strcol[7:0]	8'h00
Par2	↑	-	cr_partial_endcol[10:8]	8'h00
Par3	↑	-	cr_partial_endcol[7:0]	8'h00
Description			cr_partial_strcol [10:0]: locally displays the start column; cr_partial_endcol [10:0]: locally displays the end column.	

4.2.2.1.23 SA (33H): Scroll Area Command

8'H33	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	↑	-	cr_33_tfa[15:8]								8'h00
Par1	↑	-	cr_33_tfa[7:0]								8'h00
Par2	↑	-	cr_33_vsa[15:8]								8'h00
Par3	↑	-	cr_33_vsa[7:0]								8'h00
Par4	↑	-	cr_33_bfa[15:8]								8'h00
Par5	↑	-	cr_33_bfa[7:0]								8'h00
Description			If command table2 is cr_vscroll_en =1, the running lamp function is effective. cr_33_tfa: indicates the top fix area. cr_33_vsa: vertical scroll area. cr_33_bfa: bottom fix area.								

4.2.2.1.24 TEOFF (34H): Tearing Effect Line OFF

8'H34	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Description			Without parameters, Disables Tearing Effect output signal from TE signal line, i.e. tear_on = 0.								

4.2.2.1.25 TEON (35H): Tearing Effect Line On

8'H35	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	↑	-								cr_tear_mode	8'h00
Description			Issue 35h, i.e. tear_on = 1, Tearing Effect Line On. TEM Determining the Tearing Effect Output Line mode. 1. TEM =0: The Tearing Effect output line consists of V-Blanking information only. 2. TEM =1: The Tearing Effect output Line consists of both V-Blanking and H-Blanking information. The TE of H-blanking is determined by C5 of command table2: cr_te_gsps: start column position; cr_te_gspf: indicates the end column position.								

4.2.2.1.26 DCTR (36H): Display Control Command

8'H36	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	↑		MY	MX			BGR	MH	cr_ca	cr_gs	8'h00
Description			**cr_gs:** gip scan direction setting.								

cr_gs	cr_fv	result
0	0	FW, select D7 and D8 register groups

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0	1	BW, select D9 and DD register groups
1	0	FW, select D7 and D8 register groups
1	1	BW, select D9 and DD register groups

cr_ca: the data invert option for data_shift block.

dsh_shr = cr_dsh_shr ^ cr_ca ^ cr_mx ^ cr_mh

(when dsh_shr = 0, invert; when dsh_shr = 1, don't invert).

cr_dsh_shr	cr_ca	cr_mx=MX^ cr_36_opt[1]	cr_mh=MH^ cr_36_opt[4]	result
0	0	0	0	Normal
0	0	0	1	Left and right mirror
0	0	1	0	Left and right mirror
0	0	1	1	Normal
0	1	0	0	Left and right mirror
0	1	0	1	Normal
0	1	1	0	Normal
0	1	1	1	Left and right mirror
1	0	0（default）	0（default）	Left and right mirror
1	0	0	1	Normal
1	0	1	0	Normal
1	0	1	1	Left and right mirror
1	1	0	0	Normal
1	1	0	1	Left and right mirror
1	1	1	0	Left and right mirror
1	1	1	1	Normal

BGR：sub_pixel order of one pixel data setting，

cr_bgr_before_en	cr_bgr_en	BGR	cr_bgr_opt	result
0	0	0	0	RGB
0	0	0	1	RGB
0	0	1	0	RGB
0	0	1	1	RGB
0	1	0	0	RGB
0	1	0	1	BGR
0	1	1	0	BGR
0	1	1	1	RGB
1	0	0	0	RGB
1	0	0	1	BGR
1	0	1	0	BGR
1	0	1	1	RGB
1	1	0	0	RGB
1	1	0	1	RGB
1	1	1	0	RGB
1	1	1	1	RGB

MX：Left and right mirror, as cr ca, see cr ca for details.

MH：Left and right mirror, as cr ca, see cr ca for details.

MY：Up and down mirror

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MY	cr_36_opt[0]	result
0	0 (default)	Normal
0	1	Up and down mirror
1	0	Up and down mirror
1	1	Normal

Bit	NAME	DESCRIPTION
MY	Row Address Order	These 3bits controls MCU to memory write/read direction.
MX	Column Address Order	
MV	Row/Column Exchange	
ML	Vertical Refresh Order	LCD vertical refresh direction control '0' = LCD vertical refresh Top to Bottom '1' = LCD vertical refresh Bottom to Top
RGB	RGB-BGR ORDER	Color selector switch control '0' =RGB color filter panel, '1' =BGR color filter panel
MH	Horizontal Refresh Order	Horizontal direction '0' = Left to Right '1' = Right to Left

4.2.2.1.27 SS (37H): Scroll Start Command

8'H37	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	↑	-	cr_37_vsp[15:8]								8'h00
Par1	↑	-	cr_37_vsp[7:0]								8'h00
Description			**cr_37_vsp**：vertical scroll position.								

4.2.2.1.28 IDMOFF (38H): Idle Mode Off

8'H38	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Description			Without parameters, The Idle mode is displayed.								

4.2.2.1.29 IDMON (39H): Idle mode on

8'H39	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default

Description			Without parameters,
			The Idle mode is displayed, that is, the eight-color mode.
			Register configuration is as follows
			vedio mode: cr_idle_mode_en = 1
			command mode:
			cr_idle_mode_en=1;
			cr_corr_col_en = 0;
			cr_gray_high = 'h80.								

4.2.2.1.30 COLMOD (3AH): Interface Pixel Format

8'H3A	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default

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<table>
<tr><td>Par0</td><td>↑</td><td>-</td><td>-</td><td>dpi_format[2:0]</td><td>-</td><td>dbi_format [2:0]</td><td>8’h77</td></tr>
<tr><td>Description</td><td colspan="7">This command is used to <u>define the format of RGB picture data</u>, which is to be transferred <u>via the MCU interface</u>.
If the customized register cr_format_cmd3a_dpi_en=1 or cr_format_cmd3a_dbi_en=1, the pixel format is configured by the 3Ah command. Otherwise, the pixel format is configured by cr_pix_format
For mipi format, cr_pix_sel is set to 1
dpi_format[2:0]: DPI Pixel Format Definition（RGB）
dbi_format[2:0]: DBI Pixel Format Definition

The formats are shown in the table:
<table>
<tr><th>Bit</th><th>Description</th><th></th></tr>
<tr><td>D7</td><td>-</td><td>Set to ‘0’</td></tr>
<tr><td>D6</td><td rowspan="3">RGB interface color format</td><td rowspan="3">‘101’ = 16bit/pixel
‘110’ = 18bit/pixel</td></tr>
<tr><td>D5</td></tr>
<tr><td>D4</td></tr>
<tr><td>D3</td><td>-</td><td>Set to ‘0’</td></tr>
<tr><td>D2</td><td rowspan="3">Control interface color format</td><td rowspan="3">‘101’ = 16bit/pixel
‘110’ = 18bit/pixel
‘111’ = 24 bit/pixel</td></tr>
<tr><td>D1</td></tr>
<tr><td>D0</td></tr>
</table>
Deault: 16bits/pixel</td></tr>
</table>

4.2.2.1.31 STE (44H): Set Tear Scanline

<table>
<tr><td>8’H44</td><td>W</td><td>R</td><td>D7</td><td>D6</td><td>D5</td><td>D4</td><td>D3</td><td>D2</td><td>D1</td><td>D0</td><td>Default</td></tr>
<tr><td>Par0</td><td>↑</td><td>-</td><td colspan="5"></td><td colspan="4">cr_44te_str[11:8]</td><td>8’h00</td></tr>
<tr><td>Par1</td><td>↑</td><td>-</td><td colspan="9">cr_44te_str[7:0]</td><td>8’h00</td></tr>
<tr><td>Par2</td><td>↑</td><td>-</td><td colspan="5"></td><td colspan="4">cr_44te_end[11:8]</td><td>8’h00</td></tr>
<tr><td>Par3</td><td>↑</td><td>-</td><td colspan="9">cr_44te_end[7:0]</td><td>8’h00</td></tr>
<tr><td>Description</td><td colspan="11">-This command turns on the display module’s Tearing Effect output signal on the TE signal line when the display module reaches line N.
The TE signal is not affected by changing MV.
-The tearing effect line on has one parameter that describes the tearing effect output line mode.
-The tearing effect output line consist of V-blanking information only.
Note that set tear scanline with N=0 is equivalent to tearing effect line on with TEM=0.
The tearing effect output line shall be active low when the display module is in sleep mode。
cr_44te_str: start line;
cr_44te_end: indicates the end line
When the command table2 custom register cr_te_44en =1, the TE position is configured by the 44h command, otherwise it is determined by the cr_te_str and cr_te_end of command table2.</td></tr>
</table>

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4.2.2.1.32 GSCAN (45H): Get Scanline

8'H45	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	-	↑					cr_44te_str[11:8]				8'h00
Par1	-	↑	cr_44te_str[7:0]								8'h00
Par2	-	↑					cr_44te_end[11:8]				8'h00
Par3	-	↑	cr_44te_end[7:0]								8'h00
Description			-The display module returns the current scanline ,N, used to update the display device. The total number of scanlines on a display device is defined as VSYNC+VBP+VACT+VFP. The first scanline is defined as the first line of V Sync and is denoted as Line 0. -When in sleep in mode, the value returned by get scanline is undefined. Note: that Set Tear Scan Line with N = 0 is equivalent to Tearing Effect Line ON with M = 0. Read back the parameters of the 44 command								

4.2.2.1.33 BL (51H): Back Light Command

8'H51	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	↑	↑	cr_bl_usr								8'h80
Description			**cr_bl:**Adjust the backlight. cr_bl [7:0] is '0' when bit BCTRL of write CTRL display command (53h) is '0' cr_bl [7:0] is manual set brightness specified with write CTRL display command (53h) when bit BCTRL is '1' ; This function is to adjust the duty ratio of PWM.								

4.2.2.1.34 RBL (52H): Read Display Brightness Value

8'H52	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	-	↑	cr_bl								8'h80
Description			This command returns the brightness value of the display.								

4.2.2.1.35 WRCTRLD (53H): Write CTRL Display

8'H53	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	↑	-			cr_blctrl_en		cr_dd	bl_en			8'h00
Description			This command is used to control display brightness. cr_blctrl_en: Brightness Control Block On/Off, BCTRL=1 indicates that the brightness control function is enabled. cr_dd: Display Dimming On/Off. DD=1 indicates that the backlight climbing function is enabled. bl_en: Backlight Control On/Off, BL=1 indicates turn on the backlight. BL=0 means turn off the backlight immediately								

4.2.2.1.36 RDCTRLD (54H): Read CTRL value Display

8'H54	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	-	↑			cr_blctrl_en		cr_dd	bl_en			8'h00

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Description	This command returns ambient light and brightness control values. Read back the 53H command.

4.2.2.1.37 RDFCS (AaH):Read First Checksum

8'HAa	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	-	↑	fst_chk_sum[15:8]								8'hff
Par1	-	↑	fst_chk_sum[7:0]								8'hff
Description	This command returns the **first checksum** what has been calculated from **User's area registers** and **OTP** after the write access to those registers and/or has been done.										

4.2.2.1.38 RDCFCS (AfH):Read Continue Checksum

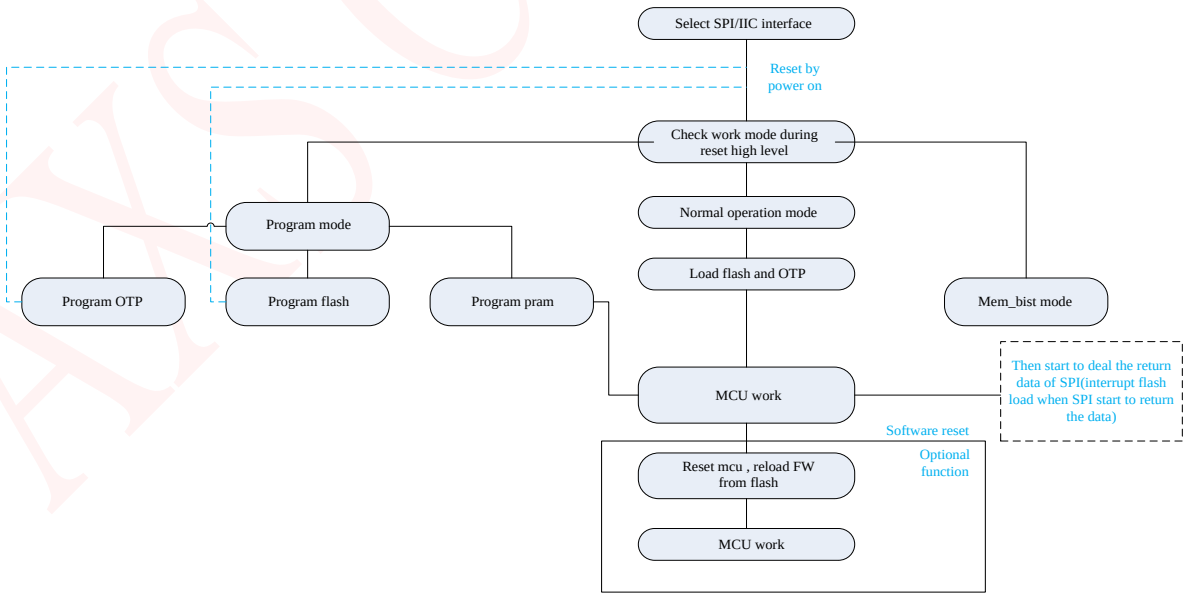
8'HAf	W	R	D7	D6	D5	D4	D3	D2	D1	D0	Default
Par0	-	↑	ctn_chk_sum[15:8]								8'hff
Par1	-	↑	ctn_chk_sum[7:0]								8'hff
Description	This command returns the **continue checksum** what has been **calculated continuously** after the first checksum has calculated from User's area registers and OTP after the write access to those registers and/or has been done.										

5 FUNCTIONS

5.1 Touch

5.1.1 TP work flow

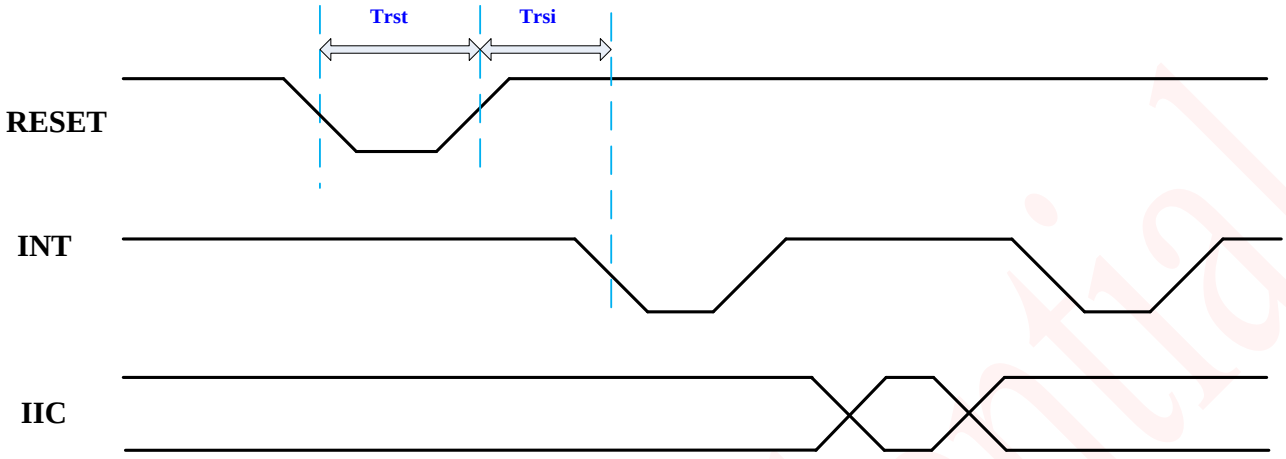
First, select the SPI or I2C interface to work through the work mode PIN. Then Write program into flash or pram with a different interface; Finally, please reset if writing program into flash or otp, and directly enter into MCU work mode if writing program into pram.



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5.1.2 Touch Reset (RSTN) input timing

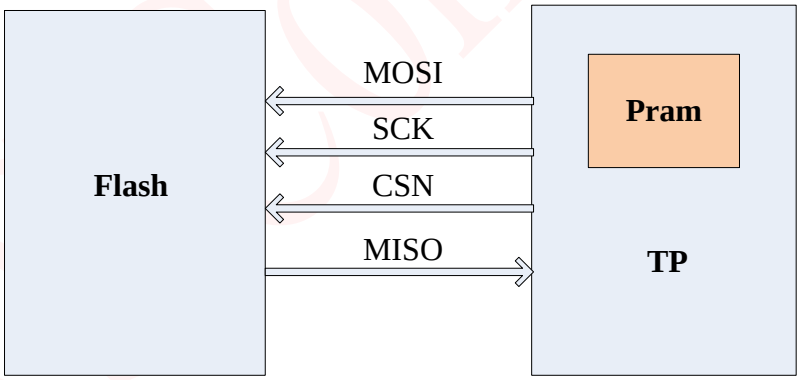
Reset time must be enough to guarantee reliable reset.



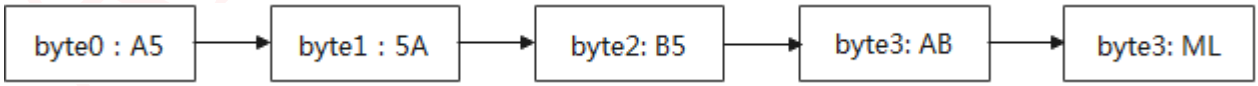
Parameter	Descrlption	Min	Max	Units
Trsi	Time of starting to report point after resetting	200	--	ms
Trst	Reset time	5	--	ms

5.1.3 Firmware booting

Touch firmware can boot from external flash or from AP optional, default is from external flash. Boot block diagram as follows.

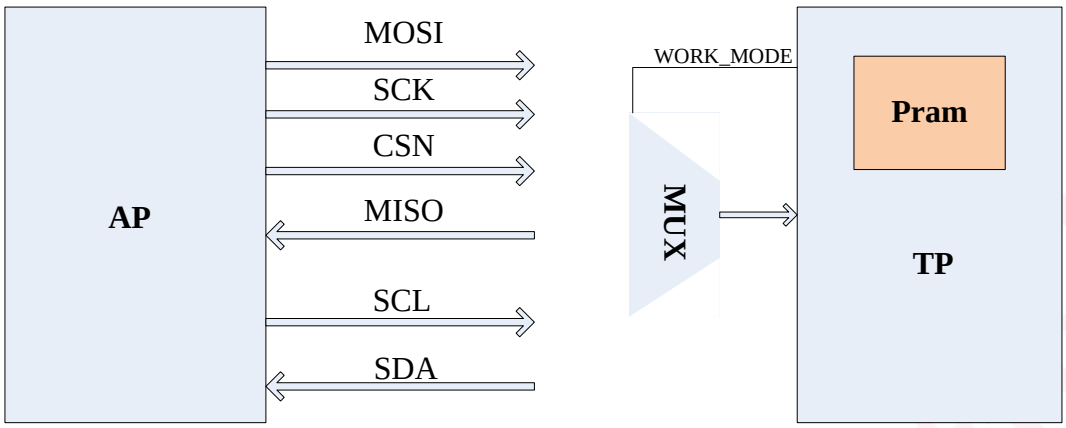


Touch controller can also boot firmware from AP through I2C or SPI interface. booting firmware protocol as follows,



byte0~byte3 is character string, it means enable booting. and ML is mode flag. when ML=1, select boot firmware from the host. it is an optional to select I2C or SPI interface through WORK_MODE pin. When WORK_MODE = 1, I2C active, otherwise, SPI active.

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5.2 Configure the color format of each interface

Color format				
Interface	cr_format_cmd3a_dpi_en=1 cr_format_cmd3a_dbi_en=X	cr_format_cmd3a_dpi_en=0 cr_format_cmd3a_dbi_en=1	cr_format_cmd3a_dpi_en=0 cr_format_cmd3a_dbi_en=0	
QSPI	iformat[2:0]=dpi_format (3ah: par0 的[6:4])	format[2:0]=dbi_format (3ah: par0 的[2:0])	format[2:0]=cr_pix_format (command table2)	format[2:0]=3‘b111/3'b100:RGB88 format[2:0]=3‘b101:RGB565 format[2:0]=3‘b010:RGB332 format[2:0]=3‘b001:RGB111 format[2:0]=3‘b000:gray mode else：reserved
SPI				format[2:0]=3‘b111:RGB888 else:RGB565
DBI				format[2:0]=3‘b111/3'b110:RGB88 format[2:0]=3‘b101:RGB565 format[2:0]=3‘b100:RGB666 else：reserved
RGB				format[2:0]=3‘b101:RGB565 format[2:0]=3‘b100:RGB666 esle:RGB88

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MIPI				format[2:0]=3‘b111:RGB888 format[2:0]=3‘b101:RGB565 format[2:0]=3‘b100:RGB666 else： reserved

5.3 MIPI-DSI Interface

Connect Pin:HS_DN,HS_DP,HS_CN,HS_CP,RSTN.The MIPI Interface is selected by setting the IM[3:0] pins as “1111” level.

5.3.1 General description

The communication can be separated into two different levels between the host and the display module:

– Interface Level: Low level communication, low power

– Packet level: High level communication, high speed

5.3.2 Interface level communication

5.3.2.1 General

The display module uses data and clock lane differential pairs for DSI. Both clock lane and data lane can be drivern in Low Power (LP) or High Speed (HS) mode.

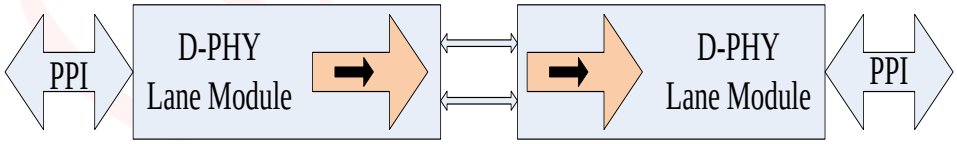
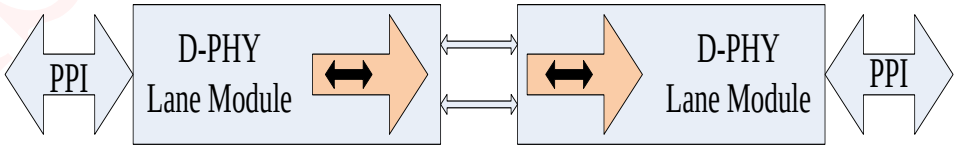
-	Lane support mode	MPU(Host) AXS15260D-S1(Slave)
Clock Lane	Unidirectional lane •High-Speed Clock only •SimplifiedEscape Mode (ULPS Only)	
Data Lane	Bi-directional lane • Forward high-speed only •Bi-directional Escape Mode •Bi-direction LPDT	

Table: Lane types and support mode

Low Power mode means that each line of the differential pair is used in single end mode and a differential receiver is disabled (The termination resistor of the receiver is disable) and it can be drivern into a low power mode.

High Speed mode means that differential pairs (The termination resistor of the receiver is

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enabled) are not used in the single end mode.

There are different modes and protocols in each mode when transferring information from the HOST to the display module and vice versa.

The State Codes of the High Speed (HS) and Low Power (LP) lane pair are defined below.

Lane Pair State Code	Line DC Voltage Levels		High Speed (HS)	Low-Power (LP)	
	Dn+ Line	Dn- Line	Burst Mode	Control Mode	Escape Mode
HS-0	Low (HS)	High (HS)	Differential-0	Note 1	Note 1
HS-1	High (HS)	Low (HS)	Differential-1	Note 1	Note 1
LP-00	Low (LP)	Low (LP)	Not Defined	Bridge	Space
LP-01	Low (LP)	High (LP)	Not Defined	HS-Request	Mark-0
LP-10	High (LP)	Low (LP)	Not Defined	LP-Request	Mark-1
LP-11	High (LP)	High (LP)	Not Defined	Stop	Note 2

Table: High Speed and Low-Power Lane Pair State Descriptions

5.3.2.2 DSI-CLK lanes

DSI-CLK+/- lanes can be drivern into three different power modes: Low Power Mode (LPM LP-11), Ultra Low Power Mode (ULPM) or High Speed Clock Mode (HSCM).

Clock lanes are in a single end mode (LP = Low Power) when there is entering or leaving Low Power Mode(LPM) or Ultra Low Power Mode (ULPM).

Clock lanes are in the single end mode (LP = Low Power) when there is entering in or leaving out High Speed Clock Mode (HSCM).

These entering and leaving protocols are using clock lanes in the single end mode to generate an entering or leaving sequences.

The principal flow chart of the different clock lanes power modes is illustrated below.

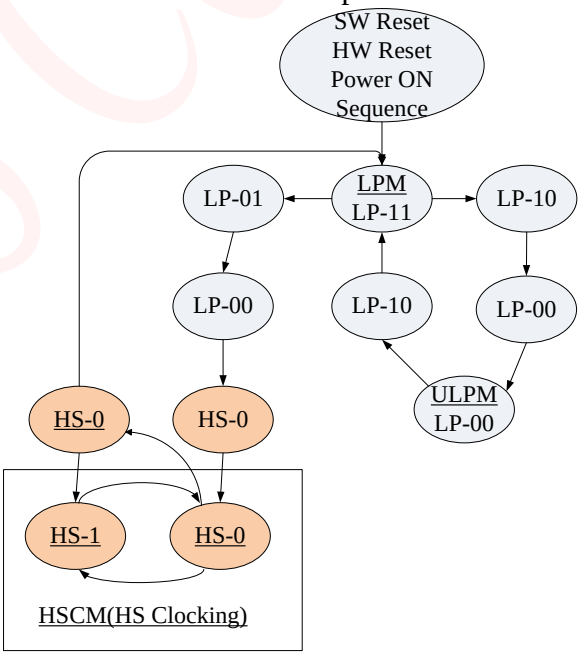


Figure: Clock Lanes Power Modes

Notes:

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1. Low-Power Receivers (LP-Rx) of the lane pair are checking the LP-00 state code, when the Lane Pair is in the High Speed (HS) mode.

2. If Low-Power Receivers (LP-Rx) of the lane pair recognizes LP-11 state code, the lane pair returns to LP-11 of the Control Mode.

Low Power Mode (LPM)

DSI-CLK+/- lanes can be drivern to the Low Power Mode (LPM), when DSI-CLK lanes are entering the LP-11 State Code, in three different ways:

1) After SW Reset, HW Reset or Power On Sequence =>LP-11

2) After DSI-CLK+/- lanes are leaving Ultra Low Power Mode (ULPM, LP-00 State Code) =>LP-10 =>LP-11 (LPM). This sequence is illustrated below.

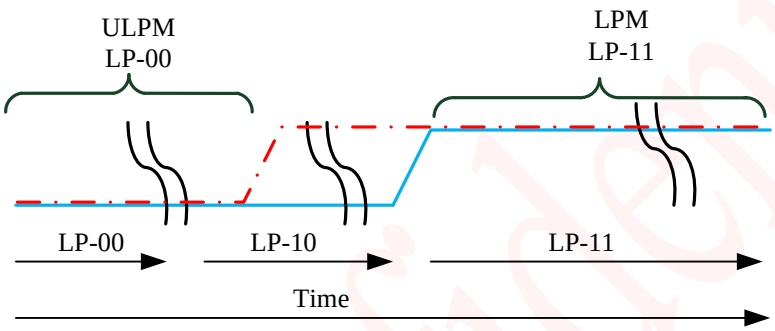


Figure:From ULPM to LPM

After DSI-CLK+/- lanes are leaving High Speed Clock Mode (HSCM, HS-0 or HS-1 State Code) =>HS-0 =>LP-11 (LPM). This sequence is illustrated below.

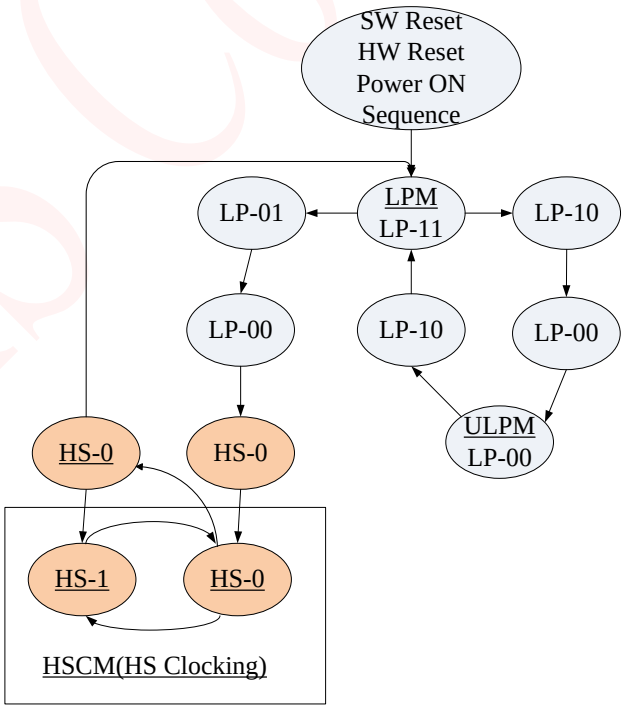


Figure: From HSCM to LPM

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All three mode changes are illustrated a flow chart below.

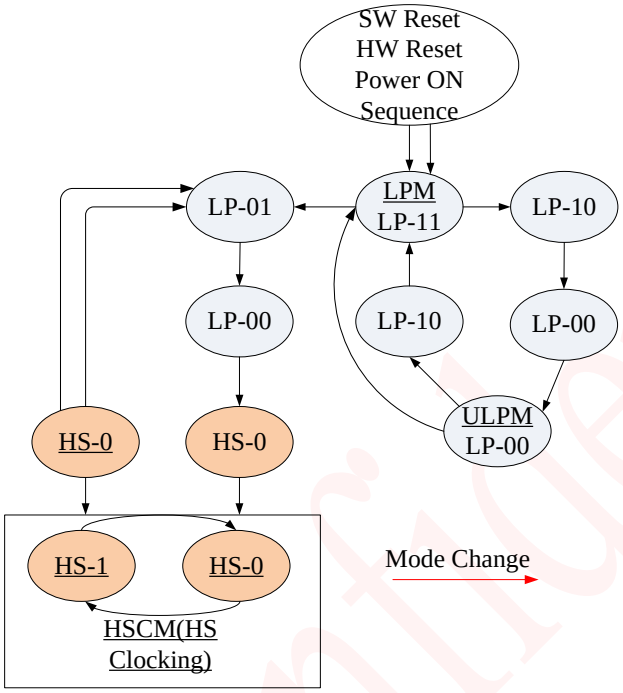


Figure: All three mode changes to LPM

Ultra Low Power Mode (ULPM)

DSI-CLK+/- lanes can be drivern to the High Speed Clock Mode (HSCM), when DSI-CLK lanes are starting to work between HS-0 and HS-1 State Codes.

The only entering possibility is from the Low Power Mode (LPM, LP-11 State Code) =>LP-01 =>LP-00 =>HS-0 =>HS-0/1 (HSCM). This sequence is illustrated below.

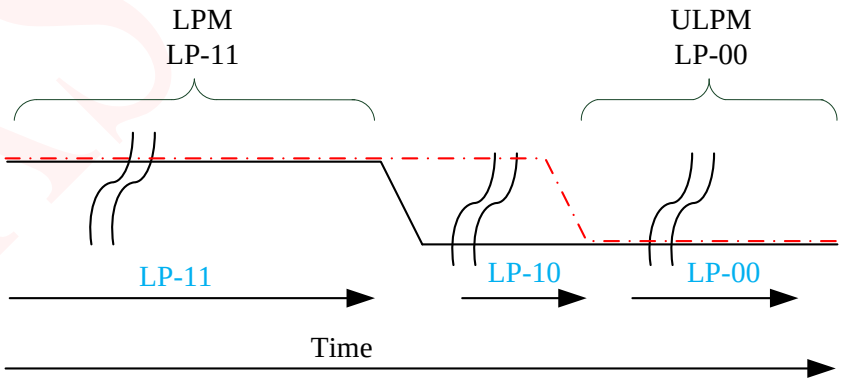


Figure: From LPM to UPLM

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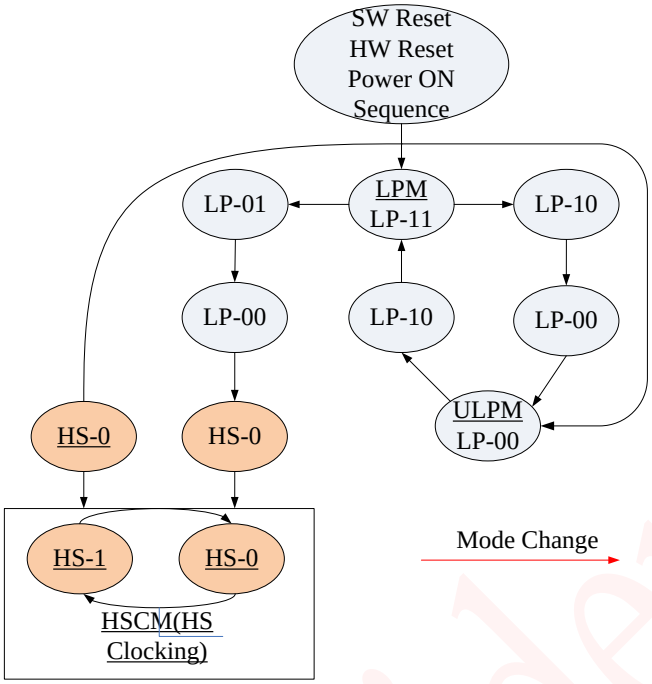


Figure:The mode change from LPM to UPLM

High-speed Clock Mode (HSCM)

DSI-CLK+/- lanes can be drivern to the High Speed Clock Mode (HSCM), when DSI-CLK lanes are starting to work between HS-0 and HS-1 State Codes.

The only entering possibility is from the Low Power Mode (LPM, LP-11 State Code) =>LP-01 =>LP-00 =>HS-0 =>HS-0/1 (HSCM). This sequence is illustrated below.

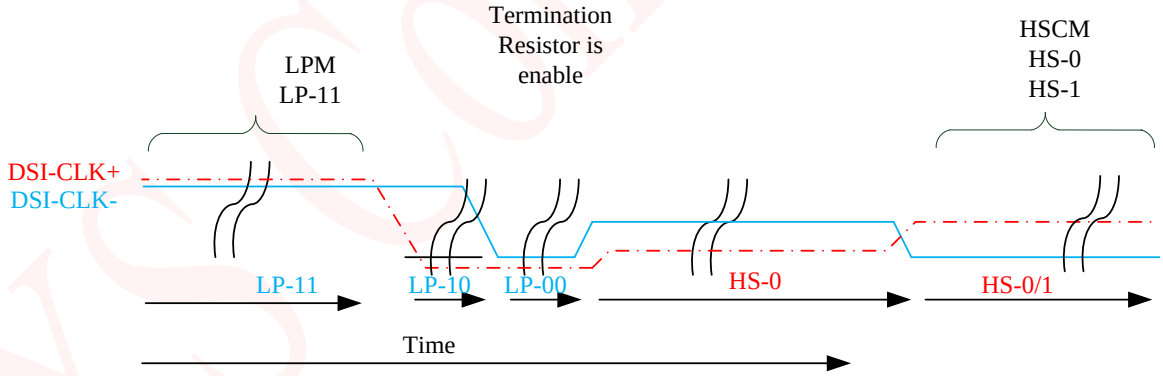


Figure: From LPM to HSCM

The mode change is also illustrated below:

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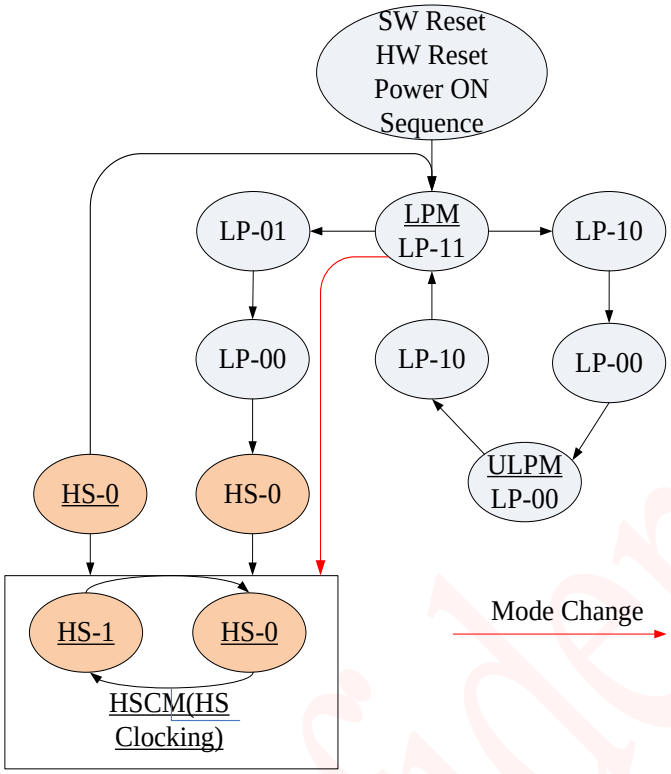


Figure:Mode change from LPM to HSCM

The high speed clock (DSI-CLK+/-) is started before high speed data is sent via DSI-Dn+/- lanes. The high speed clock continues clocking after the high speed data sending has been stopped

The burst of the high speed clock consists of:

– Even the number of transitions

– Start state is HS-0

– End state is HS-0

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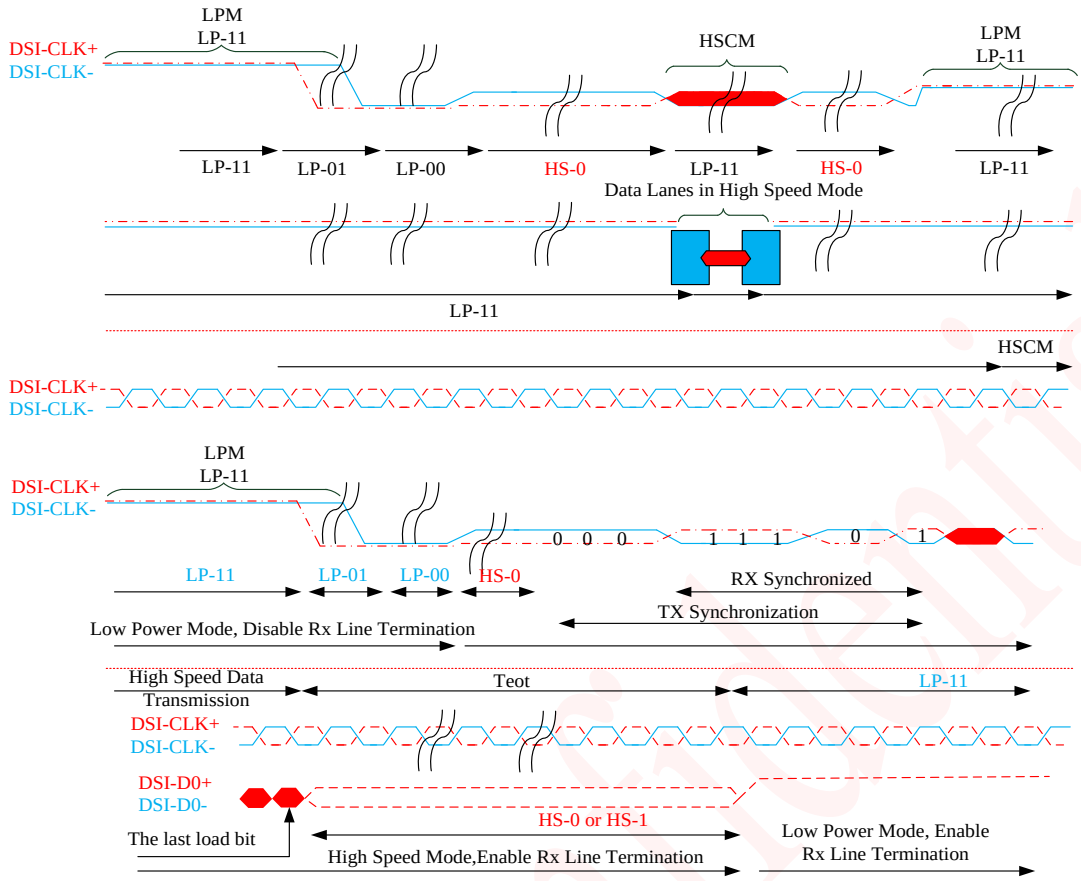


Figure: High speed clock burst

5.3.3 DSI data lanes

5.3.3.1 General

DSI-Dn+/- Data Lanes can be drivern in different modes which are:
– Escape Mode (only support DSI_D0 data lane pair)
– High-Speed Data Transmission (support all data lane pairs)
– Bus Turnaround Request (only support DSI_D0 data lane pair)
These modes and their entering codes are defined in the following table.

Mode	Entering Mode Sequence	Leaving Mode Sequence
Escape Mode	LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00	L P-00 =>LP-10 =>LP-11 (Mark-1)
High-Speed Data Transmission	LP-11 =>LP-01 =>LP-00 =>HS-0	(HS-0 or HS-1) =>LP-11
Bus Turnaround Request	LP-11 =>LP-10 =>LP-00 =>LP-10 =>LP-00	High-Z, Note

Table: Entering and leaving sequences

5.3.3.2 Escape modes

Escape mode is a special mode of operation for Data Lanes using Low-Power states. With this mode, some additional functionality becomes available. Escape mode operation shall be supported in the Forward direction and Reverse direction.

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The basic sequence of the Escape Mode is as follows

• Start: LP-11.

• Escape Mode Entry: LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00.

• Escape Command, which is coded, when one of the data lanes is changing from low-to-high-to-low, then this changed data lane is presenting a value of the current data bit.

• A payload stream if it is needed.

• Exit Escape (Mark-1) LP-00 =>LP-10 =>LP-11.

• End: LP-11.

For Data Lane0, once Escape mode is entered, the transmitter shall send an 8-bit entry command to indicate the requested action. All currently available Escape mode commands and actions are listed below.

• Send or receive “Low-Power Data Transmission” (LPDT).

• Driver data lanes to “Ultra-Low Power State” (ULPS).

• Indicate “Remote Application Reset” (RAR), which is resetting the display module (same as S/W Reset function).

• Indicate “Tearing Effect” (TEE), which is used for a TE line event from the display module to the HOST.

• Indicate “Acknowledge” (ACK), which is used for a non-error event from the display module to the HOST.

The Stop state shall be used to exit Escape mode and cannot occur during Escape mode operation because of the Spaced-One-Hot encoding. Stop state immediately returns the Lane to Control mode. If the entry command doesn’t match a supported command, that particular Escape mode action shall be ignored and the receive side waits until the transmit side returns to the Stop state. For Data Lane1 and 2, only support ULPS Escape mode commands.

• Driver data lanes to “Ultra-Low Power State” (ULPS).

The basic construction is illustrated below:

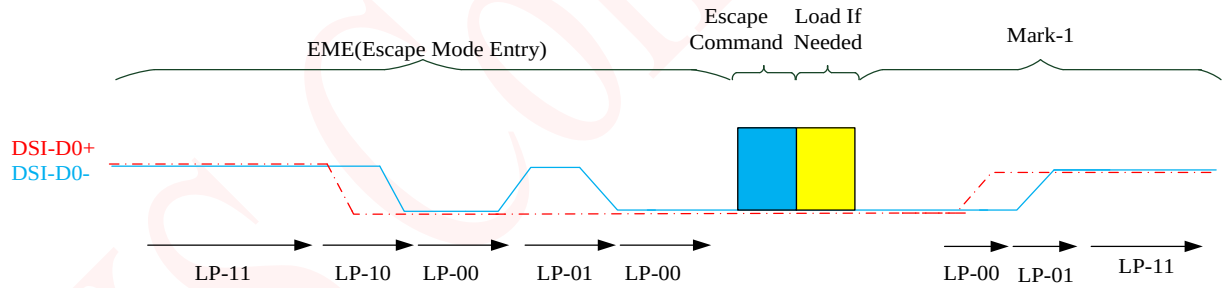


Figure: General Escape mode sequence

The number of the different Escape Commands is eight. These eight different Escape Commands can be divided to 2 different groups: Mode or Trigger. Escape command and groups are defined below.

Escape Command	Command Type Mode/Trigger	Entry Command Pattern (First Bit => Last Bit Transmitted)
Low-Power Data Transmission	Mode	1110 0001bin
Ultra-Low Power Mode	Mode	0001 1110bin
Remote Application Reset	Trigger	0110 0010 bin
Tearing Effect	Trigger	0101 1101 bin

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Acknowledge	Trigger	0010 0001 bin

Table: Escape commands

The HOST is inform the display module that it is controlling data lanes (DSI-D0+/-) with the Mode e.g. The HOST can inform the display module that it can put data lanes in the low power mode.

The HOST is waiting from the display module event information, which has been set by the HOST , with the Trigger e.g. when the display module reaches a new V-synch, the display module sent the HOST a TE trigger (TEE), if the HOST has been requested it.

Low-Power Data Transmission (LPDT)

The HOST can send data to the display module in Low-Power Data Transmission (LPDT) mode when data lanes are entering in Escape Mode and Low-Power Data Transmission (LPDT) command has been sent to the display module. The display module is also using the same sequence when it is sending data to the HOST .

The Low Power Data Transmission (LPDT) is using a following sequence:

• Start: LP-11

• Escape Mode Entry : LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00

• Low-Power Data Transmission (LPDT) command in Escape Mode: 1110 0001 (First to Last bit)

• Payload (Data):

- One or more bytes

- Data lanes are in pause mode when data lanes are stopped (Both lanes are low) between bytes

• Mark-1: LP-00 =>LP-10 =>LP-11

• End: LP-11

This sequence is illustrated for reference purposes below:

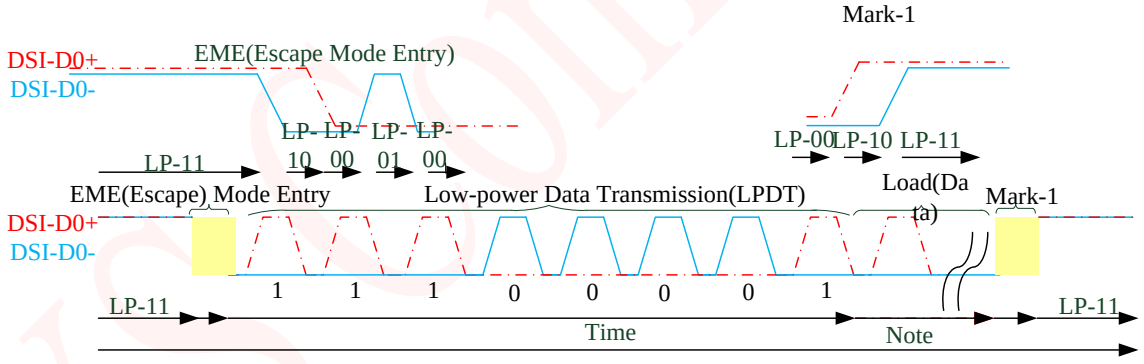


Figure: Low-power data transmission

Note: Load(Data) is presenting the first bit is logical “1” in this example

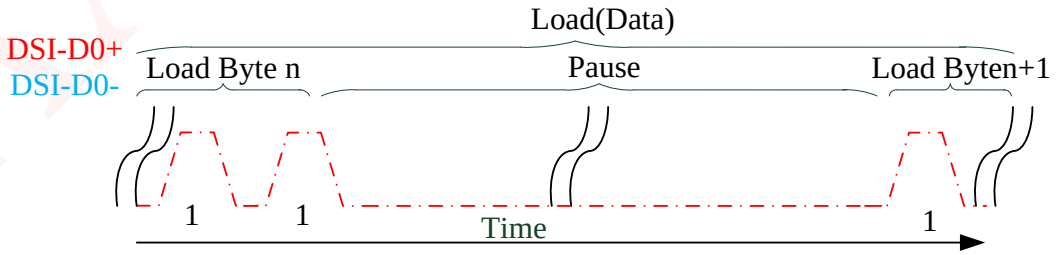


Figure: Pause (example)

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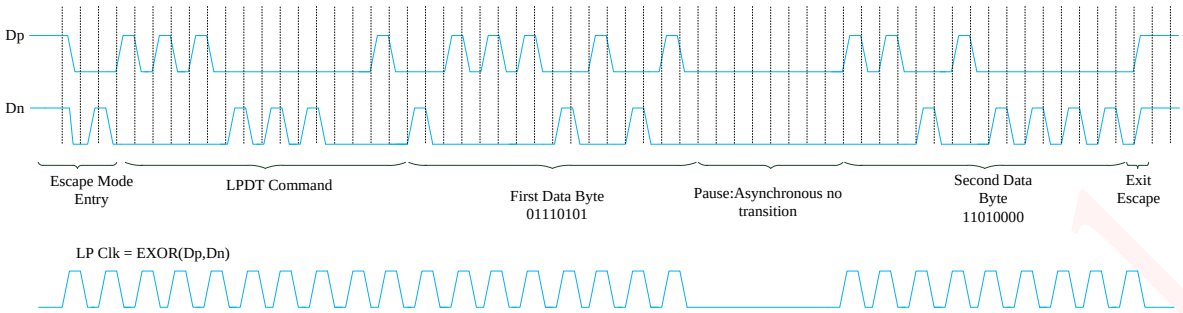


Figure: Two Data Byte Low-Power Data Transmission Example

Ultra-Low Power State (ULPS)

The HOST can force data lanes in Ultra-Low Power State (ULPS) mode when data lanes are entering in Escape Mode.

The Ultra-Low Power State (ULPS)used/uses(?) the following sequence:

- Start: LP-11
- Escape Mode Entry : LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Ultra-Low Power State (ULPS) command in Escape Mode: 0001 1110 (First to Last bit)
- Ultra-Low Power State (ULPS) when the HOST is keeping data lanes low
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:

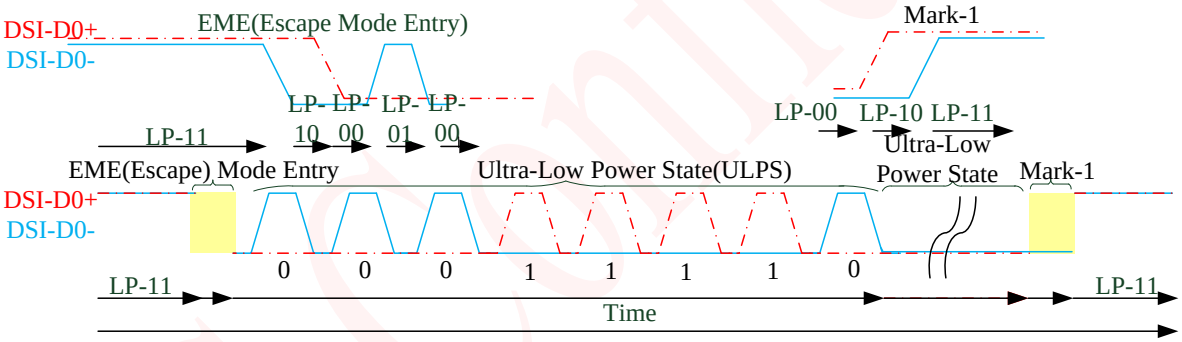


Figure: Ultra-low power state (ULPS)

Remote Application Reset (RAR)

The HOST can inform the display module that it should be reseted in Remote Application Reset (RAR) trigger when data lanes are entering in Escape Mode. The Remote Application Reset using the following sequence:

- Start: LP-11
- Escape Mode Entry : LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
- Remote Application Reset (RAR) command in Escape Mode: 0110 0010 (First to Last bit)
- Mark-1: LP-00 =>LP-10 =>LP-11
- End: LP-11

This sequence is illustrated for reference purposes below:

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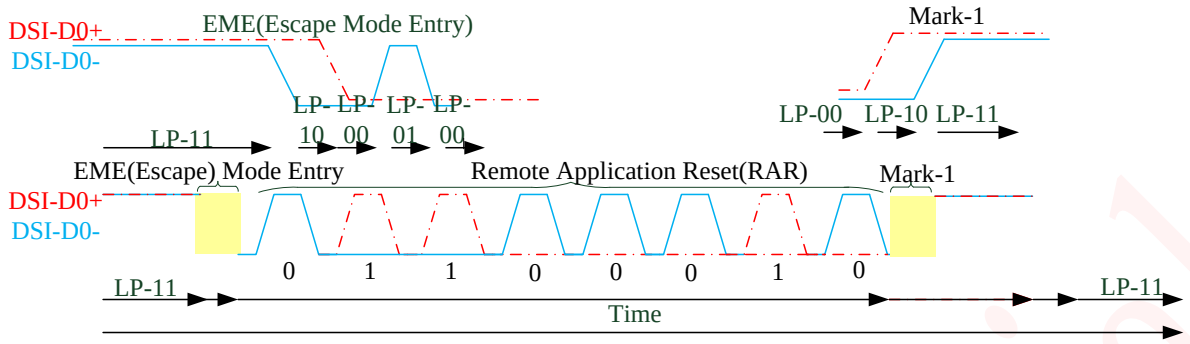


Figure: Remote Application Reset (RAR)

Tearing Effect (TEE)

The display module can inform the HOST when a tearing effect event (New V-synch) has been happen on the display module by Tearing Effect (TEE).

The Tearing Effect (TEE) is using a following sequence:

• Start: LP-11
• Escape Mode Entry: LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
• Tearing Effect (TEE) trigger in Escape Mode: 0101 1101 (First to Last bit)
• Mark-1: LP-00 =>LP-10 =>LP-11
• End: LP-11

This sequence is illustrated for reference purposes below:

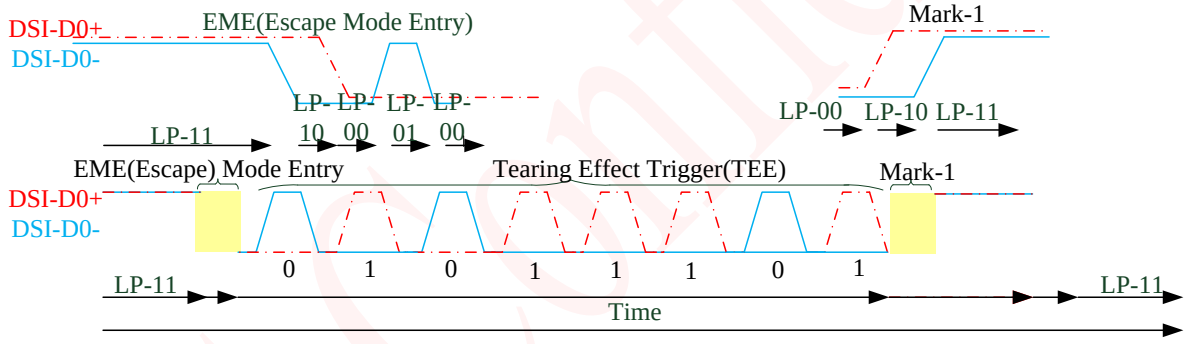


Figure: Tearing effect(TEE)

Acknowledgement (ACK)

The display module can inform the HOST when an error has not recognized on it by Acknowledge (ACK).

The Acknowledge (ACK) used/uses(?) the following sequence:

• Start: LP-11
• Escape Mode Entry: LP-11 =>LP-10 =>LP-00 =>LP-01 =>LP-00
• Acknowledge (ACK) command in Escape Mode: 0010 0001 (First to Last bit)
• Mark-1: LP-00 =>LP-10 =>LP-11
• End: LP-11

This sequence is illustrated for reference purposes below:

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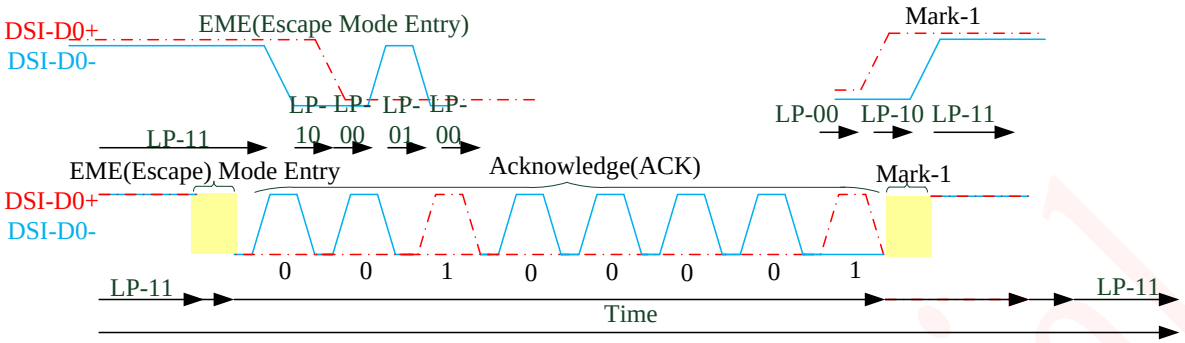


Figure: Acknowledgement (ACK)

5.3.3.3 High-Speed Data Transmission (HSDT)

Entering High-Speed Data Transmission (Tsot of HSDT)

The display module is entering High-Speed Data Transmission (HSDT) when Clock lanes DSI-CLK+/- have already been entered in the HighSpeed Clock Mode (HSCM) by the HOST. See more information on chapter “High-Speed Clock Mode (HSCM)”.

Data lanes DSI-D0+/- of the display module are entering (TSOT) in the High-Speed Data Transmission(HSDT) as follow

- Start: LP-11
- HS-Request: LP-01
- HS-Settle: LP-00 => HS-0 (Rx: Lane Termination Enable)
- Rx Synchronization: 011101 (Tx (= HOST) Synchronization: 0001 1101)
- End: High-Speed Data Transmission (HSDT) – Ready to receive High-Speed Data Load

This same entering High-Speed Data Transmission (TSOT of HSDT) sequence is illustrated below.

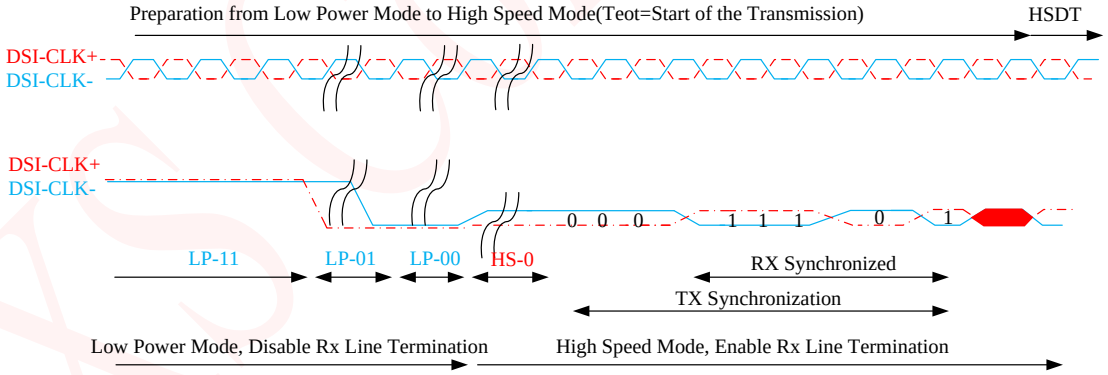


Figure: Tsot of HSDT

Leaving High-Speed Data Transmission (TEOT of HSDT)

The display module is leaving the High-Speed Data Transmission (TEOT of HSDT) when Clock lanes DSI- CLK+/- are in the High-Speed Clock Mode (HSCM) by the HOST and this HSCM is kept until data lanes DSI-D0+/- are in LP-11 mode. See more information on chapter “7.2.2 High-Speed Clock Mode (HSCM)”. Data lanes DSI-D0+/- of the display module are leaving from the High-Speed Data Transmission (TEOT of HSDT) as follow

- Start: High-Speed Data Transmission (HSDT)
- Stops High-Speed Data Transmission

-HOST changes to HS-1, if the last load bit is HS-0

-HOST changes to HS-0, if the last load bit is HS-1

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• End: LP-11 (Rx: Lane Termination Disable)

The same leaving High-Speed Data Transmission (TEOT of HSDT) sequence is illustrated below

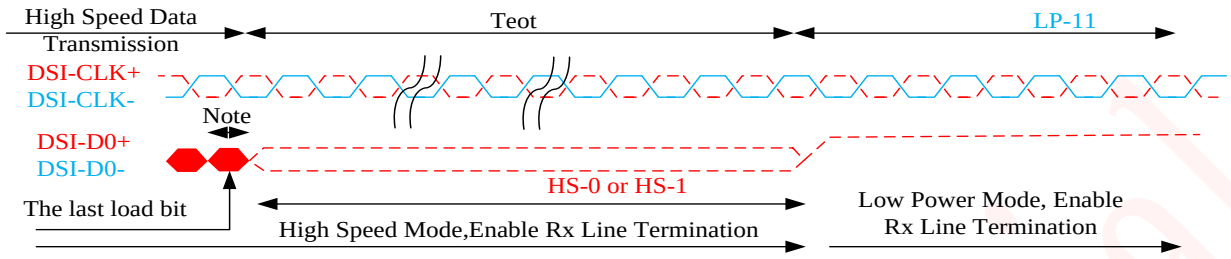


Figure: TEOT of HSDT

Note:
If the last load bit is HS0, the transmitter changes from HS0 to HS-1.
If the last load bit is HS1, the transmitter changes from HS1 to HS-0.

Burst of the High-Speed Data Transmission (HSDT)

The burst of the high-speed data transmission (HSDT) can consist of one data packet or several data packets. These data packets can be Long (LPa) or Short (SPa) packets. These packets are defined on chapter "Short Packet (SPa) and Long Packet (LPa) Structures". These different burst of the High-Speed Data Transmission (HSDT) cases are illustrated for reference purposes below.

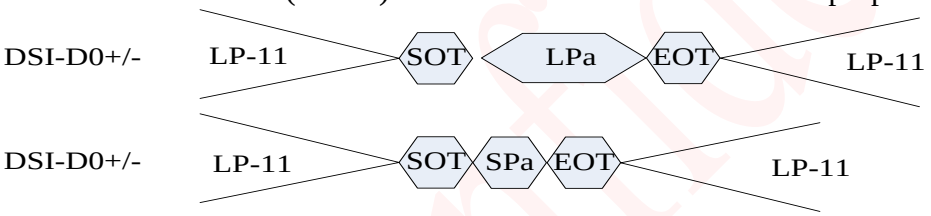


Figure: Single packet in HSDT

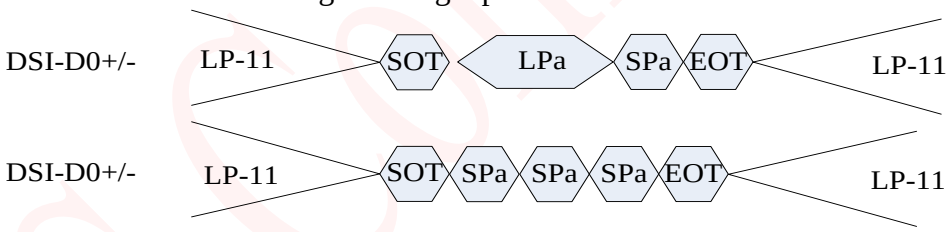


Figure: Multiple packets in HSDT

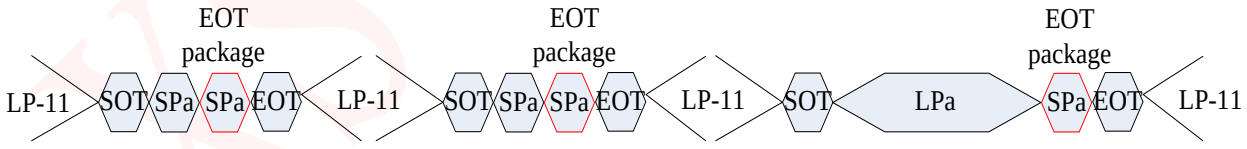


Figure: Packets with EoT package in HSDT

Abbreviation	Explanation
LP-11	Low Power Mode, Data lanes are '1's (Stop Mode)
SOT	Start of the Transmission
LPa	Long Packet
SPa	Short Packet
EOT	End of the Transmission

Table: Abbreviations

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5.3.3.4 Bus Turnaround (BTA)

The HOST or display module, which is controlling DSI-D0+/- Data Lanes, can start a bus turnaround procedure when it wants information to a receiver.

The HOST and display module can use the same sequence when this bus turnaround procedure is used.

This sequence is described for reference purposes, when the HOST wants to do the bus turnaround procedure to the display module, as follows.

• Start (HOST): LP-11

• Turnaround Request (HOST): LP-11 =>LP-10 =>LP-00

• The HOST waits until the display module is starting to control DSI-D0+/- data lanes and the HOST stops to control DSI-D0+/- data lanes (= High-Z)

• The display module changes to the stop mode: LP-00 =>LP-10 =>LP-11

The same bus turnaround procedure (From the HOST to the display module) is illustrated below.

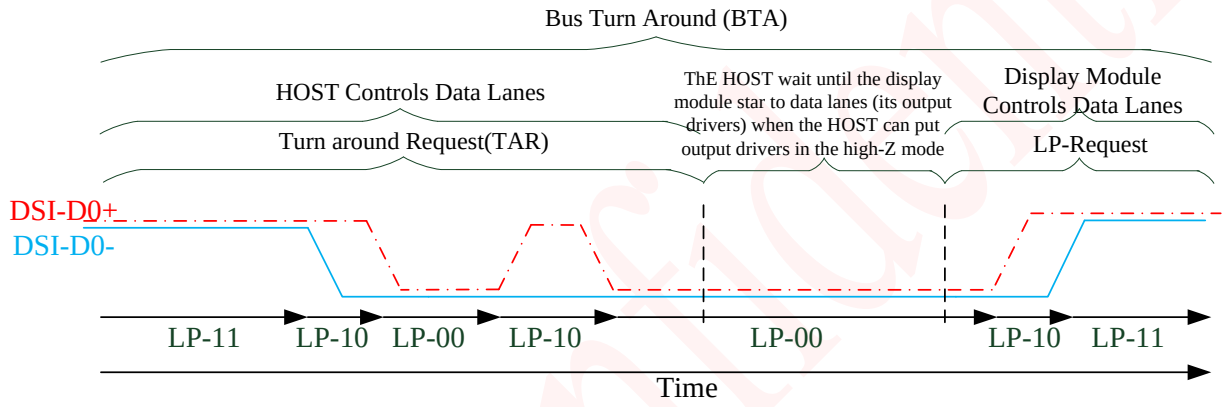


Figure: Bus turnaround procedure

5.3.4 Packet level communication

5.3.4.1 Short Packet (SPa) and Long Packet (LPa) structures

Short Packet (SPa) and Long Packet (LPa) are always used when data transmission is done in Low Power Data Transmission (LPDT) or High-Speed Data Transmission (HSDT) modes.

The lengths of the packets are

- Short Packet (SPa): 4 bytes
- Long Packet (LPa): From 6 to 65,541 bytes

The type (SPa or LPa) of the packet can be recognized from their package headers (PH).

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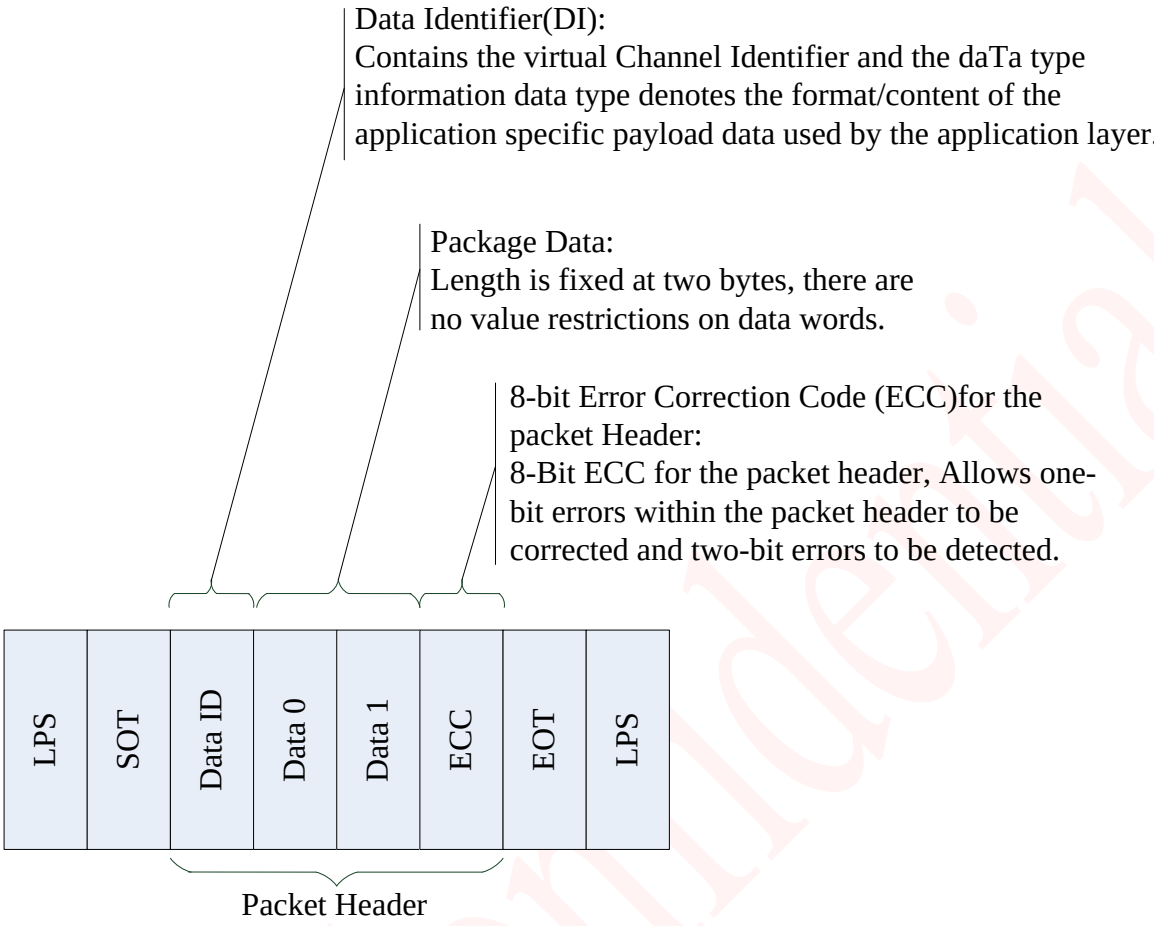


Figure: Short packet structure

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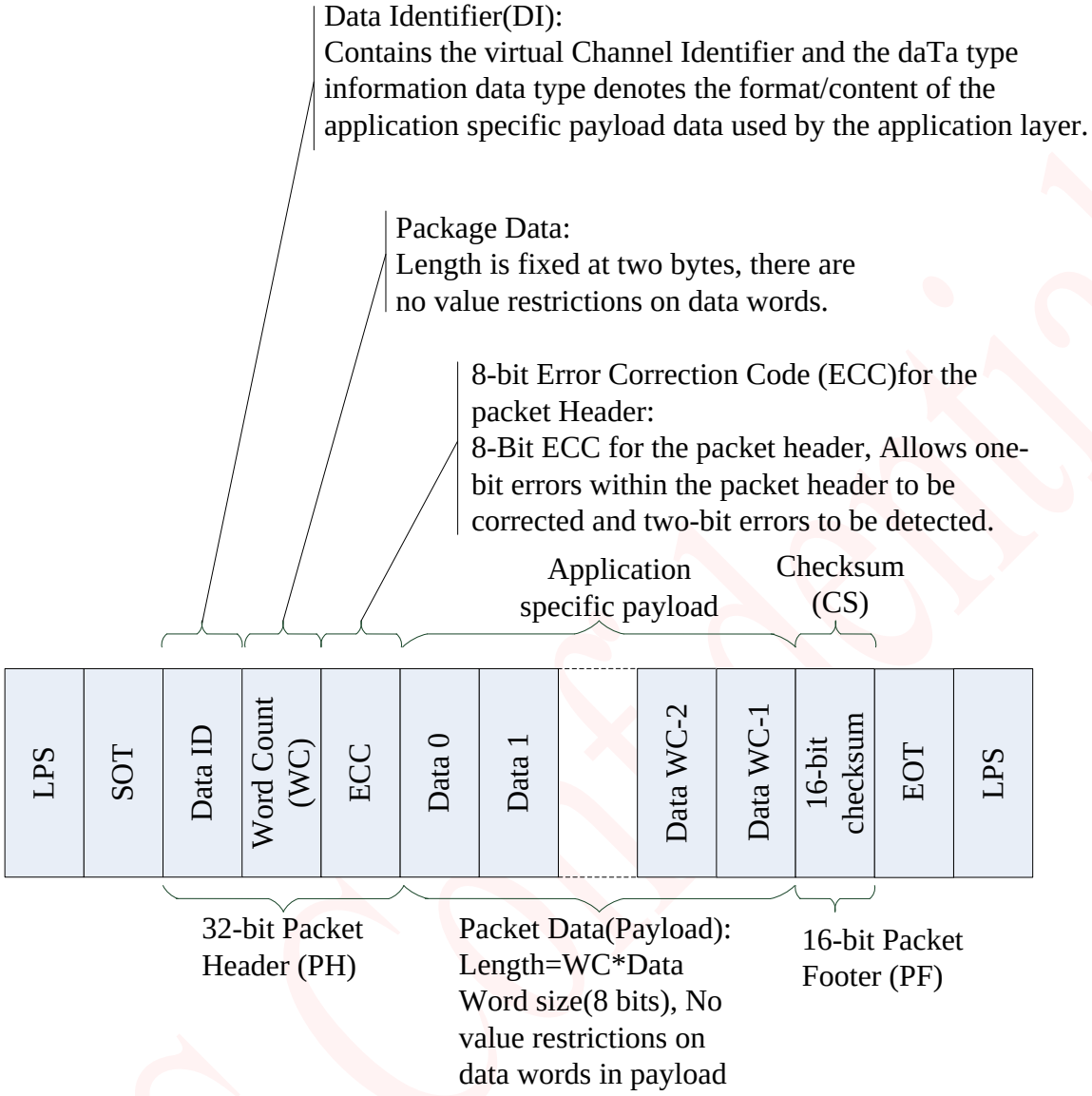


Figure:. Long packet structure

Note: Short Packet (SPa) Structure" and Long Packet (LPa) Structure" are presenting a single packet sending (= Includes LP-11, SoT and EoT for each packet sending).

The other possibility is that there is not needed SoT, EoT and LP-11 between packets if packets have sent in multiple packet format. e.g.:

- LP-11 =>SoT =>SPa =>LPa =>SPa =>SPa =>EoT =>LP-11
- LP-11 =>SoT =>SPa =>SPa =>SPa =>EoT =>LP-11
- LP-11 =>SoT =>LPa =>LPa =>LPa =>EoT =>LP-11

Bit Order of the Byte on Packets

A byte is the smallest transmission unit of a packet. Sequentially, a transmitter shall send data LSB first, MSB last. For packets with multi-byte fields, the least significant byte shall be transmitted first unless otherwise specified. Figure below shows a complete Long packet data transmission. Note, the figure shows the byte values in standard positional notation, i.e. MSB on the left and LSB on the right, while the bits are shown in chronological order with the LSB on the left, the MSB on the right and time increasing left to right.

DI	WC(Least Significant Byte)	ECC	Data	CRC(LS Byte)	CRC(MS Byte)

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39hex								01hex								15hex								01hex								0Ehex								1Ehex							
1	0	0	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	1	1	1	1	0	0	0
LSB							MSB	LSB							MSB	LSB							MSB	LSB							MSB	LSB							MSB	LSB							MSB

Figure: Bit order of the byte on packets

Byte Order of the Multiple Byte Information on Packets

Byte order of the multiple bytes information, what is used on packets, is that the Least Significant (LS) Byte of the information is sent in the first and the Most Significant (MS) Byte of the information is sent in the last. e.g. Word Count (WC) consists of 2 bytes (16 bits) when the LS byte is sent in the first and the MS byte is sent in the last.

This same order is illustrated for reference purposes below.

WC(Least Significant Byte)								WC(Most Significant Byte)							
01hex								00hex							
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	B14	B15
LSB							MSB	LSB							MSB

Figure: Byte order of the multiple byte information on packets

Packet Head (PH)

The packet header is always consisting of 4 bytes. The content of these 4 bytes are different when it is used to Short Packet (SPa) or Long Packet (LPa).

Short Packet (SPa):

- 1st byte: Data Identification (DI) => Identification that this is Short Packet (SPa)
- 2nd and 3rd bytes: Packet Data (PD), Data 0 and 1
- 4th byte: Error Correction Code (ECC)

DI								Data 0								Data 1								ECC							
15hex								3Ahex								07hex								18hex							
1	0	1	0	1	0	0	0	0	1	0	1	1	1	0	0	1	1	1	0	0	0	0	0	0	0	0	1	1	0	0	0
B0	B1	B2	B3	B4	B5	B6	B7	B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	B14	B15	B0	B1	B2	B3	B4	B5	B6	B7
LSB							MSB	LSB							MSB	LSB							MSB	LSB							MSB

Figure: Packet head on short packet

Long Packet (LPa):

- 1st byte: Data Identification (DI) => Identification that this is Long Packet (LPa)
- 2nd and 3rd bytes: Word Count (WC)
- 4th byte: Error Correction Code (ECC)

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DI								WC(Least Significant Byte)								WC(Most Significant Byte)								ECC							
39hex								01hex								00hex								15hex							
1	0	0	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	0	0
B0	B1	B2	B3	B4	B5	B6	B7	B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	B14	B15	B0	B1	B2	B3	B4	B5	B6	B7
LSB							MSB	LSB							MSB	LSB							MSB	LSB							MSB

Figure: Packet head on long packet

Data Identification

Data Identification (DI) is a part of Packet Header (PH) and it consists of 2 parts:

• Virtual Channel (VC), 2 bits, DI[7...6]

• Data Type (DT), 6 bits, DI[5…0]

The Data Identification (DI) structure is illustrated on a table below.

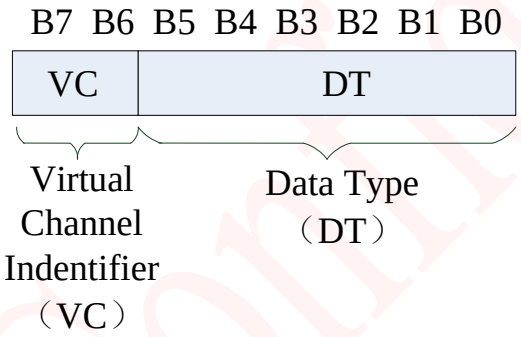


Table: Data identification structure

Data Identification (DI) is illustrated on Packet Header (PH) for reference purposes below.

DI								WC(Least Significant Byte)								WC(Most Significant Byte)								ECC							
39hex								01hex								00hex								15hex							
1	0	0	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	0	0
B0	B1	B2	B3	B4	B5	B6	B7	B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	B14	B15	B0	B1	B2	B3	B4	B5	B6	B7
LSB							MSB	LSB							MSB	LSB							MSB	LSB							MSB

Figure: Data identification of the packet head

Virtual Channel (VC)

A processor may service up to four peripherals with tagged commands or blocks of data, using the Virtual Channel ID field of the header for packets targeted at different peripherals.

Virtual Channel (VC) is a part of Data Identification (DI[7…6]) structure and it is used to address where a packet is wanted to send from the MCU. Bits of the Virtual Channel (VC) are

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illustrated for reference purposes below.

AXS15260D-S1 only support VC code=00, package with other VC code(01/10/11) will be filter out.

DI								WC(Least Significant Byte)								WC(Most Significant Byte)								ECC							
39hex								01hex								00hex								15hex							
1	0	0	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	0	0
B0	B1	B2	B3	B4	B5	B6	B7	B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	B14	B15	B0	B1	B2	B3	B4	B5	B6	B7
LSB							MSB	LSB							MSB	LSB							MSB	LSB							MSB

Figure: Virtual channel on the packet head

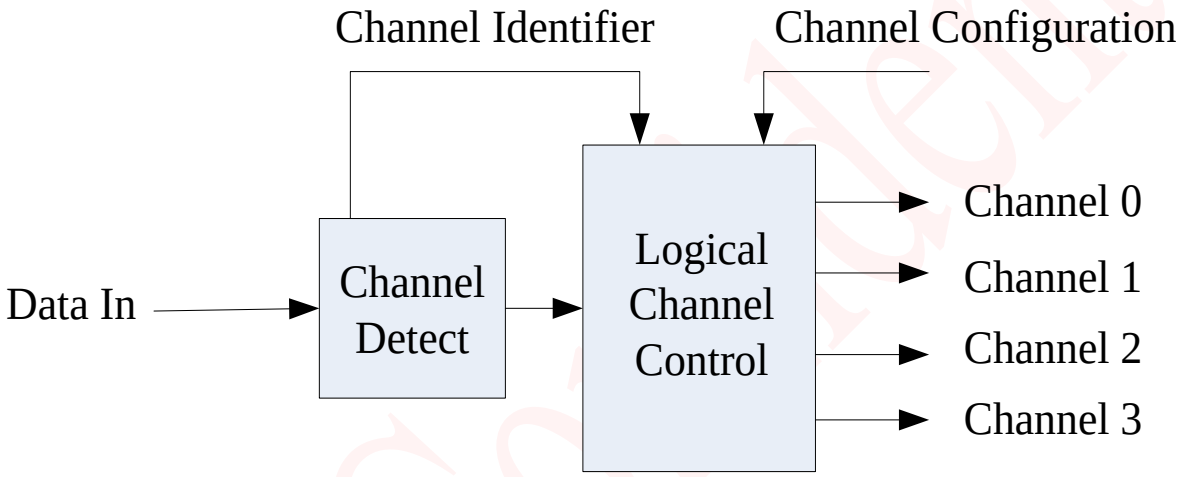


Figure: Virtual channel block diagram (receiver case)

Data Type (DT)

Data Type (DT) is a part of Data Identification (DI[5…0]) structure and it is used to define a type of the used data on a packet.

Bits of the Data Type (DT) are illustrated for reference purposes below.

DI								WC(Least Significant Byte)								WC(Most Significant Byte)								ECC							
39hex								01hex								00hex								15hex							
1	0	0	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	0	0
B0	B1	B2	B3	B4	B5	B6	B7	B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	B14	B15	B0	B1	B2	B3	B4	B5	B6	B7
LSB							MSB	LSB							MSB	LSB							MSB	LSB							MSB

Figure: Data type on the packet head

This Data Type (DT) also defines what the used packet is: Short Packet (SPa) or Long Packet (LPa). Data Types (DT) are different from the MCU to the display module (or other devices) and vice versa. These Data Type (DT) are defined on tables below.

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From the MCU to the Display module		
Data Type(HEX)	Data Type(Binary)	Description
01h	00 0001	Sync Event, V Sync Start
11h	01 0001	Sync Event, V Sync End
21h	10 0001	Sync Event, H Sync Start
31h	11 0001	Sync Event, H Sync End
08h	00 1000	End of Transmission (EoT)packet
03h	00 0011	Generic Short WRITE, no parameters
13h	01 0011	Generic Short WRITE, 1 parameters
23h	10 0011	Generic Short WRITE, 2 parameters
04h	00 0100	Generic READ, no parameters
14h	01 0100	Generic READ, 1 parameters
24h	10 0100	Generic READ, 2 parameters
05h	00 0101	DCS WRITE, no parameters
15h	01 0101	DCS WRITE, 1 parameters
06h	00 0110	DCS READ, no parameters
37h	11 0111	Set Maximum Return Packet Size
09h	00 1001	Null Packet, no data
19h	01 1001	Blanking Packet, no data
29h	10 1001	Generic Long Write
39h	11 1001	DCS Long Write/Write_LUT Command packet
3Eh	11 1110	Packet Pixel Streem, 24-bit RGB, 8-8-8 Format
0Eh	001110	Packed Pixel Stream,16-bit RGB,5-6-5 Format
1Eh	011110	Packed Pixel Stream,18-bit RGB,6-6-6 Format
2Eh	101110	Loosely Packed Pixel Stream,18-bit RGB,6-6-6 Format

Table: Data type from the MCU to the display module

From the Display module to the MCU		
Data Type(HEX)	Data Type(Binary)	Description
1Ch	01 1100	DCS Long Read Response
21h	10 0001	DCS Short Read Response, 1 Byte returned
22h	10 0010	DCS Short Read Response, 2 Byte returned
1Ah	011010	Genaric Long READ Response
11h	01 0001	Generic Short Read Response, 1 Byte returned
12h	01 0010	Generic Short Read Response, 2 Byte returned

Table: Data type from the display module to the MCU

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The receiver will ignore other Data Type (DT) if they are not defined on tables above. Host send "Generic Read" data type, AXS15260D-S1 will return Generic Read package to Host.

Packet data on the short packet

Packet Data (PD) of the Short Packet (SPa) is defined after Data Type (DT) of the Data Identification (DI) has indicated that Short Packet (SPa) is wanted to send. Packet Data (PD) of the Short Packet (SPa) consists of 2 data bytes: Data 0 and Data 1. Packet Data (PD) sending order is that Data 0 is sent in the first and the Data 1 is sent in the last. Bits of Data 1 are set to 00h, if the information length is 1 byte. Packet Data (PD) of the Short Packet (SPa), when the length of the information is 1 or 2 bytes are illustrated for reference purposes below.

DI								Data 0								Data 1								ECC							
15hex								35hex								01hex								1Ehex							
1	0	1	0	1	0	0	0	1	0	1	0	1	1	0	0	1	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0
B0	B1	B2	B3	B4	B5	B6	B7	B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	B14	B15	B0	B1	B2	B3	B4	B5	B6	B7
LSB							MSB	LSB							MSB	LSB							MSB	LSB							MSB

Figure: Packet data on the short packet, 2 bytes information

Packet Data (PD) information:

- Data 0: 10hex
- Data 1: 00hex (Null)

DI								Data 0								Data 1								ECC							
05hex								10hex								00hex								2Chex							
1	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0
B0	B1	B2	B3	B4	B5	B6	B7	B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	B14	B15	B0	B1	B2	B3	B4	B5	B6	B7
LSB							MSB	LSB							MSB	LSB							MSB	LSB							MSB

Figure: Packet data on the short packet, 1 bytes information

Word count on the long packet

Word Count (WC) of the Long Packet (LPa) is defined after Data Type (DT) of the Data Identification (DI)has indicated that Long Packet (LPa) is wanted to send. Word Count (WC) indicates a number of the data bytes of the Packet Data (PD) what is wanted to send after Packet Header (PH) versus Packet Data (PD) of the Short Packet (SPa) is placed in the Packet Header (PH). Word Count (WC) of the Long Packet (LPa) consists of 2 bytes.

These 2 bytes of the Word Count (WC) sending order is that the Least Significant (LS) Byte is sent in the first and the Most Significant (MS) Byte is sent in the last.

Word Count (WC) of the Long Packet (LPa) is illustrated for reference purposes below.

DI								WC(Least Significant Byte)								WC(Most Significant Byte)								ECC							
39hex								01hex								00hex								15hex							
1	0	0	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	0	0

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B0	B1	B2	B3	B4	B5	B6	B7	B0	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	B13	B14	B15	B0	B1	B2	B3	B4	B5	B6	B7
LSB							MSB	LSB							MSB	LSB							MSB	LSB							MSB

Figure: Word count on the long packet

Error Correction Code (ECC)

Error Correction Code (ECC) is a part of Packet Header (PH) and its purpose is to identify an error or errors:

- Short Packet (SPa): Data Identification (DI) and Packet Data (PD) bytes (24 bits: D[23…0])
- Long Packet (LPa): Data Identification (DI) and Word Count (WC) bytes (24 bits: D[23…0])

D[23…0] is illustrated for reference purposes below.

DI								Data 0								Data 1								ECC							
05hex								10hex								00hex								2Chex							
1	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0
D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23	P0	P1	P2	P3	P4	P5	P6	P7
LSB							MSB	LSB							MSB	LSB							MSB	LSB							MSB

Figure: D[23:0] and P[7:0] on the short packet

DI								WC(Least Significant Byte)								WC(Most Significant Byte)								ECC							
39hex								01hex								00hex								15hex							
1	0	0	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	0	0	0
D0	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20	D21	D22	D23	P0	P1	P2	P3	P4	P5	P6	P7
LSB							MSB	LSB							MSB	LSB							MSB	LSB							MSB

Figure: D[23:0] and P[7:0] on the long packet

Error Correction Code (ECC) can recognize one error or several errors and makes correction in one bit error case.

Bits (P[7…0]) of the Error Correction Code (ECC) are defined, where the symbol '^' is presenting XOR function (Pn is '1' if there is odd number of '1's and Pn is '0' if there is even number of '1's), as follows.

ecc_parity = {P7,P6,P5,P4,P3,P2,P1,P0};

- P7 = 0
- P6 = 0
- P5 = D10^D11^D12^D13^D14^D15^D16^D17^D18^D19^D21^D22^D23
- P4 = D4^D5^D6^D7^D8^D9^D16^D17^D18^D19^D20^D22^D23
- P3 = D1^D2^D3^D7^D8^D9^D13^D14^D15^D19^D20^D21^D23

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• P2 = D0^D2^D3^D5^D6^D9^D11^D12^D15^D18^D20^D21^D22
• P1 = D0^D1^D3^D4^D6^D8^D10^D12^D14^D17^D20^D21^D22^D23
• P0 = D0^D1^D2^D4^D5^D7^D10^D11^D13^D16^D20^D21^D22^D23

P7 and P6 are set to '0' because Error Correction Code (ECC) is based on 64 bit value ([D63…0]), but this implementation is based on 24 bit value (D[23…0]). Therefore, there is only needed 6 bits (P[5…0]) for Error Correction Code (ECC).

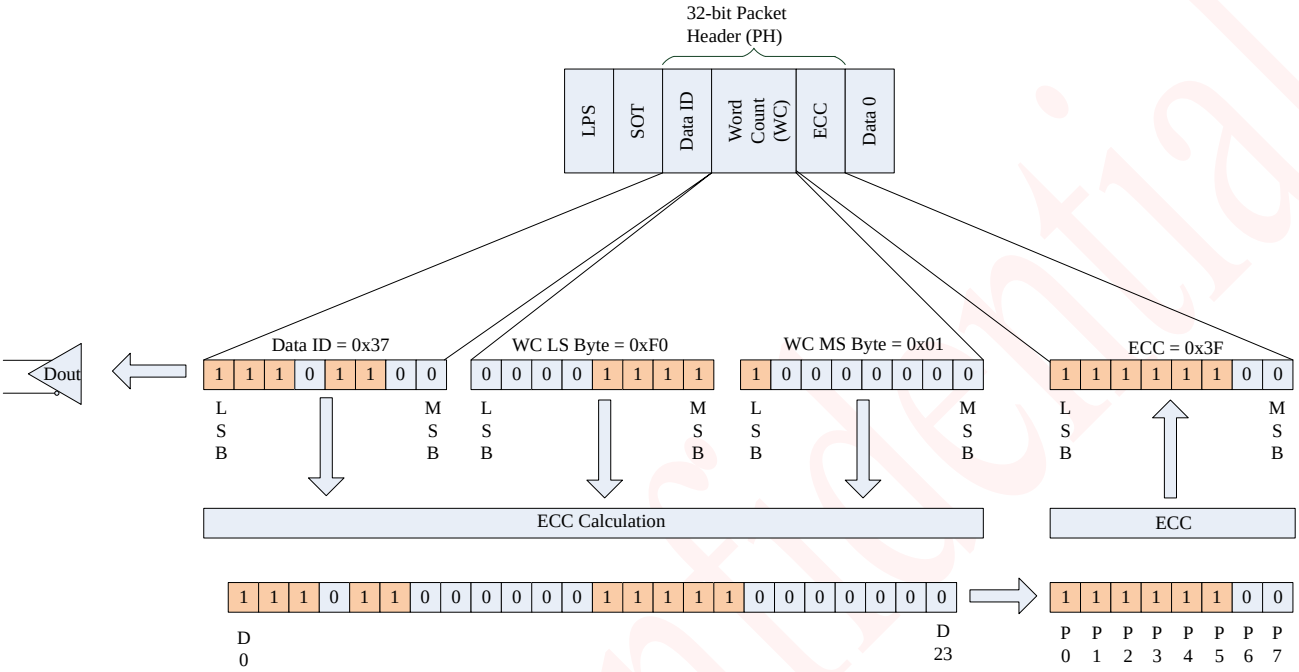


Figure: 24-bit ECC generation on TX side (Example)

Packet footer on the long packet

Packet Footer (PF) of the Long Packet (LPa) is defined after the Packet Data (PD) of the Long Packet (LPa). The Packet Footer (PF) is a checksum value what is calculated from the Packet Data of the Long Packet (LPa).

The checksum is using a 16-bit Cyclic Redundancy Check (CRC) value which is generated with a polynomial X16+X12+X5+X0 as it is illustrated below.

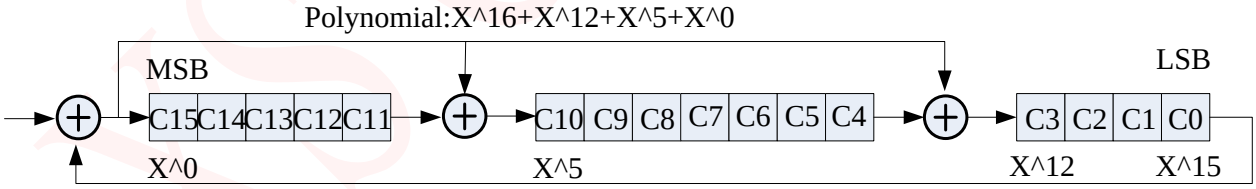


Figure: 16-bit cyclic redundancy check (CRC) calculation

The 16-bit Cyclic Redundancy Check (CRC) generator is initialized to FFFFh before calculations. The Most Significant Bit (MSB) of the data byte of the Packet Data (PD) is the first bit what is inputted into the 16-bit Cyclic Redundancy Check (CRC). The receiver is calculated own checksum value from received Packet Data (PD). The receiver compares own checksum and the Packet Footer (PF) what the transmitter has sent. The received Packet Data (PD) and Packet Footer (PF) are correct if the own checksum of the receiver and Packet Footer (PF) are equal and vice versa the received Packet Data (PD) and Packet Footer (PF) are not correct if the own checksum of the receiver and Packet Footer (PF) are not equal.

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5.3.4.2 Packet transmissions

Packet from the MCU to the display module

Display Command Set (DCS), which is defined on chapter “Instructions” is used from the MCU to the display module. This Display Command Set (DCS) is always defined on the Data 0 of the Packet Data (PD), which is included in Short Packet (SPa) and Long packet (LPa) as these are illustrated below.

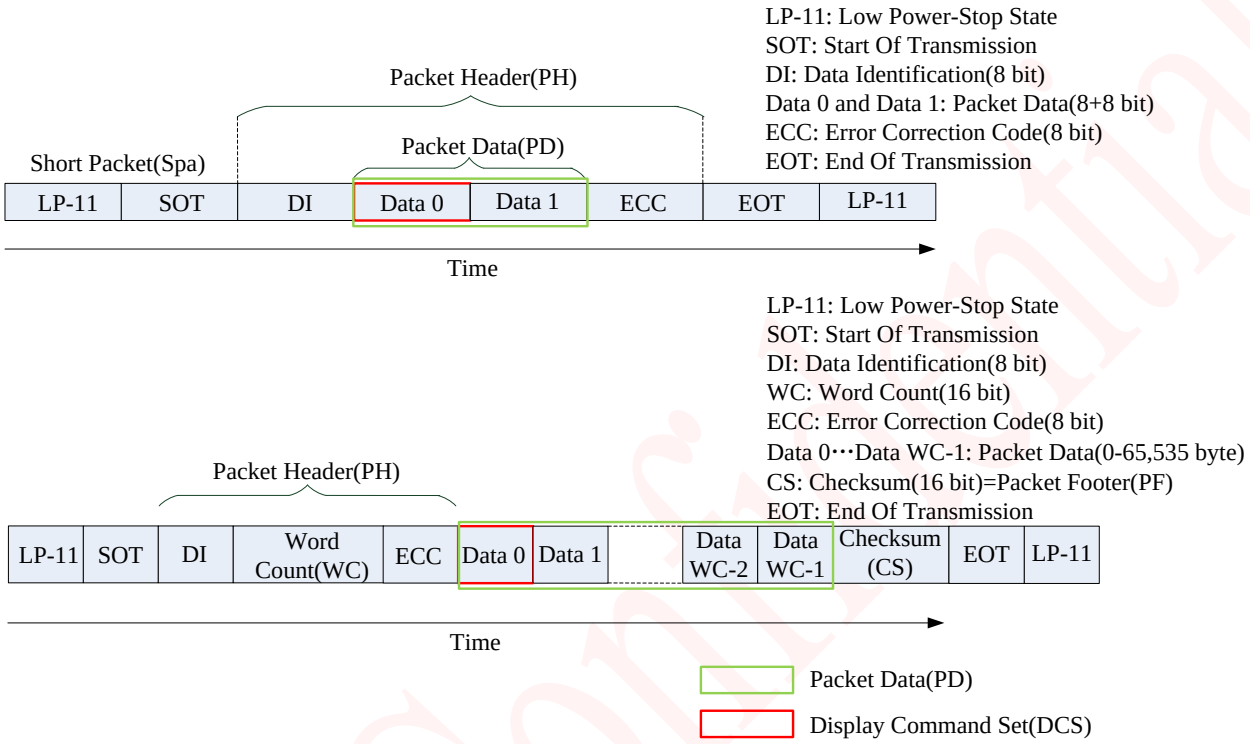


Figure: DCS on the short packet and long packet

Packet from the display module to the MCU

Used packet types

The display module is always using Short Packet (SPa) or Long Packet (LPa), when it is returning information to the MCU after the MCU has requested information from the Display Module. This information can be a response of the Display Command Set (DCS).

The used packet type is defined on Data Type (DT).

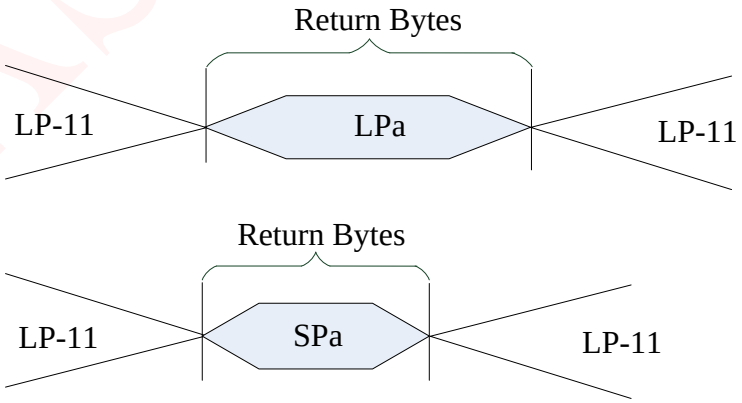


Figure: Return bytes on single packet

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5.3.5 Customer-defined generic read data type format

The short packet of Data Type 14h (Generic READ, 1 parameters) specifies the register content for read and the Nth parameter that will begin reading. After Data Type 14h is received, BTA is executed. Then, the Nth parameter becomes the first data, and the number of data of WC (word count) value is output.

Packet Structure(processor → peripheral)

		P 0	P 1		
	Data Type 24h	Manufacturer Command	Start Parameter N	ECC	

Low Power Data Transfer(peripheral → processor)

					Data 0	Data 1		Data WC-1		
	Data Type 1Ah	WC 0	WC 1	WC 2	1th Parameter	2th Parameter		Nth Parameter	CRC 0	CRC 1

Figure: Generic read data type format

5.3.6 MIPI video parameter

In the MIPI video mode, the frame is transmitted from the host processor to a display module as a sequence of pixels, with each horizontal line of the image data sent as a group of consecutive pixels.

Vsync (VS) indicates the beginning of each frame of the displayed image.

Hsync (HS) signals the beginning of each horizontal line of pixels.

Each pixel value (16-, 18-, or 24-bit data) is transferred from the host processor to the display module during one pixel period. The rising edge of PCLK is used by the display module to capture pixel data. Since PCLK runs continuously, control signal DE is required to indicate when valid pixel data is being transmitted on the pixel data signals.

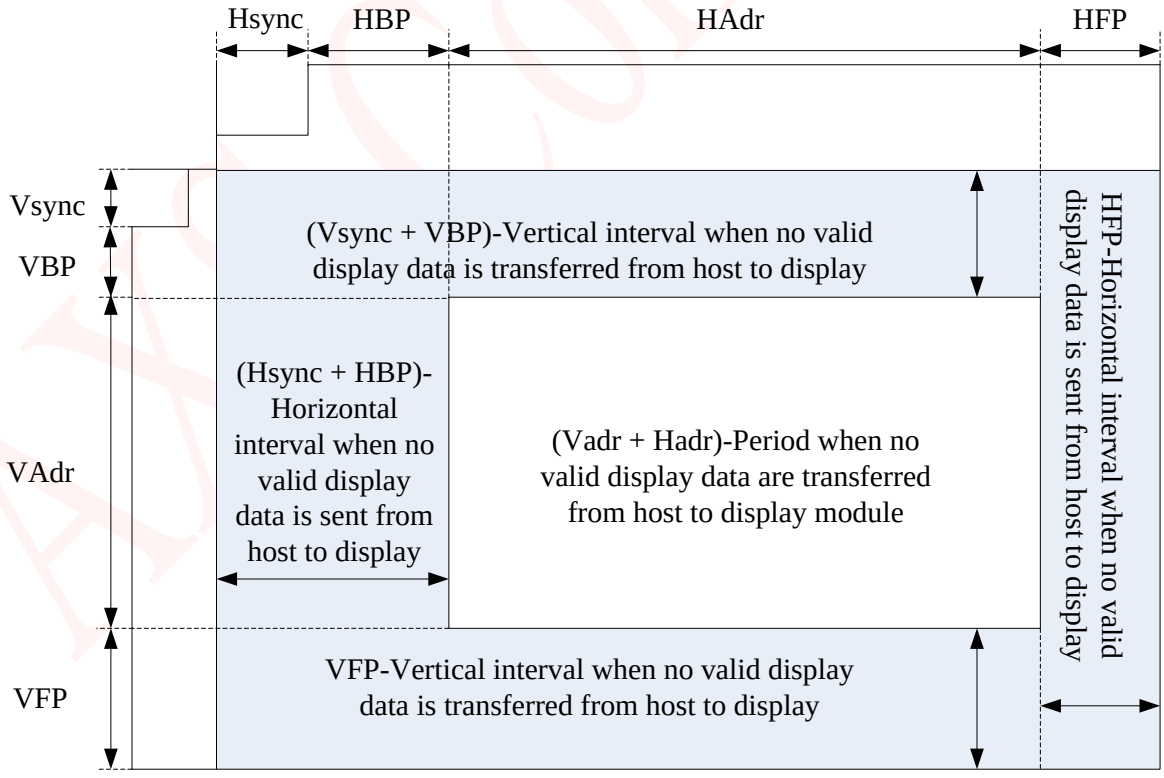


Figure 6.2.6.1 define timing parameter for MIPI video operation.

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(Resolution for 320 horizontal x 480 vertical display with Frame-Rate of 60 Hz)

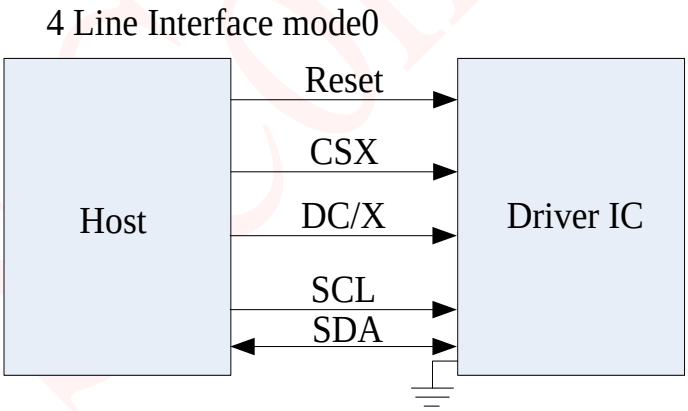
Parameters	Symbols	Min.	Typ	Max.	Unit
Horizontal Synchronization	Hsync	2	2	-	PCLK
Horizontal Back Porch	HBP	2	60	-	PCLK
Horizontal Front Porch	HFP	2	60	-	PCLK
Hsync+ HBP+ HFP	-	6	122	-	PCLK
Horizontal Address (Display area)	HAdr	-	320	-	PCLK

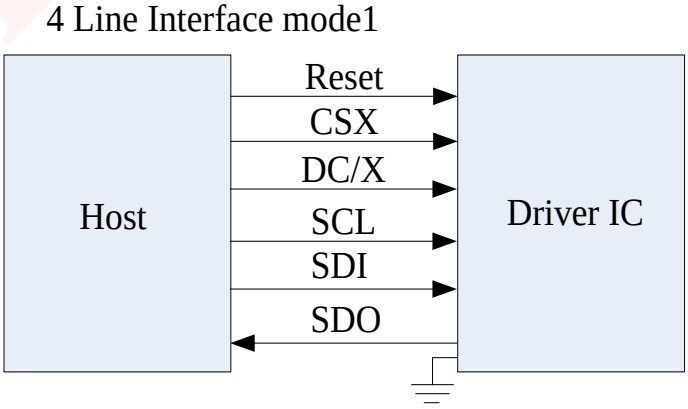
5.4 Serial Interface (SPI)

5.4.1 4-Line Interface

Connect Pin:DIN_SDA(SDI),DIN_SDA_DUAL(SDO),RS(D/CX),SCL,CSX,RSTN

The Serial Peripheral Interface (SPI) is selected by setting the IM[3:0] pins as “1101” level. The chip select pin (CSX), the serial transfer clock pin (SCL), the display data/command selection (D/CX), the serial data input/output pin (SDA or SDI/SDO) are used in SPI mode. The ID pin sets the least significant bit of the identification code. The DB[23:0] pins, which are not used, must be tied to GND

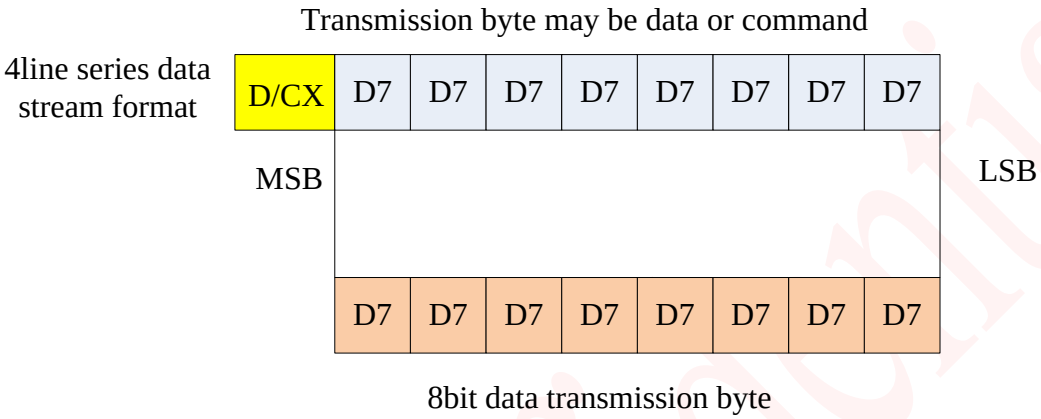




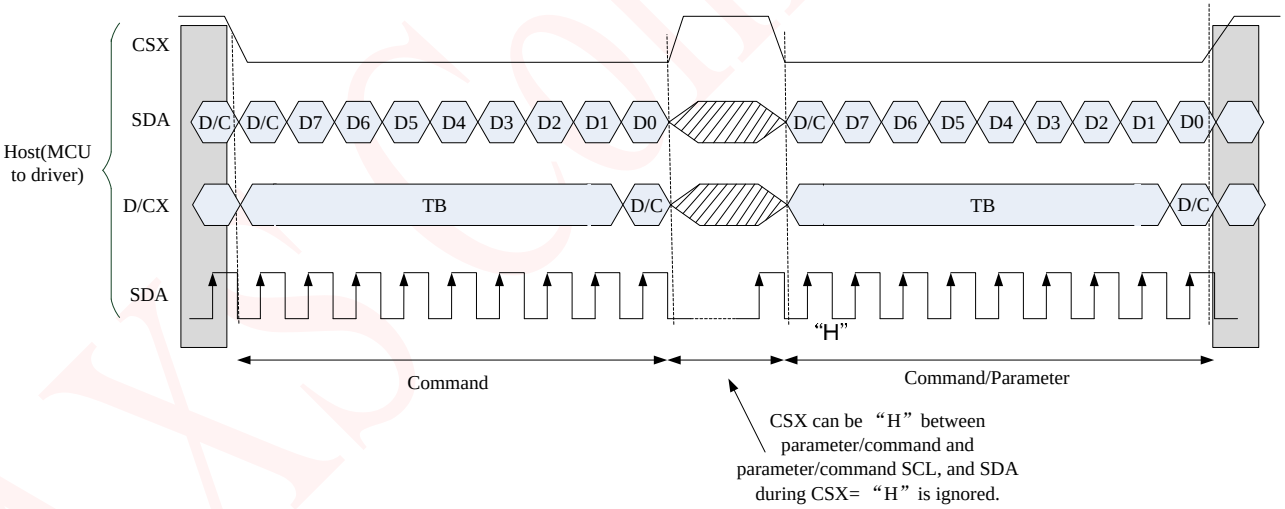
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5.4.1.1 Write Sequence

The write mode of the interface means the host writes commands and data to AXS15260D-S1. The 4-lines serial data packet contains a data/command and a transmission byte. If D/CX is “low”, the transmission byte is interpreted as a command byte. If D/CX is “high”, the transmission byte is stored in the display data RAM (Memory write command), or command register as parameter.



The host drives the CSX pin to low and the MSB data bit (D7) is set on SDA by the host. On the next falling edge of SCL the next bit (D6) is set on SDA. If the optional D/CX signal is used, a byte is eight read cycle long. The 4-line serial interface writes sequence described in the Figure as below.

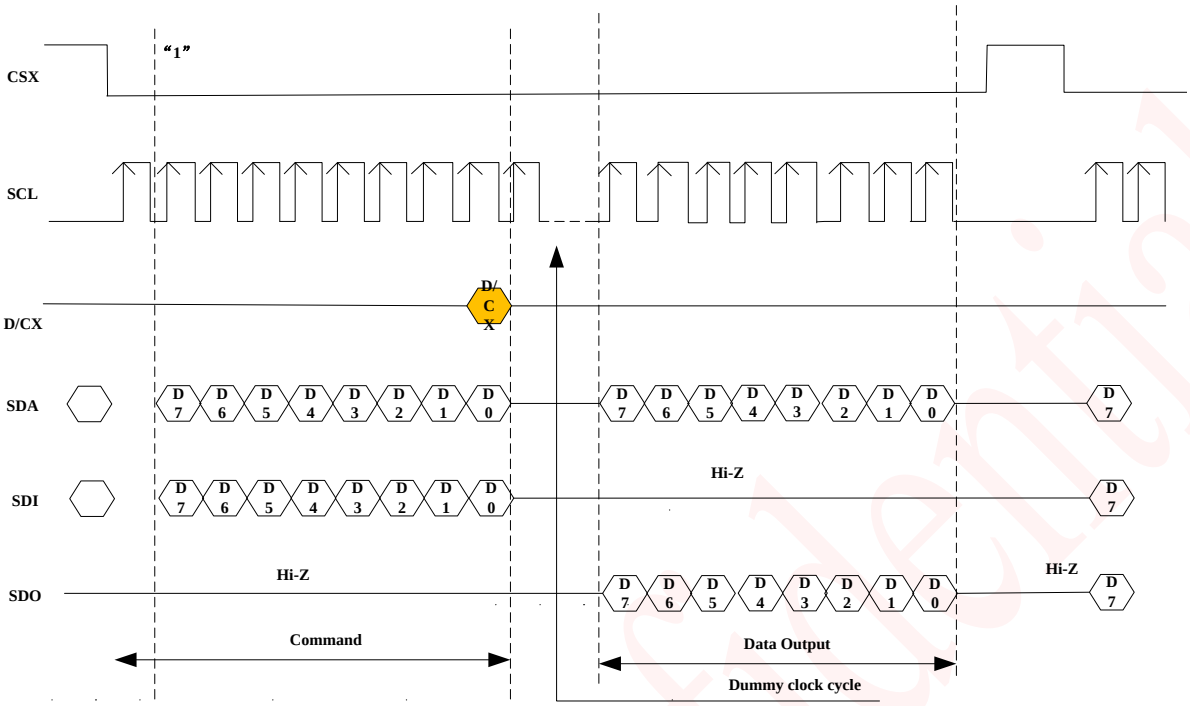


5.4.1.2 Read Sequence

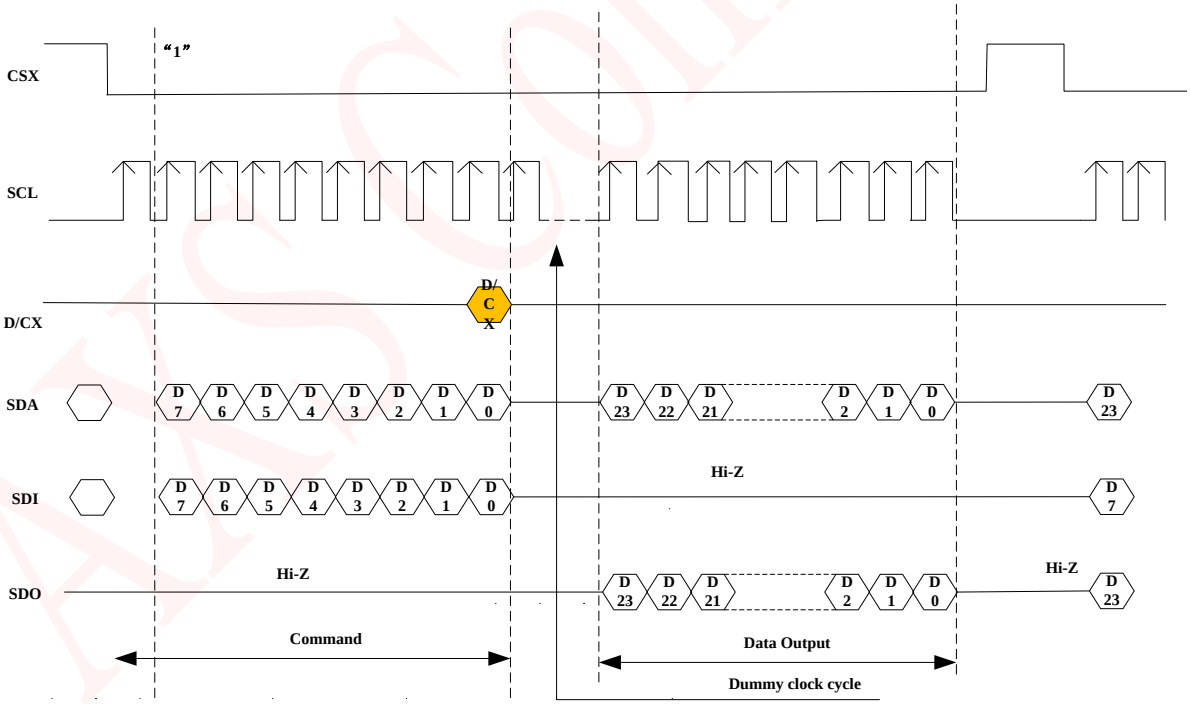
The read mode of the interface means that the micro controller reads register value from the driver. To achieve read function, the micro controller first has to send a command (read ID or register command) and then the following byte is transmitted in the opposite direction. After that CSX is required to go to high before a new command is send (see the below figure). The driver samples the SDA (input data) at rising edge of SCL, but shifts SDA (output data) at the falling edge of SCL. Thus the micro controller is supported to read at the rising edge of SCL.

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5.4.1.2.1 4-line serial protocol (8-bit read, cr_spi_rd_en=1):

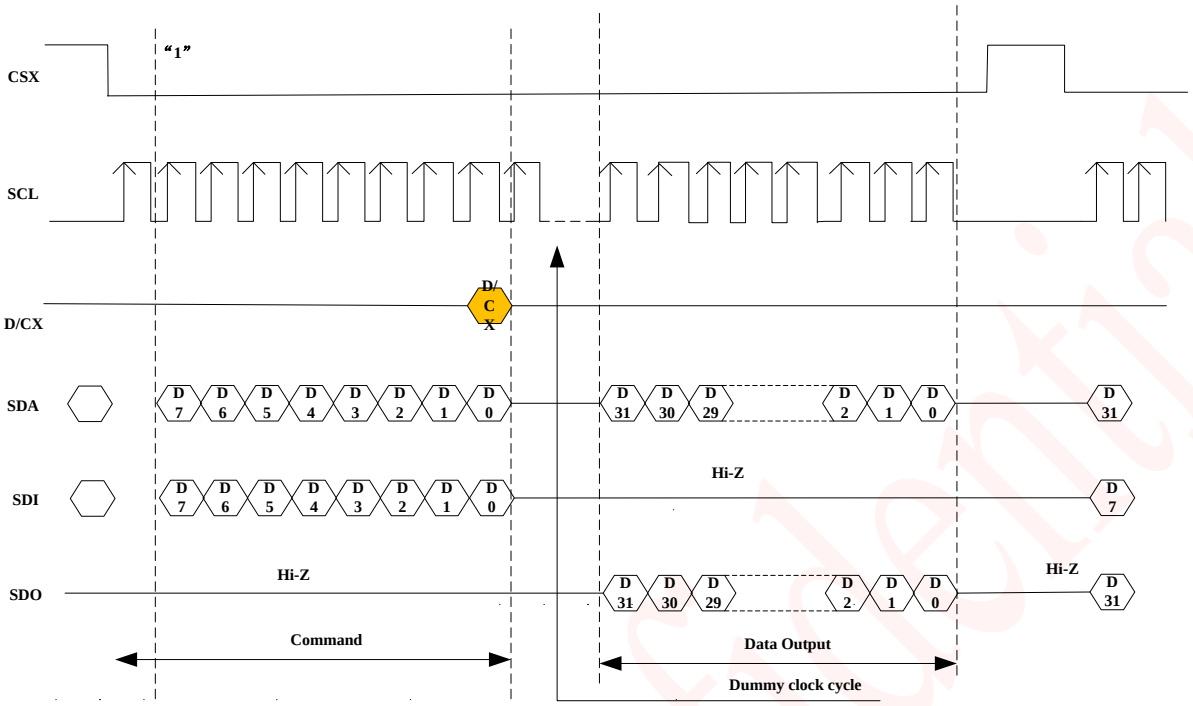


5.4.1.2.2 4-line serial protocol (24-bit read,cr_spi_rd_en=1)



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5.4.1.2.3 4-line serial protocol (32-bit read,cr_spi_rd_en=1)



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5.5 RGB Interface

5.5.1 RGB Color Format

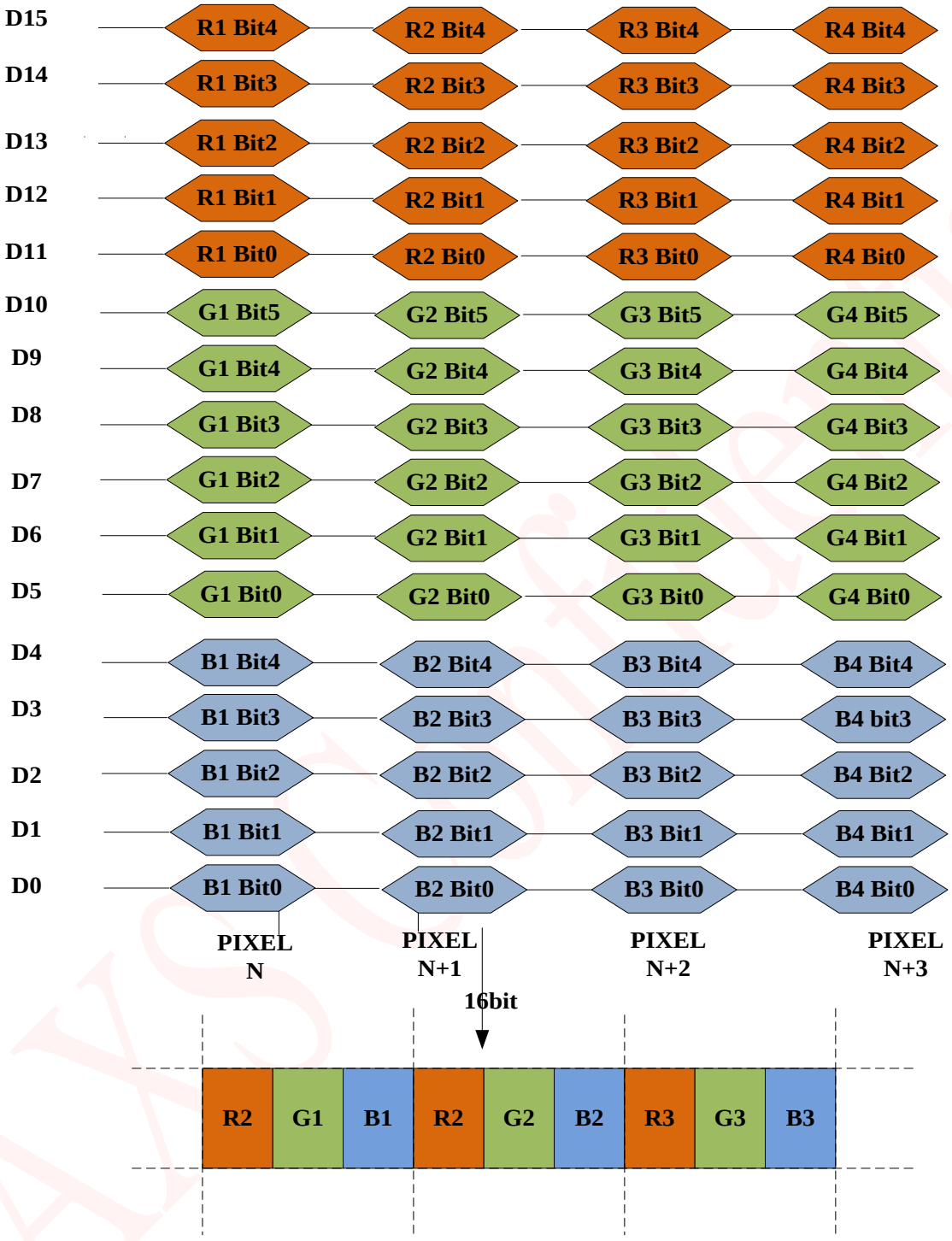
Connect Pin:DB[15/17/23:0],VSYNC_QSPI_DIN2,HSYNC,DE_QSPI_DIN3,PCLK

AXS15260D-S1 supports two kinds of RGB interface, DE mode ， and 16bit/18bit/24bit data format. When DE mode is selected and the VSYNC, HSYNC, DOTCLK, DE, DB[15/17/23:0] pins can be used; When using RGB interface, only serial interface can be selected. IM [3:0] as “1110”。

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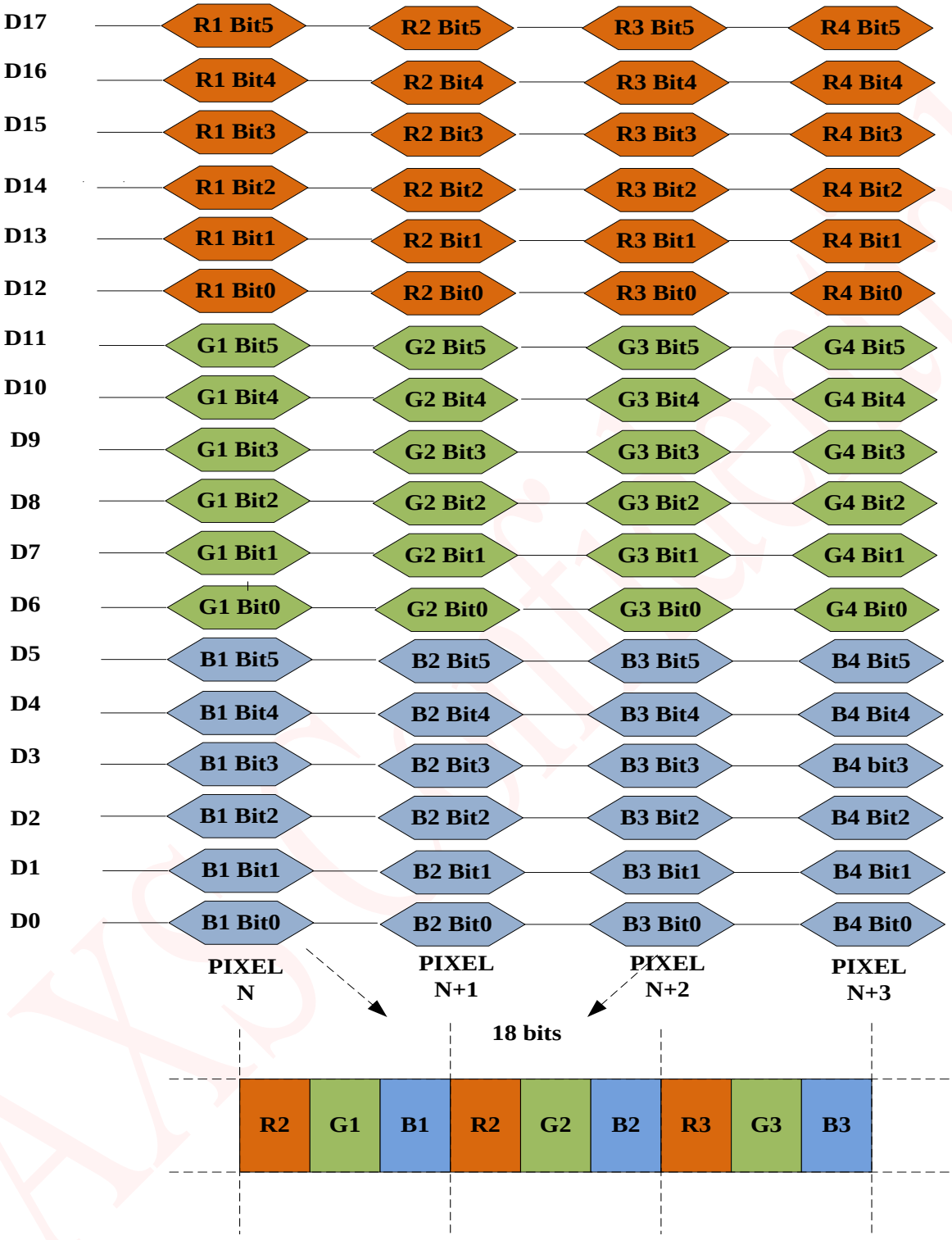
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5.5.1.1 Write data for 16-bit/pixel (RGB 5-6-5-bit input)65K-Colors



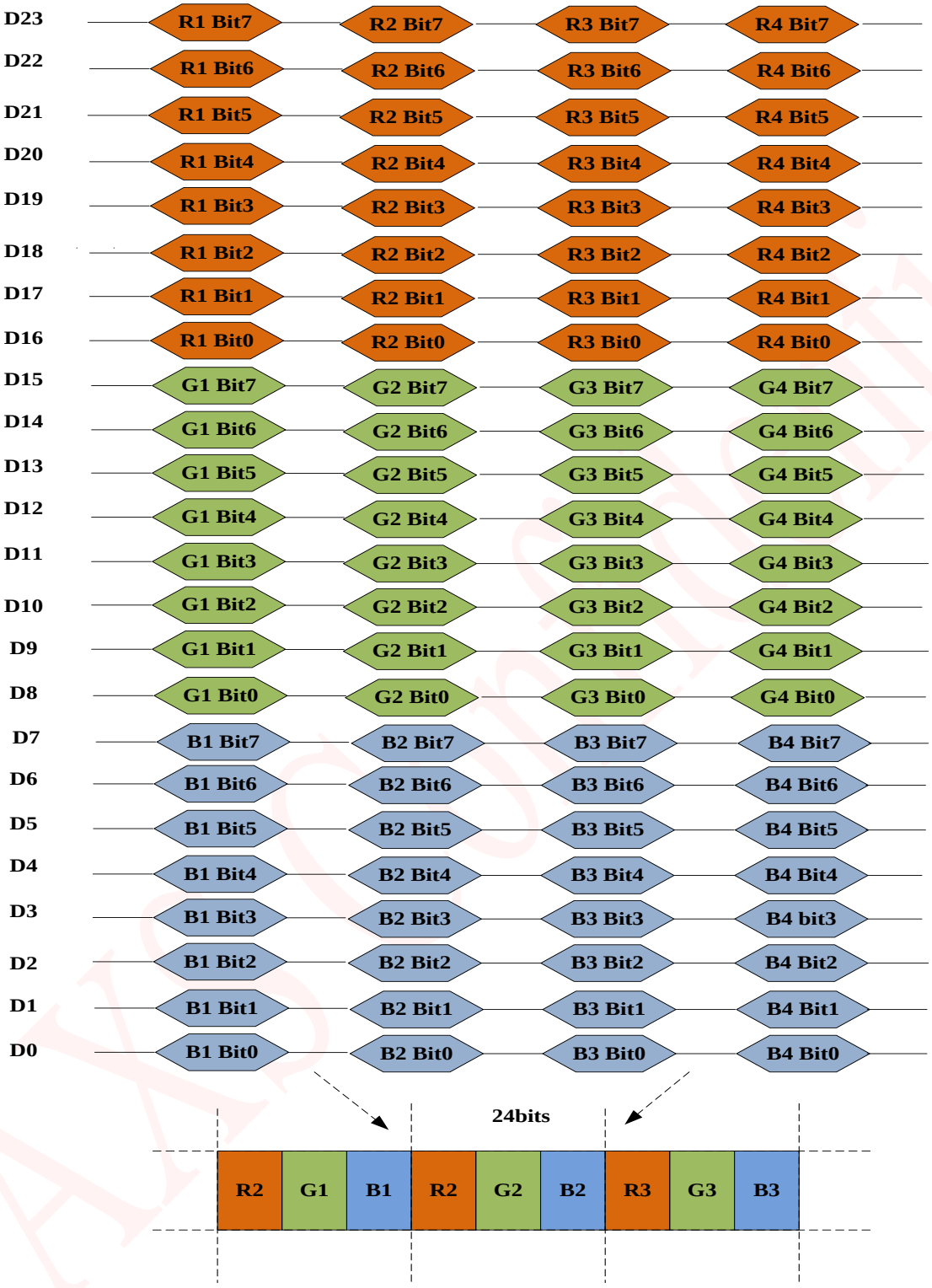
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5.5.1.2 Write data for 18-bit/pixel (RGB 6-6-6-bit input)262K-Colors



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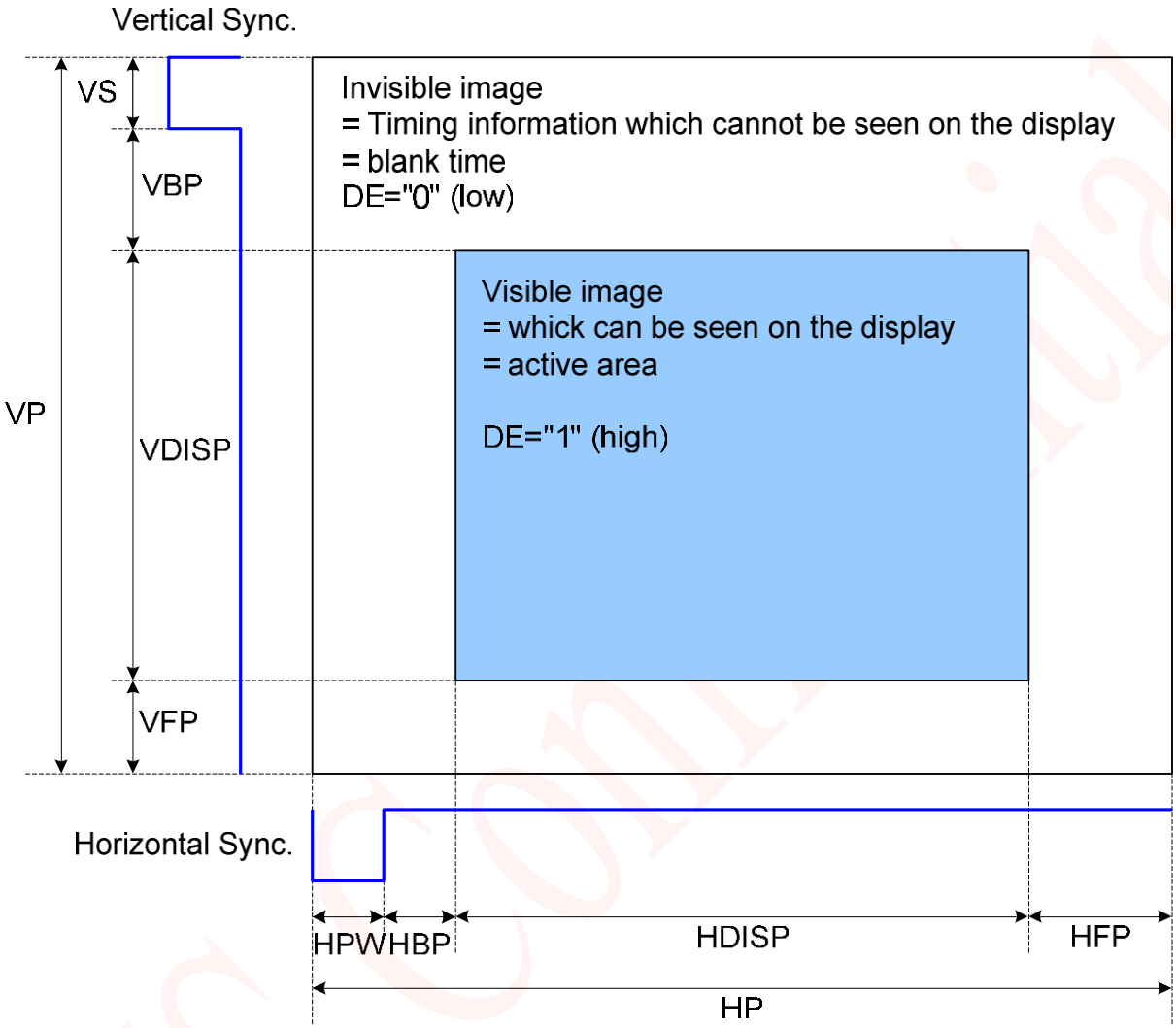
5.5.1.3 Write data for 24-bit/pixel (RGB 8-8-8-bit input)16.7M-Colors



5.5.2 RGB Interface Definition

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The display operation via the RGB interface is synchronized with the VSYNC, HSYNC, and DOTCLK signals. The data can be written only within



the specified area with low power consumption by using window address function. The back porch and front porch are used to set the RGB interface timing.

DRAM Access Area by RGB Interface

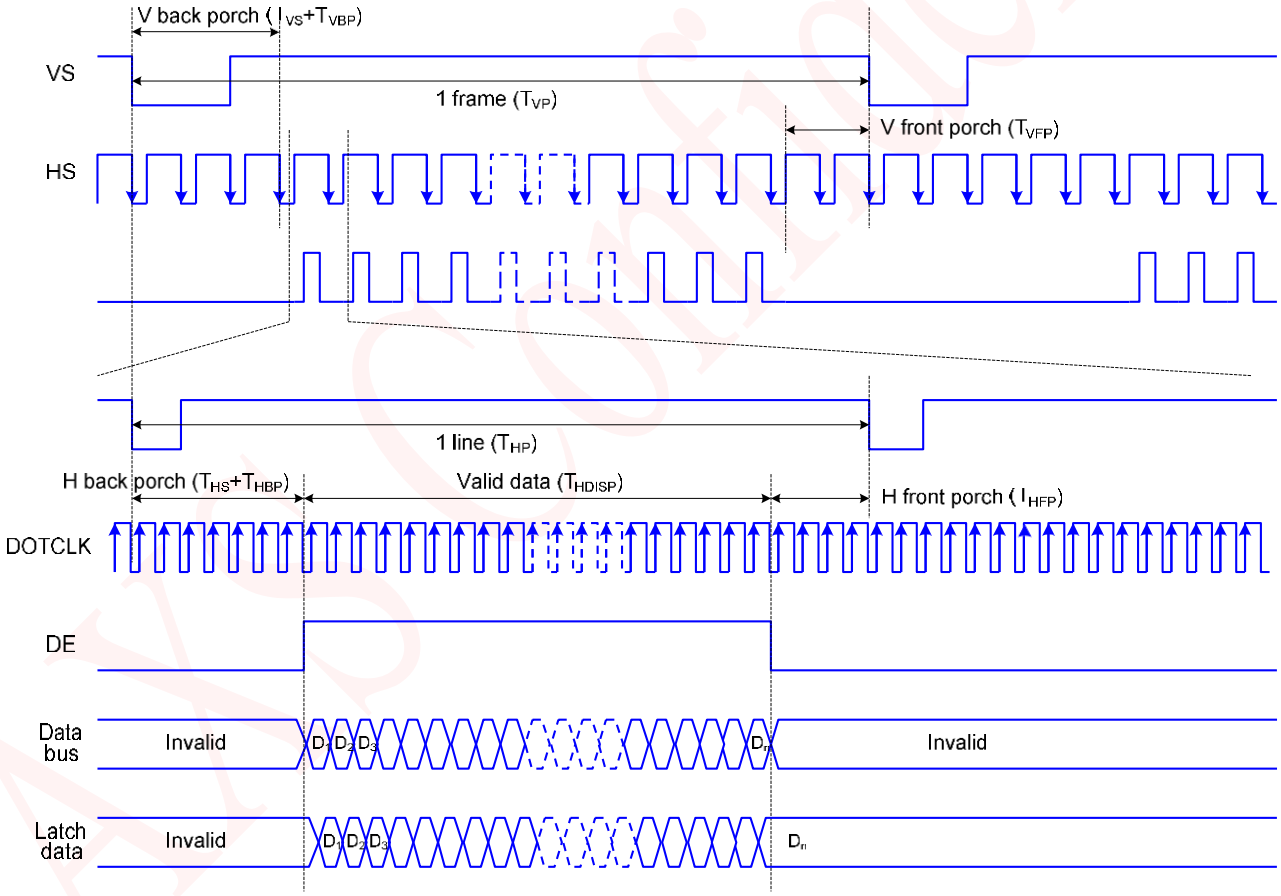
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Please refer to the following table for the setting limitation of RGB interface signals.

Parameter	Symbol	Min.	Typ.	Max.	Unit
Horizontal Sync. Width	hpw	-	2	hpw+hbp=75	Clock
Horizontal Sync. Back Porch	hbp	-	30		Clock
Horizontal Sync. Front Porch	hfp	-	30	-	Clock
Vertical Sync. Width	vs	-	2	-	Line
Vertical Sync. Back Porch	vbp	-	254		Line
Vertical Sync. Front Porch	vfp	-	10	-	Line

5.5.3 RGB Interface Timing

The timing chart of RGB interface DE mode is shown as follows.



Note: The setting of front porch and back porch in host must match that in IC as this mode.

VSYNC, HSYNC, and DOTCLK signals must be supplied during a display operation period. In RGB interface mode, the panel controlling signals are generated from DOTCLK, not the internal clock generated from the internal oscillator.

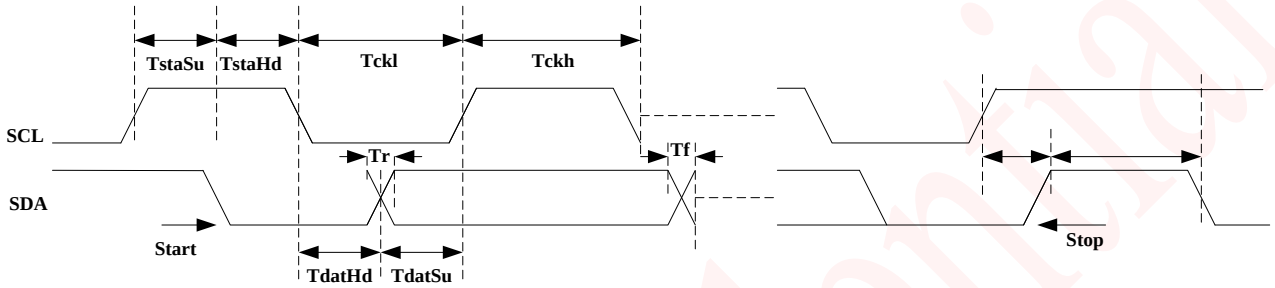
When switching between the internal operation mode and the external display interface operation mode, follow the sequences below in setting instruction.

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In RGB interface mode, the front porch period continues until the next VSYNC input is detected after drawing one frame.In RGB interface mode, a RAM address is set in the address counter every frame on the falling edge of VSYNC.

5.6 Touch part interface

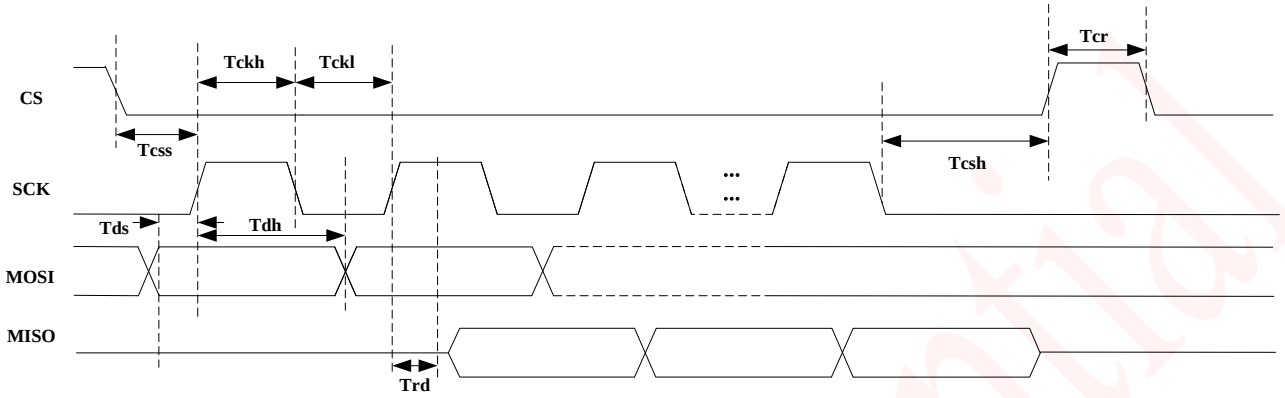
5.6.1 I2C Interface



Parameter	Symbol	Min.	Typ	Max.	Unit
Working Frequency	F_{clk}	20	-	400	Khz
I2C Clock Low	T_{CKL}	1300	-	-	ns
I2C Clock High	T_{CKH}	600	-	-	ns
I2C Clock and Data rising time	T_r	-	-	300	ns
I2C Clock and Data falling time	T_f	-	-	300	ns
I2C Data hold time	T_{DatHd}	0	-	-	ns
I2C Data setup time	T_{DatSu}	100	-	-	ns
I2C Start Condition hold time	T_{StaHd}	600	-	-	ns
I2C Start Condition setup time	T_{StaSu}	600	-	-	ns
I2C Stop Condition setup time	T_{StpSu}	600	-	-	ns
I2C Bus free time	$T_{BusFree}$	1300	-	-	ns

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5.6.2 SPI Interface



SPI Timing

Parameter	Symbol	Min.	Typ	Max.	Unit
SCK Frequency	f_{SCK}	-	-	18	Mhz
CS Set-up Time	t_{CSS}	200	-	-	ns
CS Hold Time	t_{CSH}	200	-	-	ns
CS Recovery Time	t_{CR}	1	-	-	ns
SCK clock High Time	t_{CKH}	27.8	-	-	ns
SCK clock Low Time	t_{CKL}	27.8	-	-	ns
Data Output Delay Time	t_{RD}	-	-	-	ns
Input Data Set-up Time	t_{DS}	25	-	-	ns
Input Data Hold Time	t_{DH}	25	-	-	ns

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5.7 Display Reference Clock Function

The AXS15260D-S1 provides a function to decide internal oscillator for display clock reference of driver IC.

Relationship between Liquid Crystal Driver Duty and the Frame Frequency

The formula below is used to calculate the relationship between the liquid crystal driver duty and the frame frequency. The frame frequency is determined by setting the 1H period adjustment (RTN) bit.

RTN setting for 1H period:

Step1: To decide real one line period in Command Mode:

$$RTN = 1H(period) = \frac{Fosc - 5\%}{(Line + BP + FP) * FrameRate(Hz)}$$

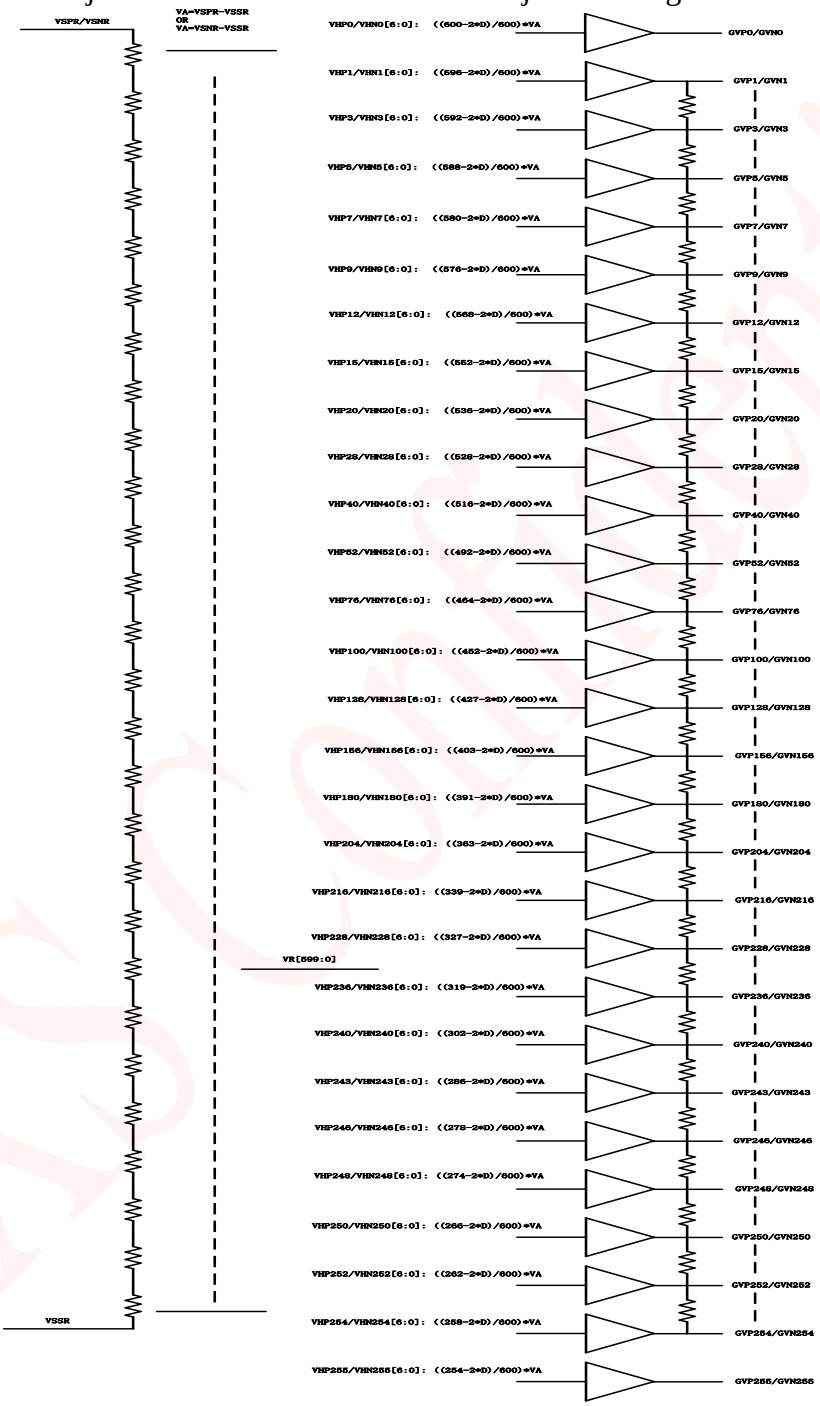
$$RTN = 1H(period) = \frac{F_{pll} - 5\%}{(Line + BP + FP) * FrameRate(Hz)}(us)$$ (Note.1, Note.2, Note.3)

RTN: Number of clocks per line.
Line: Display Line Number
FP: Number of lines for the front porch.
BP: Number of lines for the back porch.
Note.1: The RTN formula can cover full temp range variation
Note.2: When Touch function ON (must take account of DP/TP ratio)
Note.3: For detailed RTN calculation method, please contact AXS.

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5.8 Gamma Function

The structure of grayscale amplifier is shown as below. The 30 voltage levels between VSPR/VSNR and GND determined by the gradient adjustment register, the reference adjustment register, the amplitude adjustment resister and the micro-adjustment register.



Figure：Gamma Architecture for AXS15260D-S1

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5.9 Reset Function

The RESET function of AXS15260D-S1 is triggered by a RESX input. After reset function triggered, the AXS15260D-S1 enter a reset period, and the duration of this period must be at least 5ms. During this period, the AXS15260D-S1A-S1 and its power circuit will be initialized.

Initial States of Output Pins

The following table represents the output pins and its initial state.

Output Pins	Initial State
S<810:1>	GND
SX<480:1>	GND
VCOM	Disabled (GND level output)
GOUT_L[17:1], GOUT_R[17:1]	Disabled (GND level output)
LED_PWM	Disabled (GND level output)
VGH	Disabled
VGL	Disabled
VGHO	Disabled
VGLO	Disabled
VREF_TP	Disabled (Hi-z)
VCG_TP	Disabled (Hi-z)

5.10 Driver Operation Mode

AXS15260D-S1 driver can work in four operation modes of Reset State, Power Off state, Power On state and Display On State.

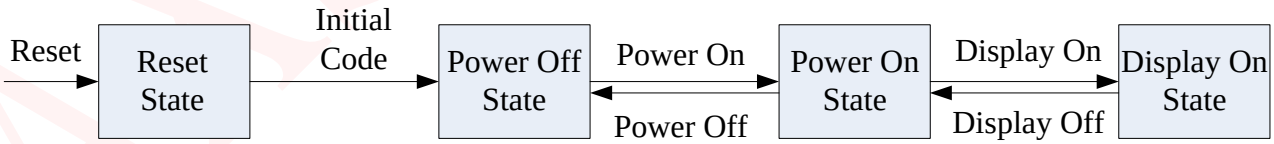
After reset, AXS15260D-S1 driver is in Reset state. In this state, all configurations are not initialized. Driver powers are not setup.

Initial code can be loaded from internal OTP or external flash or sent by host via MIPI interface. After the Initial code is loaded, AXS15260D-S1 driver enters Power Off state. AXS15260D-S1 driver is configured but powers are not up.

Power on command is sent from the Host. AXS15260D-S1 driver changes to Power On state on receiving power on command. All powers will be generated internally.

In Power on state, display on command can turn on the data path for display.

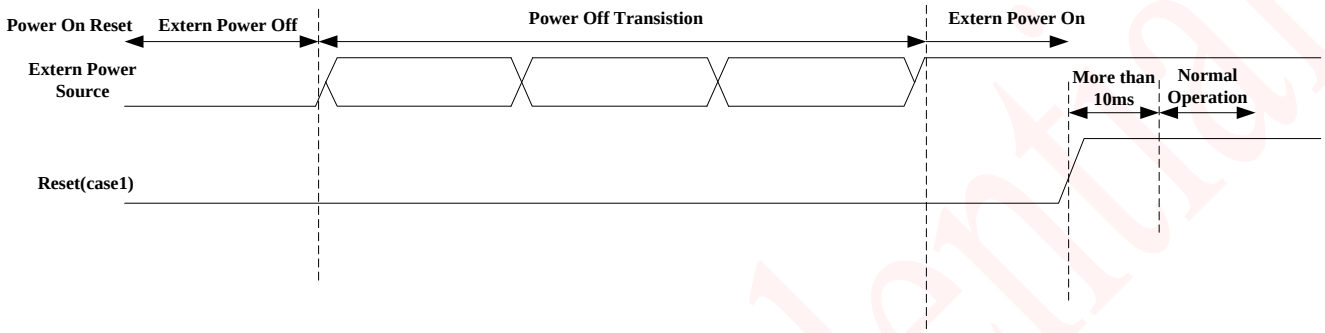
In Display on state, display off command can turn AXS15260D-S1 driver to Power On state and power off command can turn it to Power Off state.



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5.10.1 Timing of Reset Pin

AXS15260D-S1 provides H/W pin to do driver IC initialization. For power-on reset, one-finger reset (Case1) methods can be applied to do driver IC initialization. The detailed H/W reset pin timing is shown as below.
Of the two methods, RESET(Case 2) is recommended.



H/W reset pin timing for power-on reset

5.11 Tear Effect Information

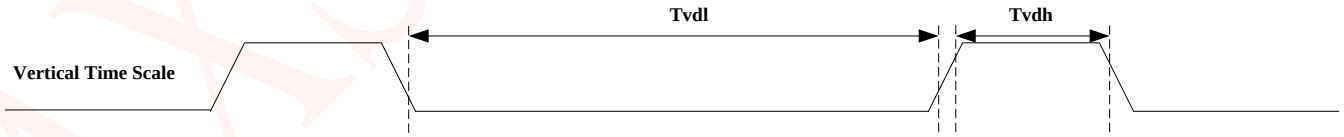
5.11.1 General

Tearing Effect line supplies to the MCU a Panel synchronization signal and this signal can be enabled or disabled by the Tearing Effect Line Off & On commands. The mode of the Tearing Effect Signal is defined by the Parameter of the Tearing Effect Line On command

5.11.2 Tearing effect line models

The Tearing Effect line supplies to the MCU a Panel synchronization signal and this signal can be enabled or disabled by the Tearing Effect Line Off & On commands. The mode of the Tearing Effect Signal is defined by the Parameter of the Tearing Effect Line On command.

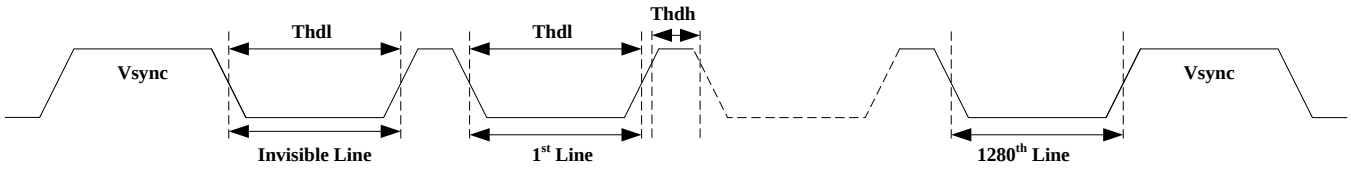
Mode1：The Tearing Effect Output signal consists of V-Sync information only:



tvdh = The display panel is not updated from the Frame Memory.
tvdl = The display panel is updated from the Frame Memory (except Invisible Line – see below).

Mode 2： The Tearing Effect Output signal consists of V-Sync and H-Sync information; There is one V-sync and 4 H-sync pulses per field:

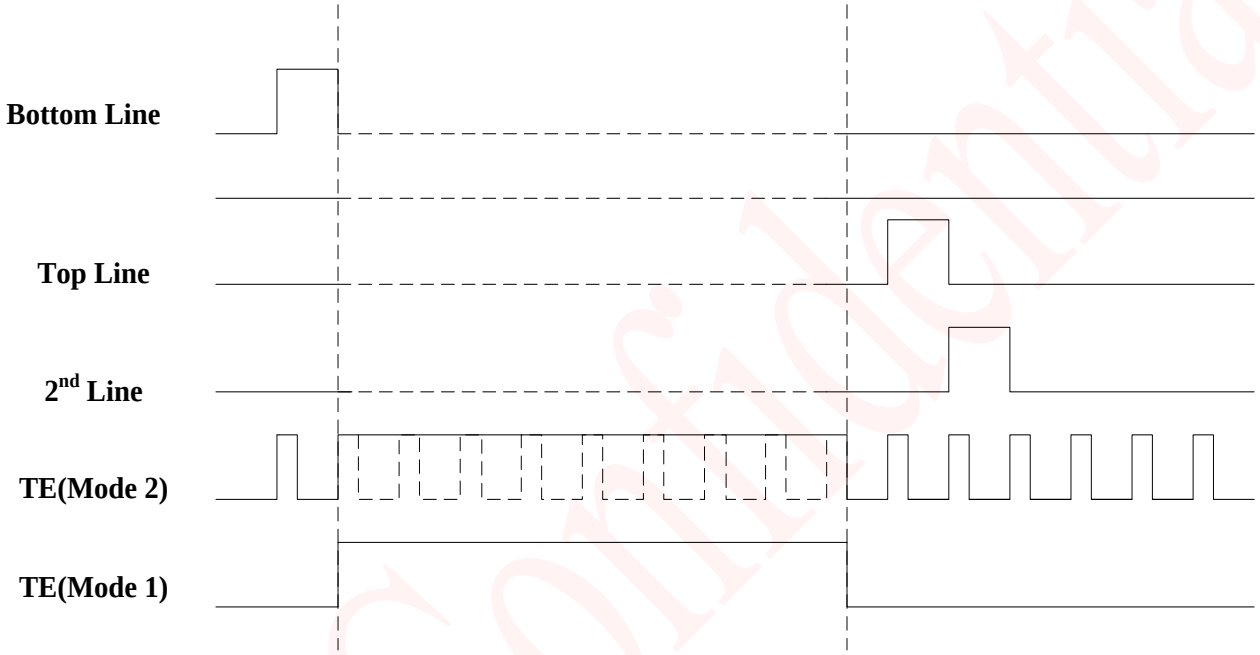
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t_{vdh} = The display panel is not updated from the Frame Memory.

$t_{vd}l$ = The display panel is updated from the Frame Memory (except Invisible Line – see below).

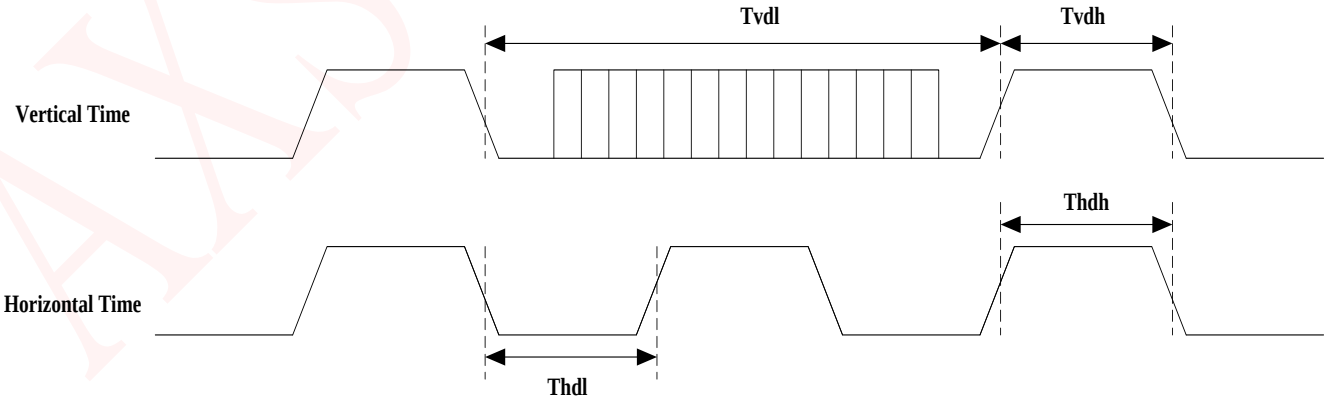
TE Line mode1 and Mode2 is shown as below graph:



Note: During Sleep In Mode, the Tearing Effect Output Pin is active Low.

5.11.2.1 Tearing effect line timing

The Tearing Effect signal is described below:



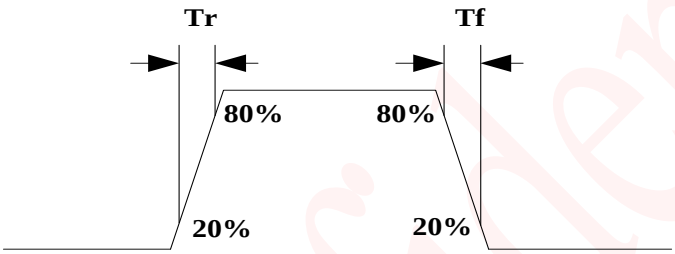
Symbol	Parameter	min	max	unit	description

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tvdl	Vertical Timing Low Duration	13	-	ms	
tvdh	Vertical Timing High Duration	1000	-	µs	
thdl	Horizontal Timing Low Duration	16	-	µs	
thdh	Horizontal Timing Low Duration	-	500	µs	

Idle Mode Off/On

The TE signal rising and falling timing is described below:



5.12 OTP Programming Procedure

5.12.1 Power function description

The OTP control signals control data reading and writing, only load once when reset. According to the request of the OTP, reset signal need delay at least 20ns to generate control signal. And signal transition should be less than 1ns.

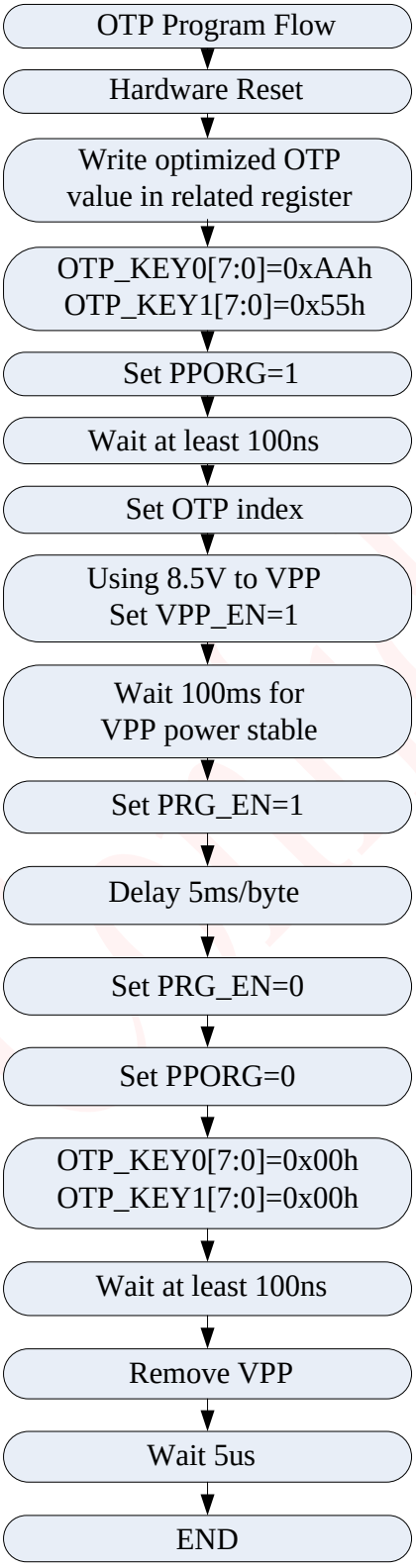
STEP: Write the register parameters, then use index read all index, if both consistent, to write read content into the OTP. This method requires each register readout parameters contend is the same as written contend before. Each number of parameter is the same. And it may waste OTP resources if did not in byte. In order to prevent the wrong, the rest of resource as a reserve. The content behind can cover the front.

In the program, Write the index and parameters, matching read contend with register number, and loading the content behind in register.

In addition, you can also load OTP after reset release, and programming OTP by external interface generating timing.

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5.12.2 OTP program flow chart

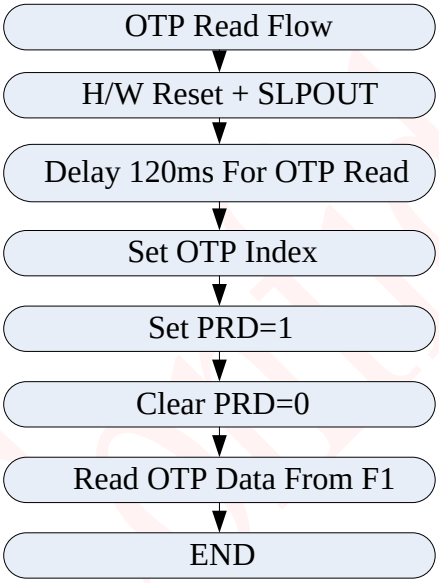


Step	External Power OTP Program Sequence
1	Power on and reset the module
2	Write optimized OTP value in related register

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3	OTP_KEY 0xAA 0x55
4	Set PPORG=1, wait 100ns
5	Set OTP index
6	Use 8.5V to VPP, Set VPP_EN=1, wait 100ms
7	Set PRG_EN=1, wait 5ms/byte
8	Set PRG_EN=0
9	Set PPORG=0, wait 100ns
10	OTP_KEY 0x00 0x00
11	Remove VPP, wait 5us

5.12.3 OTP read program flow chart with External Power

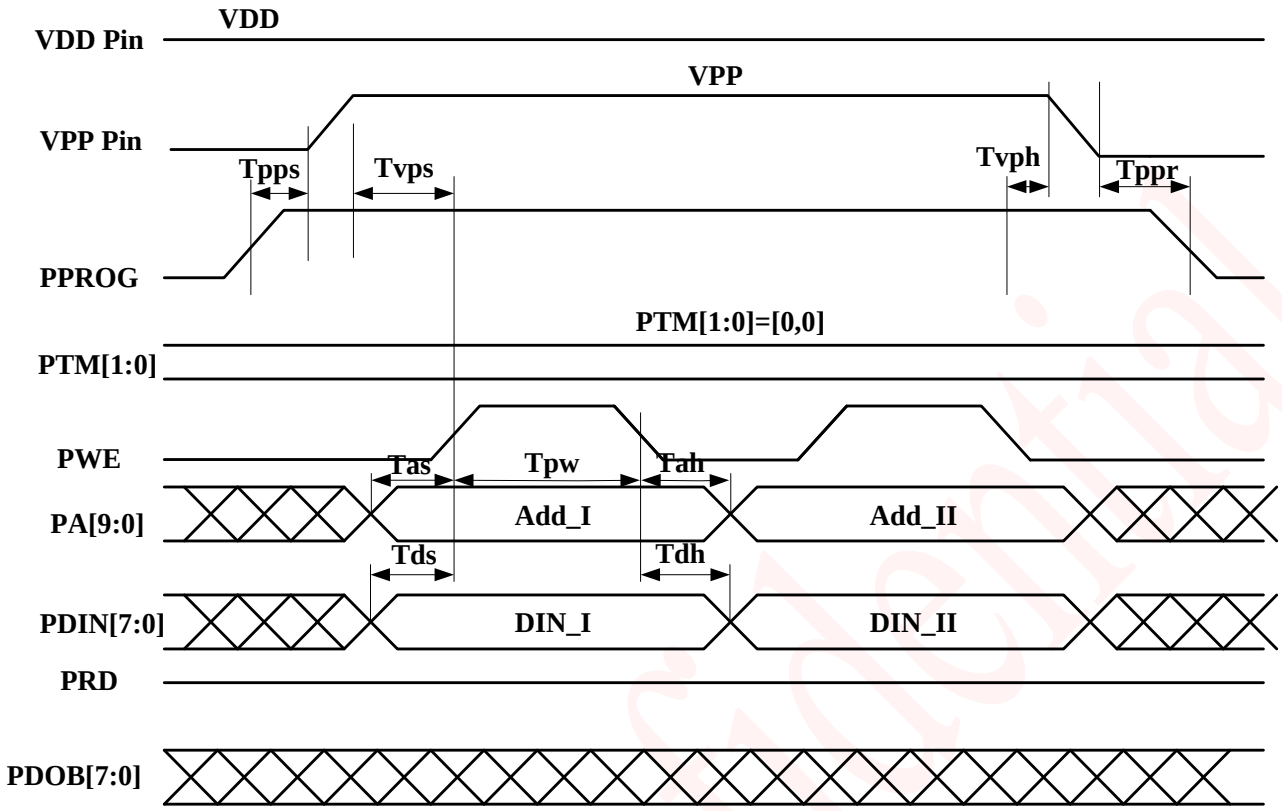


When testing, connect signal with the test port, then observe the contend.

5.12.4 Timing description

Rogram timing：

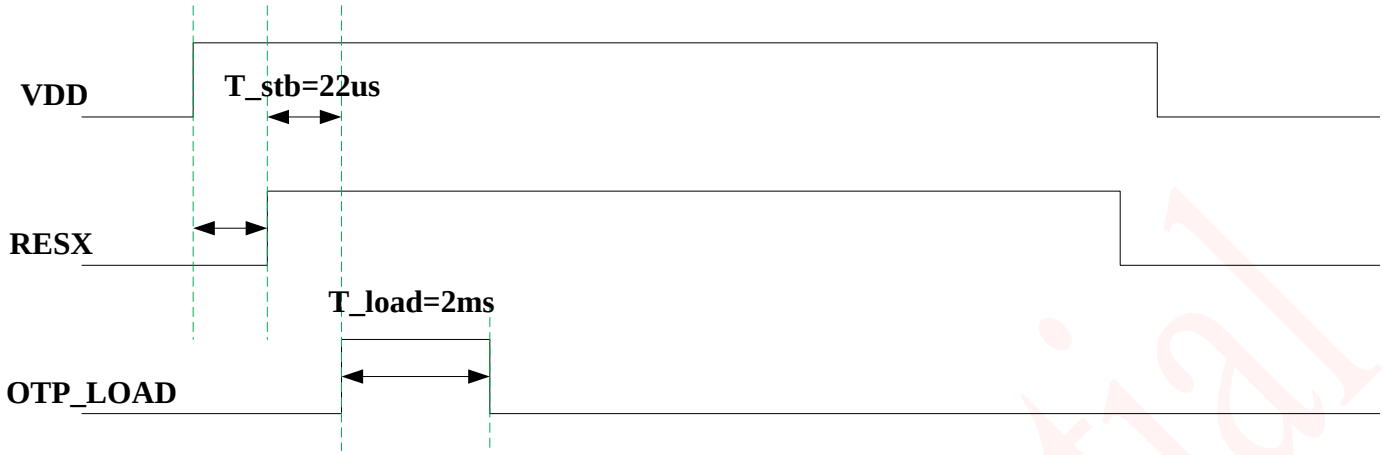
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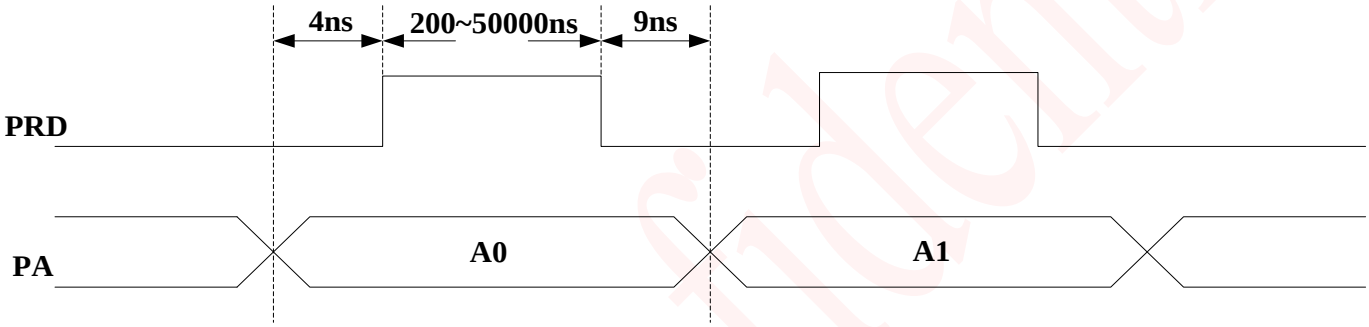
Parameter	Symbol	Min	Max	Unit
Address Setup Time	Tas	4	-	ns
Address Hold Time	Tah	9	-	ns
Data Setup Time	Tds	4	-	ns
Data Hold Time	Tdh	9	-	ns
Program Mode Setup Time	Tpps	10	-	ns
Program Mode Recovery Time	Tppr	10	-	ns
External VPP Setup Time	Tvps	0	-	ns
External VPP Hold Time	Tvph	0	-	ns
Program Pulse Width Time	Tpw	300	350	ns

load otp timing description, time is default value.

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when read, tprd should set as short as possible:



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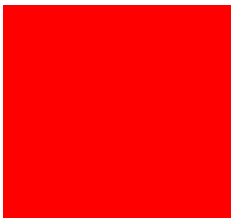
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5.13 BIST Function

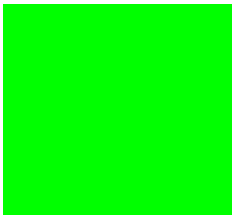
The BIST (Build In Self Test) is used for inspection, fabrication process and reliability test without external interface operation. this function is controlled by A1 command.

The following BIST patterns are built in the AXS15260D-S1, every pattern and its definition number is also defined as follows:

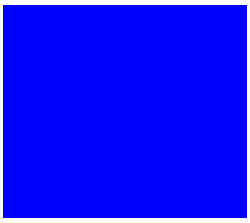
Pattern0：Red
Cr_pat_sel[0]=1



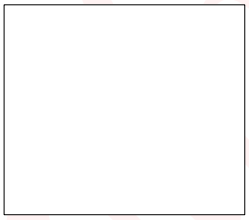
Pattern1：Green
Cr_pat_sel[1]=1



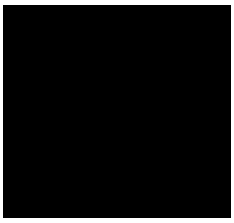
Pattern2：Blue
Cr_pat_sel[2]=1



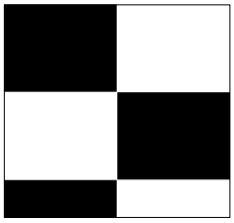
Pattern3：White
Cr_pat_sel[3]=1



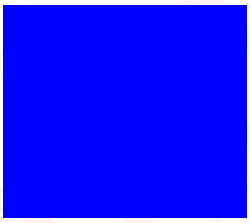
Pattern4：Black
Cr_pat_sel[4]=1



Pattern5：Chese
Cr_pat_sel[5]=1



Pattern6：Mid gray
Cr_pat_sel[6]=1



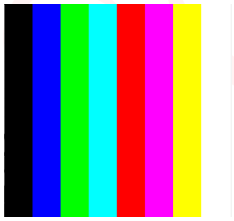
Pattern7：Gray col
Cr_pat_sel[7]=1



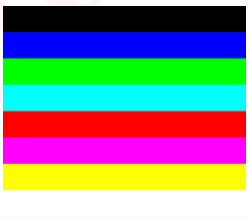
Pattern8：Gray row
Cr_pat_sel[1]=1



Pattern9：strp col
Cr_pat_sel[1]=1



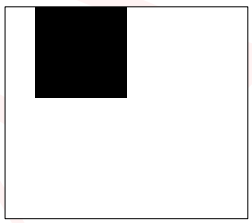
Pattern10：strp row
Cr_pat_sel[1]=1



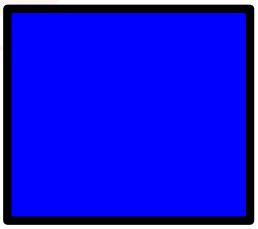
Pattern11：data partial
Cr_pat_sel[11]=1



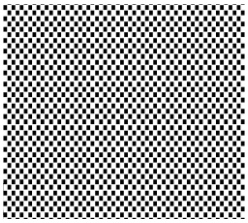
Pattern12：partial inverse
Cr_pat_sel[12]=1



Pattern13：Display rec
Cr_pat_sel[13]=1



Pattern14：data one
Cr_pat_sel[14]=1



Pattern15：crosstalk Grayscale partial mode
Cr_pat_sel[15]=1



Note:
1.cr_bist_en_mode: Bist mode enable
2.cr_pat_sel: display pattern select
3.Pattern 6(Mid Gray): RGB value(gray_red,gray_g,gray_b)can be defined by the user;
The default color display blue(gray_red=gray_g=0,gray_b=8'h FF);

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4.Pattern 13(Display rec):
RGB value (gray_red,gray_g, gray_b)can be defined by the user;
the default color display blue (gray_red=gray_g=0, gray_b=8'h FF);
but the whole picture will add a black border.

6 ELECTRICAL CHARACTERISTICS

6.1 Absolute Operation Range

VDDI : VDDI VDDI_TP VDDI_F VDDI_LDO VDDI_DRV VDDI_M

Item	Symbol	Min	Max	Unit
Supply Voltage(Analog)	VSP	-0.3	+7.8	V
Supply Voltage(Analog)	VSN	-7.8	+0.3	V
Supply Voltage(I/O)	VDDI	-0.3	+4.0	V
Driver Supply Voltage	VGH-VGL	-0.3	+32	V
Logic Input Voltage Range	VIN	-0.3	VDDI+0.3	V
Logic Output Voltage Range	VO	-0.3	VDDI+0.3	V
Operating Temperature Range	TOPR	-30	+85	℃
Storage Temperature Range	TSTG	-40	+125	℃

6.2

6.2 DC characteristic

6.2.1 Basic DC characteristic

(VSP=4.5V~6.5V, VSN=-6.5V~-4.5V, VDDI = 1.65V~3.6V, Ta = -30°C ~ 70°C)

Parameter	Symbol	Conditions	Specification			Unit	Notes
			MIN	TYP	MAX		
Power & Operation Voltage							
Analog Power	VSP	supply voltage	4.5	5.5	6.5	V	
Analog Power	VSN	supply voltage	-6.5	-5.5	-4.5	V	
Logic Operating voltage（1.8V）	VDDI	supply voltage	-	1.8	-	V	
Logic Operating voltage（3.3V）	VDDI	supply voltage	-	3.3	-	V	
VGH charge pump	VGHOUT1	-	12	15	16	V	
VGH charge pump	VGHOUT2	-	TBD	TBD	TBD	V	
VGL charge pump	VGLOUT1	-	-7.6	-10	-13	V	
VGL charge pump	VGLOUT2	-	TBD	TBD	TBD	V	
VCOM Operation							
VCOM voltage	VCOM	-	-2.5	-	0	V	0
Source Driver							

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Source positive output range	Vsout	-	0.5	-	VSP-0.3	V	
Source negative output range	Vsout	-	-0.3	-	VSN+0.3	V	
Reference Voltage							
Internal reference voltage	VREF	-	-	1.8	-	V	
Current Consumption							
Sleep-IN mode(LP-11)	IIOVCC	RESX=High		TBD	TBD	uA	
	IVDD			TBD	TBD	uA	
Sleep-IN mode(ULPS)	IIOVCC	RESX=High		TBD	TBD	uA	
	IVDD			TBD	TBD	uA	

Note:
VGHOUT1:When the external on-load current is below 1.25mA, the maximum voltage can reach 16V, and the register can be adjusted normally;
VGLOUT1:When the external on-load current is below 1.35mA, the minimum voltage can reach -13V, and the register can be adjusted normally.

6.2.2 MIPI DC character

DC characteristics for MIPI-DSI
(VSP=4.5V~6.5V, VSN=-6.5V~-4.5V, VDDI = 1.65V~3.6V, Ta = -30ºC ~ 70ºC)

Parameter	Symbol	Conditions	Specification			Unit
			MIN	TYP	MAX	
Power supply voltage for MIPI Interface						
Power supply voltage for MIPI interface	VDDI_HS	-		1.8		V
LPDT Input Characteristics						
Pad signal voltage range	VI	-	-50	-	1350	mV
Ground Shift	HS_VSS	-	-50	-	50	mV
Logic 0 input threshold	VIL	-	0	-	550	mV
Logic 1 input threshold	VIH	-	880	-	VDDI_M	mV
Input hysteresis	VHYST	-	25	-	-	mV
LPDT Output Characteristics						
Output low level	VOL	-	-50	-	50	mV
Output high level	VOH	-	1.1	1.2	1.3	V
Logic 1 contention threshold	VIHCD,MIN	-	450	-	HS_LDO	mV
Logic 0 contention threshold	VILCD,MAX	-	0	-	200	mV
Output impedence of LPDT	ZOLP	-	80	-	125	ohm

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Hi-speed Input/Output Characteristics						
Single-end input low voltage	VILHS	-	-40	-	-	mV
Single-end input high voltage	VIHHS	-	-	-	460	mV
Common mode voltage	VCMRXDC	-	70	-	330	mV
Hi-speed transmit voltage	VOD	-	140	200	250	mV
Differential input impedence	ZID	-	80	100	125	ohm

6.3 AC characteristic

6.3.1 Reset

The part of touch

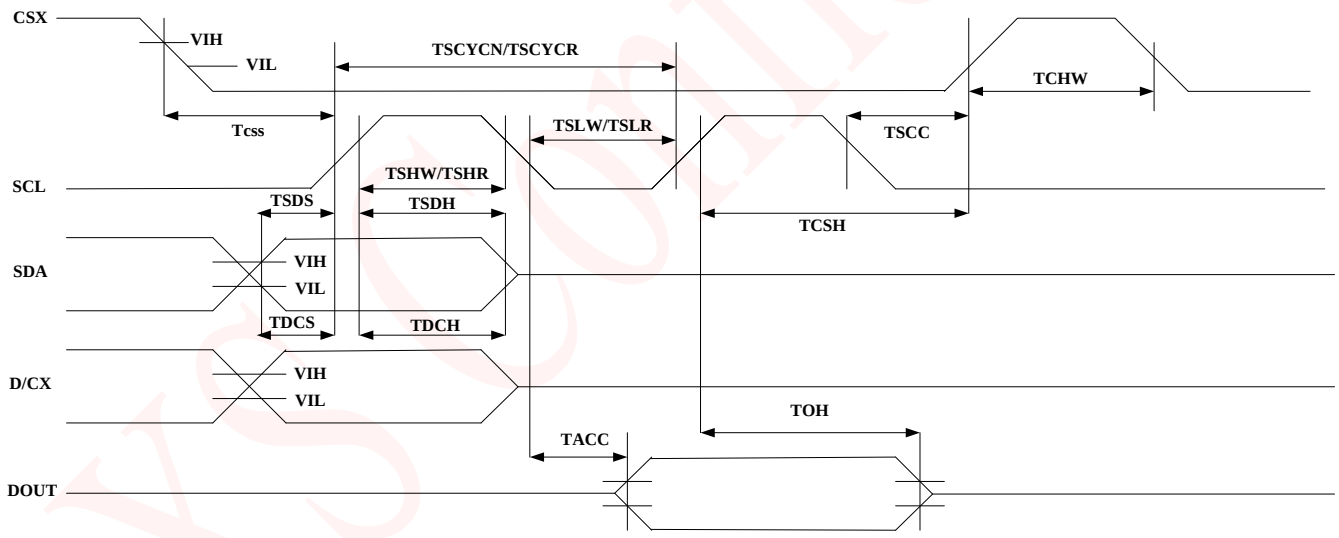
External Reset: 0~VDDI

Soft reset

Host can issue reset instructions to reset the system

6.3.2

6.3.2 Serial interface characteristics (4-line SPI)



Signal	Symbol	Parameter	MIN	MAX	Unit	Description
CSX	T_{CSS}	Chip select setup time (write)	15	-	ns	
	T_{CSH}	Chip select hold time (write)	15	-	ns	
	T_{CSS}	Chip select setup time (read)	60	-	ns	
	T_{SCC}	Chip select hold time (read)	65	-	ns	
	T_{CHW}	Chip select “H” pulse width	40	-	ns	
	T_{SCYCW}	Serial clock cycle (Write)	25	-	ns	-write command & data
	T_{SHW}	SCL “H” pulse width (Write)	12	-	ns	

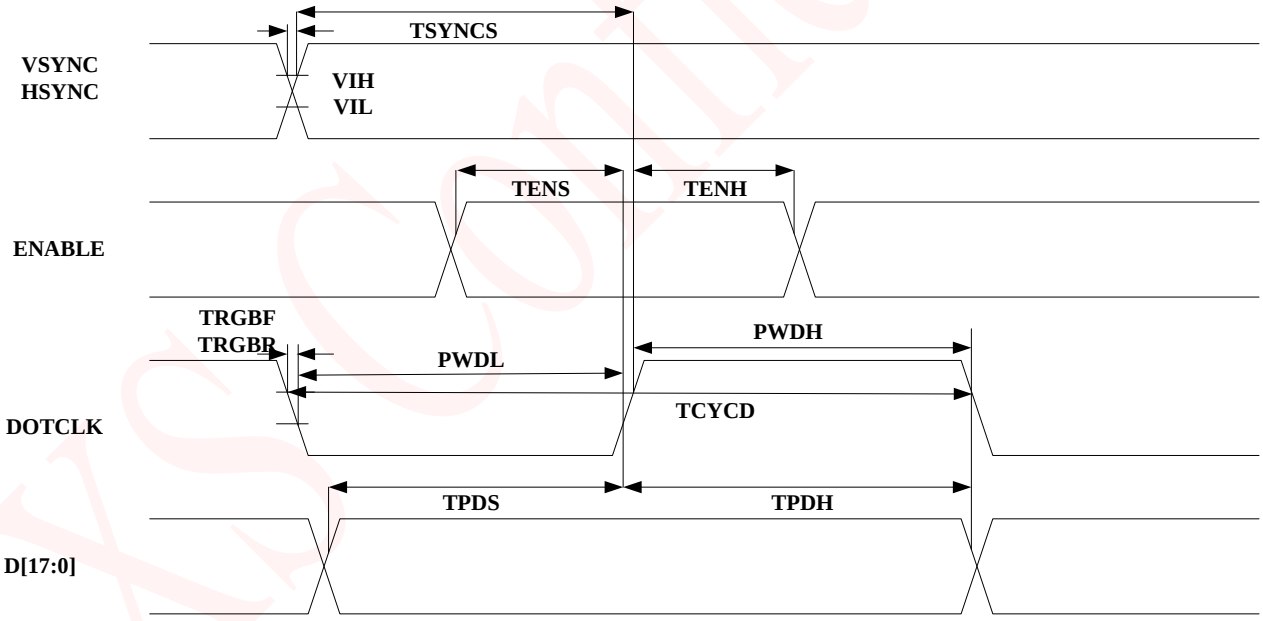
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SCL	T_{SLW}	SCL “L” pulse width (Write)	12	-	ns	ram
	T_{SCYCR}	Serial clock cycle (Read)	25	-	ns	-read command & data ram
	T_{SHR}	SCL “H” pulse width (Read)	12	-	ns	
	T_{SLR}	SCL “L” pulse width (Read)	12	-	ns	
D/CX	T_{DCS}	D/CX setup time	10	-	ns	
	T_{DCH}	D/CX hold time	10	-	ns	
SDA (DIN)	T_{SDS}	Data setup time	10	-	ns	
	T_{SDH}	Data hold time	10	-	ns	
DOUT	T_{ACC}	Access time	10	50	ns	For maximum CL=30pF
	T_{OH}	Output disable time	15	50	ns	For minimum CL=8pF

Note 1: VSP=4.5V~6.5V, VSN=-6.5V~-4.5V, VDDI = 1.65V~3.6V, Ta = -30°C ~ 70°C.
Note 2: The input signal rise time and fall time (tr, tf) is specified at 15 ns or less.
Logic high and low levels are specified as 30% and 70% of IOVCC for Input signals.

6.3.3

6.3.3 RGB interface characteristics



RGB Interface Characteristics

Signal	Symbol	Parameter	Min	Max	Unit	Description
VSYNC/HSYNC	tsyncs	VSYNC/HSYNC setup time	15	-	ns	24/18/16-bit
	tsynch	VSYNC/HSYNC hold time	15	-	ns	
DE	tens	DE setup time	15	-	ns	
	tenh	DE hold time	30	-	ns	
D[23:0]	tpds	Data setup time	15	-	ns	
	tpdh	Date hold time	15	-	ns	

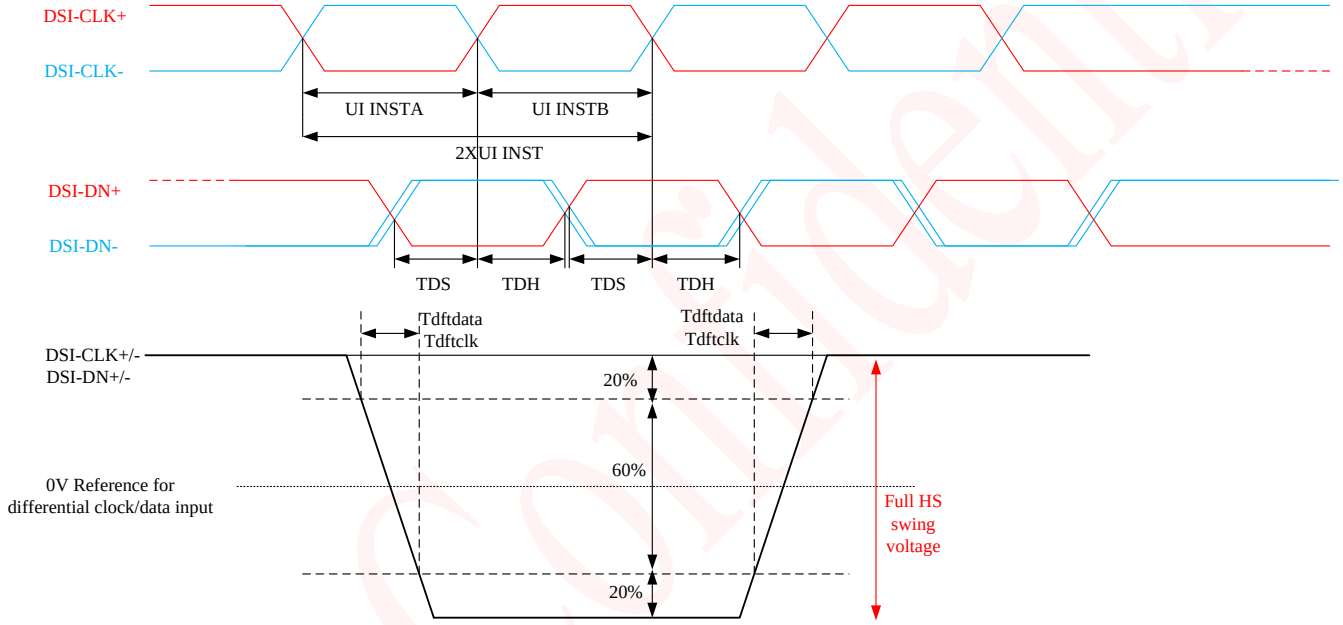
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DOTCLK	PWDH	DOTCLK high-level period	20	-	ns	bus RGB interface mode
	PWDL	DOTCLK low-level period	20	-	ns	
	tcycd	DOTCLK cycle time	40	-	ns	
	trgbr,trgbf	DOTCLK,HSYNC,VSYNC rise/fall time	-	-	ns	

6.3.4

6.3.4 MIPI-DSI characteristics

6.3.4.1 High speed mode



Parameter	Symbol	Parameter	Specification			Unit	Description
			MIN	TYP	MAX		
DSI-CLK+/-	$2xUI_{INSTA}$	Double UI instantaneous	1.6		25	ns	
DSI-CLK+/-	UI_{INSTA} UI_{INSTB}	UI instantaneous halfs	0.8		12.5	ns	$UI=UI_{INSTA}=UI_{INSTB}$
DSI-D0+/-	T_{DS}	Data to clock setup time	0.15	-		UI	
DSI-D0+/-	T_{DH}	Data to clock hold time	0.15	-		UI	

Figure: AC characteristics for MIPI-DSI High speed mode

6.3.4.2 Low power mode

Parameter	Symbol	Parameter	Specification			Unit
			MIN	TYP	MAX	

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Low Power Mode						
DSI-D0+/-	T_{LPXM}	Length of LP-00, LP-01, LP-10 or LP -11 periods MPU Display Module	50	-	-	ns
DSI-D0+/-	T_{LPXD}	Length of LP-00, LP-01, LP-10 or LP -11 periods Display Module MPU	58	-	-	ns
DSI-D0+/-	$T_{TA-SURED}$	Time-out before the MPU start driving	T LPXD	-	2XT LPXD	ns
DSI-D0+/-	$T_{TA-GETD}$	Time to driver LP-00 by display module	5XT LPXD	-	-	ns
DSI-D0+/-	T_{TA-GOD}	Time to driver LP-00 after turnaround request - MPU	4XT LPXD	-	-	ns
DSI-D0+/-	Ratio T_{LPX}	Ratio of T_{LPXM} / T_{LPXD} between MCU and display module	2/3	-	3/2	

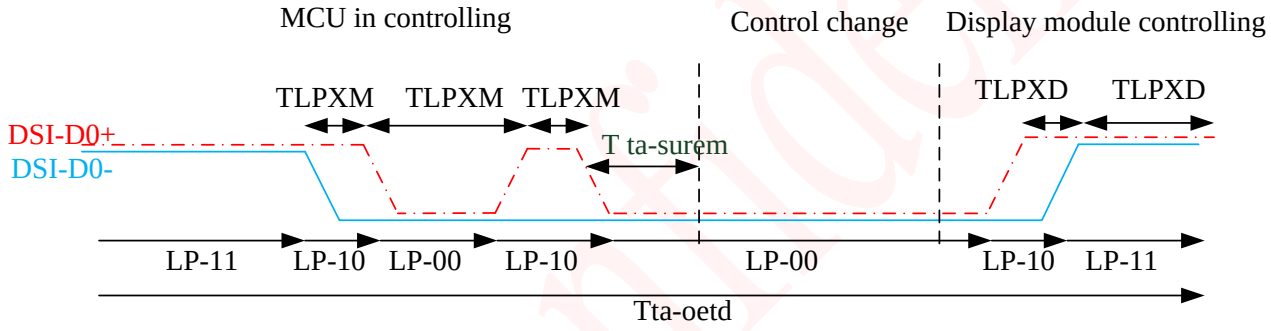


Figure: BTA from the MCU to the Display Module

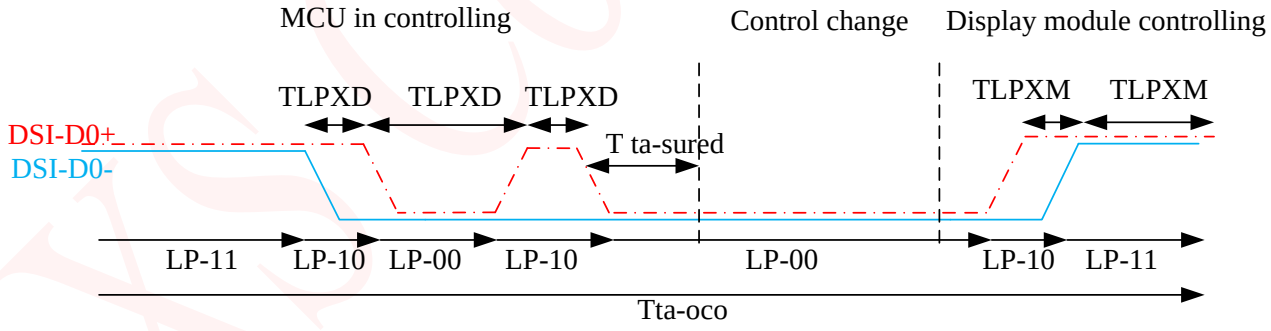


Figure: BTA from the Display Module to the MCU

6.3.4.3 Bursts

Parameter	Symbol	Parameter	Specification			Unit
			MIN	TYP	MAX	
High Speed Data Transmission Bursts						
DSI-Dn+/-	T_{LPX}	Length of any low-power state period	50	-	-	ns
DSI-Dn+/-	T_{HS-}	Time to driver LP-00 to prepare for HS	40ns + 4UI	-	85ns +	ns

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	PREPARE	transmission			6UI	
DSI-Dn+/-	$T_{HS-PREPARE}$ +$T_{HS-ZERO}$	$T_{HS-PREPARE}$ + time to driver HS-0 before the sync sequence	145ns + 10UI	-	-	ns
DSI-Dn+/-	$T_{D-TERM-EN}$	Time to enable Data Lanereceiver line termination measured from when Dn crosses $V_{IL(max)}$	Time for Dn to reach $V_{TERM-EN}$	-	35ns + 4UI	ns
DSI-Dn+/-	$T_{HS-SKIP}$	Time-out at RX to ignore transition period of EoT	40	-	55ns + 4UI	ns
DSI-Dn+/-	$T_{HS-TRAIL}$	Time to driver flipped differential state after last payload data bit of a HS transmission burst	max (8UI, 60ns+4UI)	-	-	ns
DSI-Dn+/-	$T_{HS-EXIT}$	Time to driver LP-11 after HS burst	100	-	-	ns
DSI-Dn+/-	T_{EoT}	Time from start of $T_{HS-TRAIL}$ Period to start of LP-11 state	-	-	105ns +12UI	ns

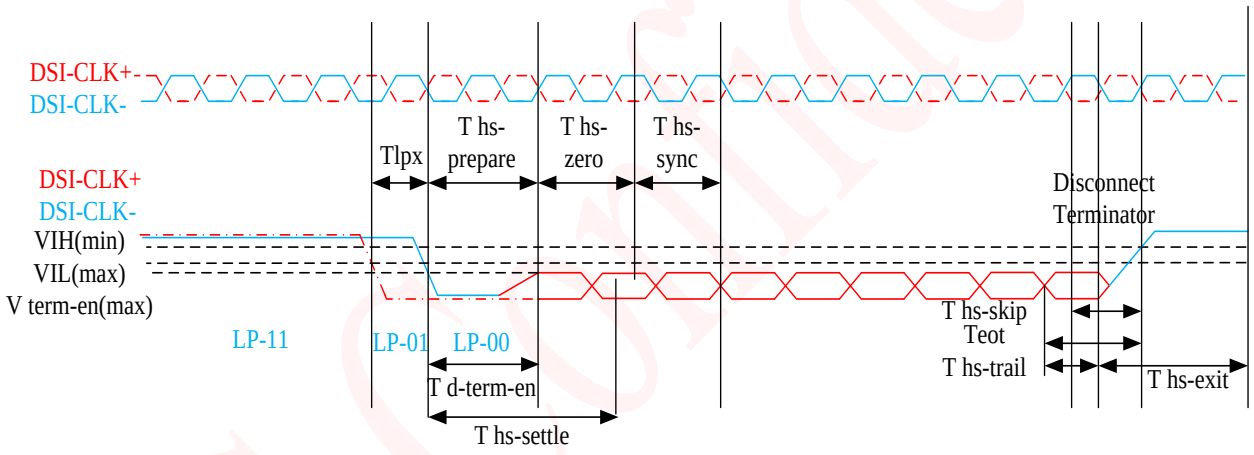


Figure: High Speed Data Transmission Bursts

Parameter	Symbol	Parameter	Specification			Unit
			MIN	TYP	MAX	
Switching the clock Lane between clock Transmission and Low Power Mode						
DSI-CLK+/-	$T_{CLK-POST}$	Time that the transmitter shall continue sending HS clock after the last associated Data Lanehas transitioned to LP mode	60ns + 52UI	-	-	ns
DSI-CLK+/-	$T_{CLK-PRE}$	Time that the HS clock shall be drivern prior to any associated Data Lanebeginning the transition from LP to HS mode	8	-	-	UI
DSI-CLK+/-	$T_{CLK-PREPARE}$	Time to driver LP-00 to prepare	38	-	95	ns

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		for HS clock transmission				
DSI-CLK+/-	$T_{CLK-TERM-EN}$	Time to enable Clock Lanereceiver line termination measured from when Dn crosses $V_{IL(max)}$	Time for Dn to reach $V_{TERM-EN}$	-	38	ns
DSI-CLK+/-	$T_{CLK-PREPARE}$ +$T_{CLK-ZERO}$	$T_{CLK-PREPARE}$+ time for lead HS-0 driver period before starting Clock	300	-	-	ns
DSI-CLK+/-	$T_{CLK-TRAIL}$	Time to driver HS differential state after last payload clock bit of a HS transmission burst	60	-	-	ns
DSI-CLK+/-	T_{EoT}	Time from start of $T_{CLK-TRAIL}$ periodto start of LP-11 state	-	-	105ns + 12UI	ns

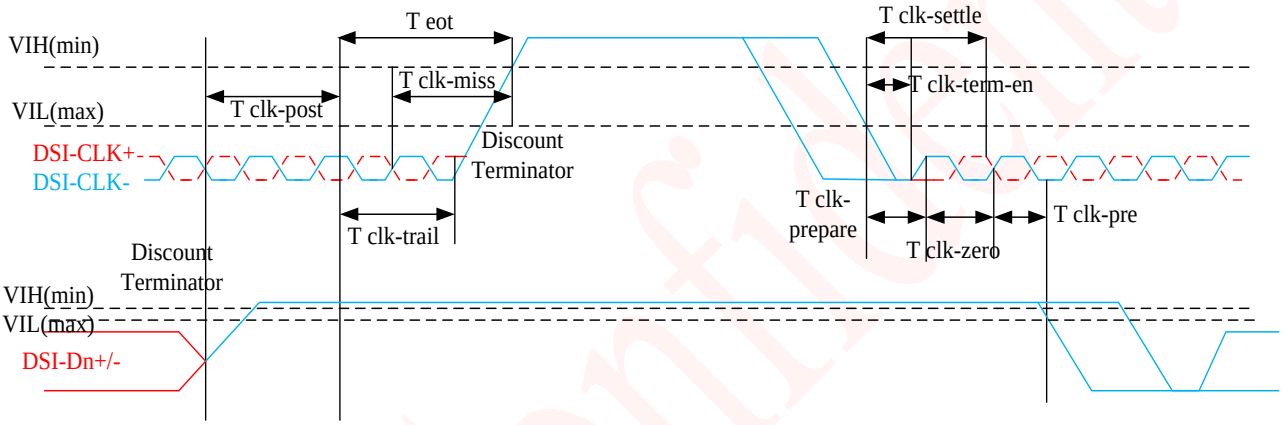


Figure: Switching the clock Lane between clock Transmission and Low Power Mode

6.3.4.4 LP-11 between High Speed and Low Power Modes

DSI-D0 High Speed or Low Power modes are starting or finishing from/to Stop State (SS, LP-11) when

different combinations, what are listed below, are possible:

1. High Speed Mode => Stop State (SS, LP-11) => High Speed Mode
2. High Speed Mode => Stop State (SS, LP-11) => Low Power Mode
3. Low Power Mode => Stop State (SS, LP-11) => High Speed Mode
4. Low Power Mode => Stop State (SS, LP-11) => Low Power Mode

The Low Power Mode is also including 2 different functions:

1. Escape
2. Bus Turnaround (BTA)

Stop State (SS, LP-11) Timings from Previous mode to Next mode

Next / Previous	Escape mode	HSDT	BTA

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	Min	Max	Min	Max	Min	Max
Escape mode	100ns	-	100ns	-	100ns	-
HSDT	60ns+52UI	-	60ns+52UI	-	60ns+52UI	-
BTA	100ns	-	100ns	-	100ns	-

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7 Power Definition

7.1 Start-up time

The part of touch

Power on：

Flash boot≤100mS

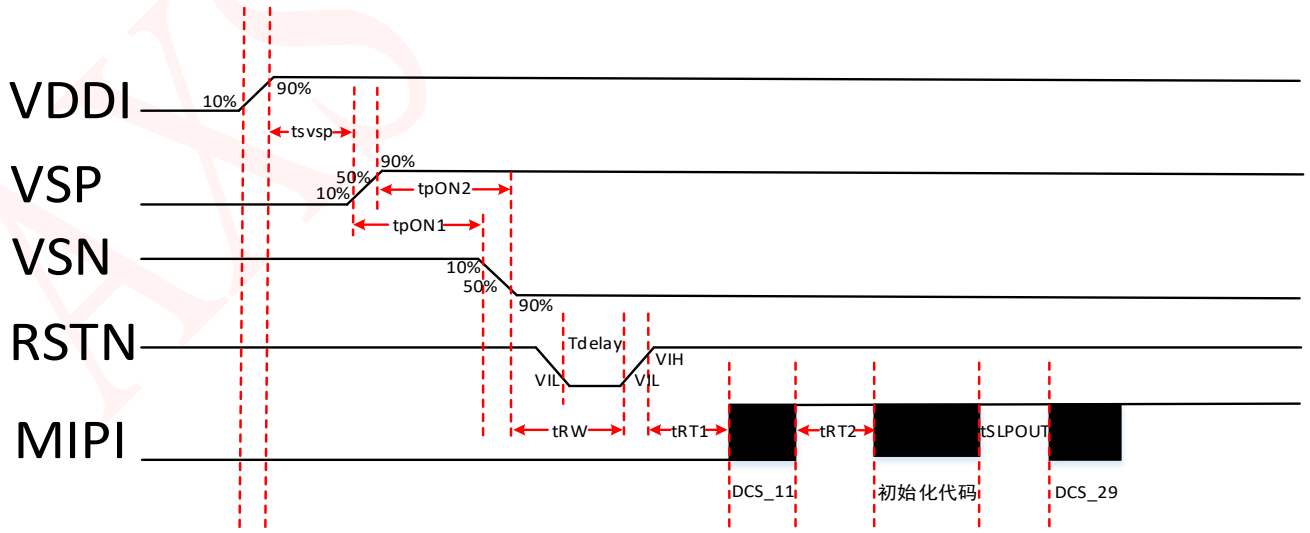
Wake up≤50ms

7.2 Power On/Off Sequence

The power on/off sequence is illustrated below:

7.2.1 Power on Sequence

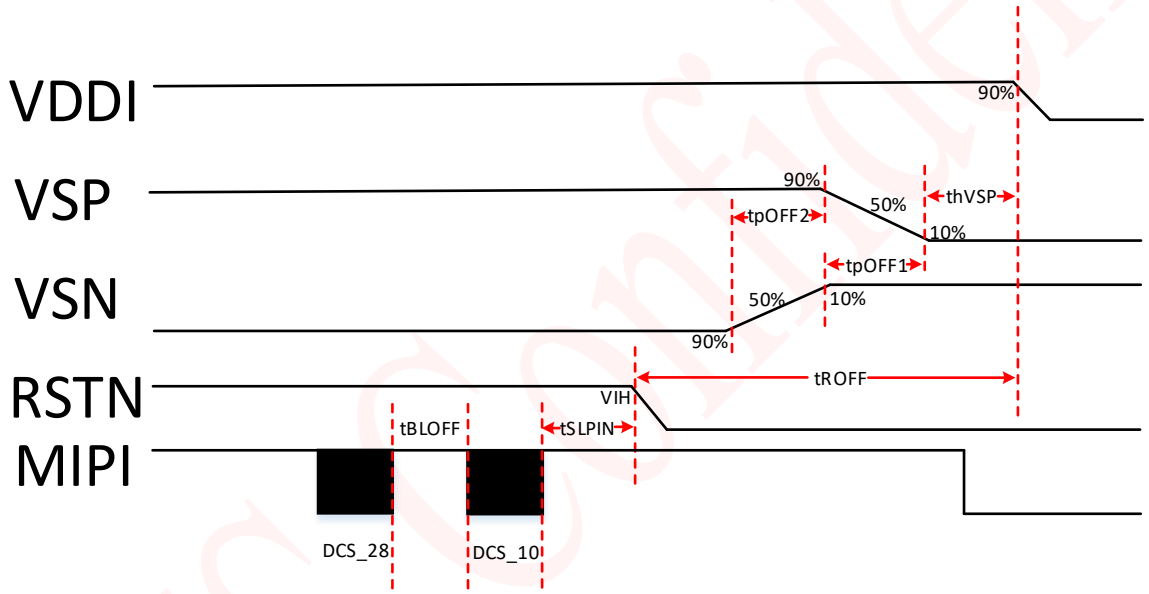
Description	Signal	Condition	Typ	Unit
RST Hi-level (VIH) to DCS_11 time	tRT1	Power On	160	ms
SLPOUT Sequence Request time	tSLPOUT	Power On	13.5	ms
VDDI On to VSP On time (90% to 10%)	ts_vsp	Power On	14.7	ms
VSP-VSN delay time （10%-10%)	tPON1	Power On	7.9	ms
VSP-VSN delay time （90%-90%)	tPON2	Power On	8	ms
All Power On to RSTN Hi-level time (90%-VIL)	tRW	Power On	28.7	ms
DCS_11 to 1st Command time	tRT2	Power On	9.9	ms
RSTN delay time	Tdelay	Power On	10	ms



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7.2.2 Power off Sequence

Description	Signal	Condition	Typ	Unit
VSP-VSN delay time （90%-90%)	tPOFF2	Power Off	8.77	ms
VSP-VSN delay time （10%-10%)	tPOFF1	Power Off	3.4	ms
VSP Off to VDDI Off time （10%-90%)	thVSP	Power Off	8.84	ms
RSTN Low to VDDI Off time (VIH to 90%)	tROFF	Power Off	167	ms
SLPIN Sequence Request time	tSLPIN	Power Off	140	ms
DCS_28 to DCS_10 delay time	tBLOFF	Power Off	9.9	ms

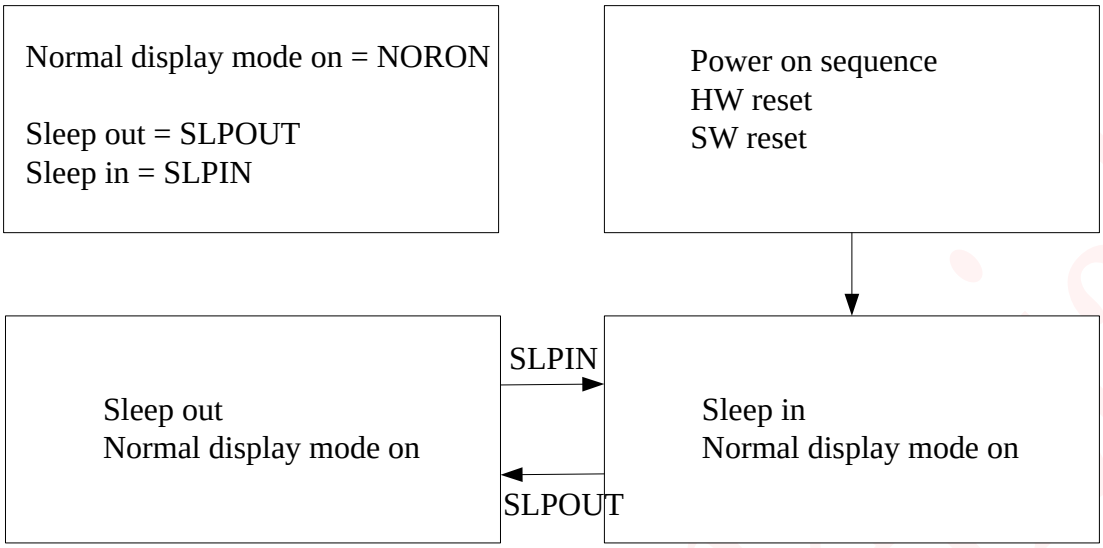


7.2.3 Uncontrolled power off

The uncontrolled power off means a situation when e.g. there is removed a battery without the controlled power off sequence. There will not be any damage for the display module or the display module will not cause any damage to the host or lines of the interface. At an uncontrolled power off event, the display will go blank and there will not be any visible effects within (TBD) second on the display (blank display) and remains blank until “Power On Sequence” powers it up.

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7.3 Power flow chart



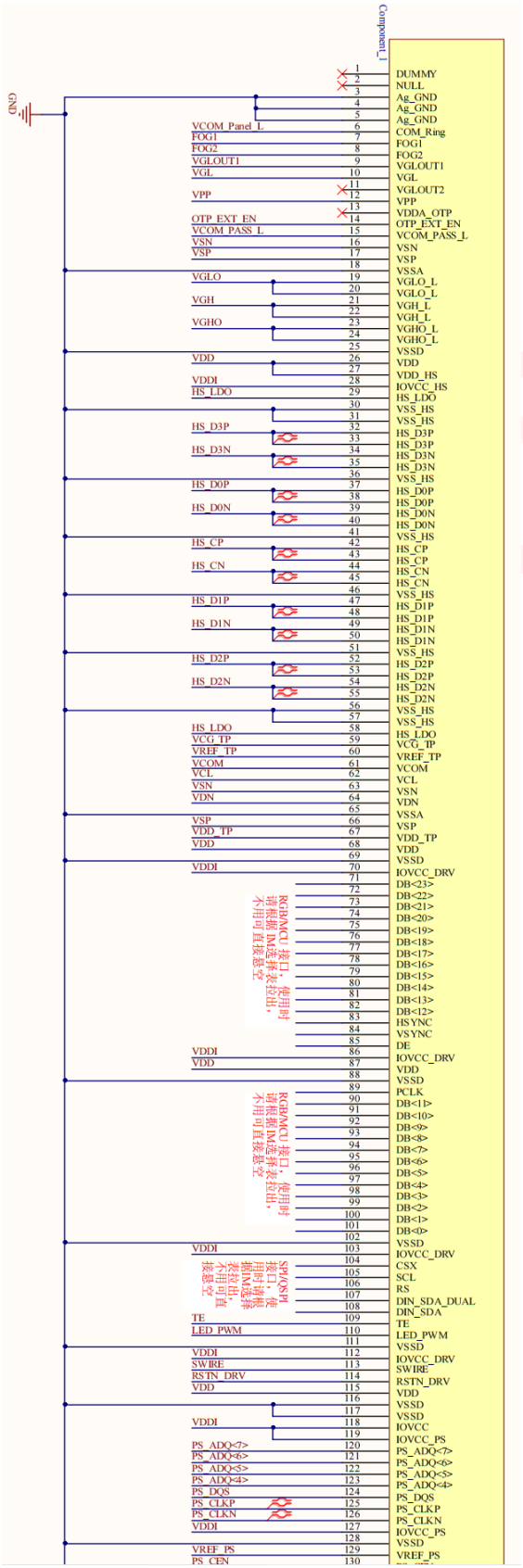
Note 1: There is not any abnormal visual effect when there is changing from one power mode to another power mode.

Note 2: There is not any limitation, which is not specified by this spec, when there is changing from one power mode to another power mode.

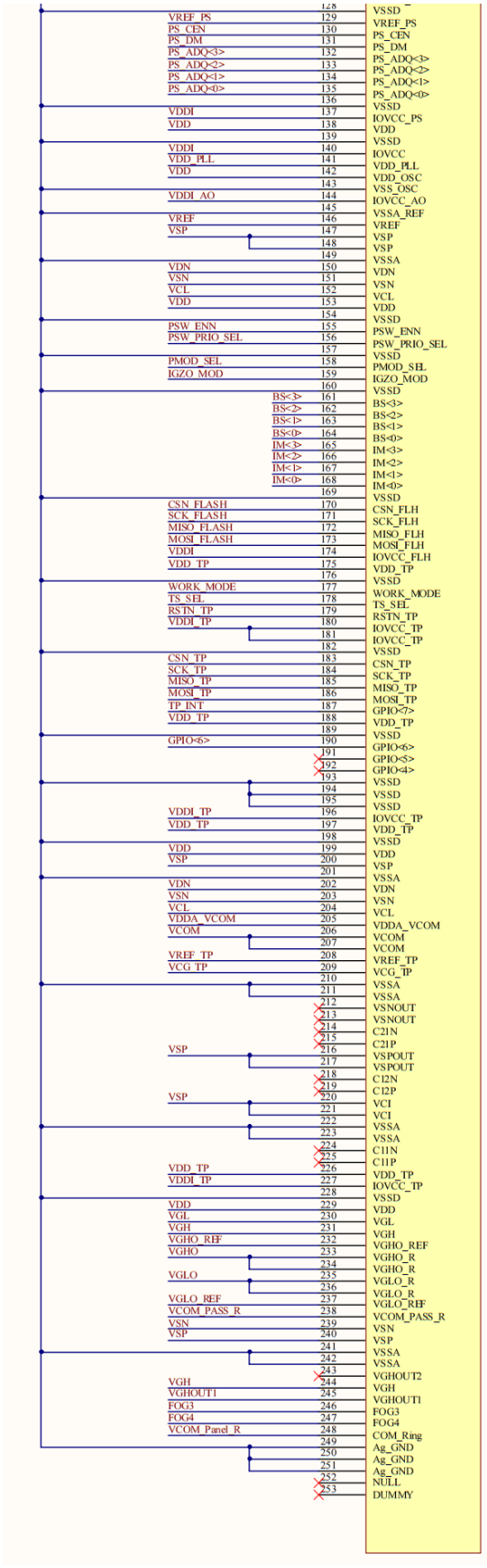
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8 APPLICATION CIRCUIT



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VGHOUT1/2 、VGLOUT1/2 ： panel是a-si 工艺用OUT1，LTPS 工艺用OUT2

PSW_ENN is power exchanger set pin: 0:enable ;1:disable,Default=1

PMOD_SEL is power voltage set pin: 0:when VDDI=3.3v 1:when VDDI=1.8v,Default=1

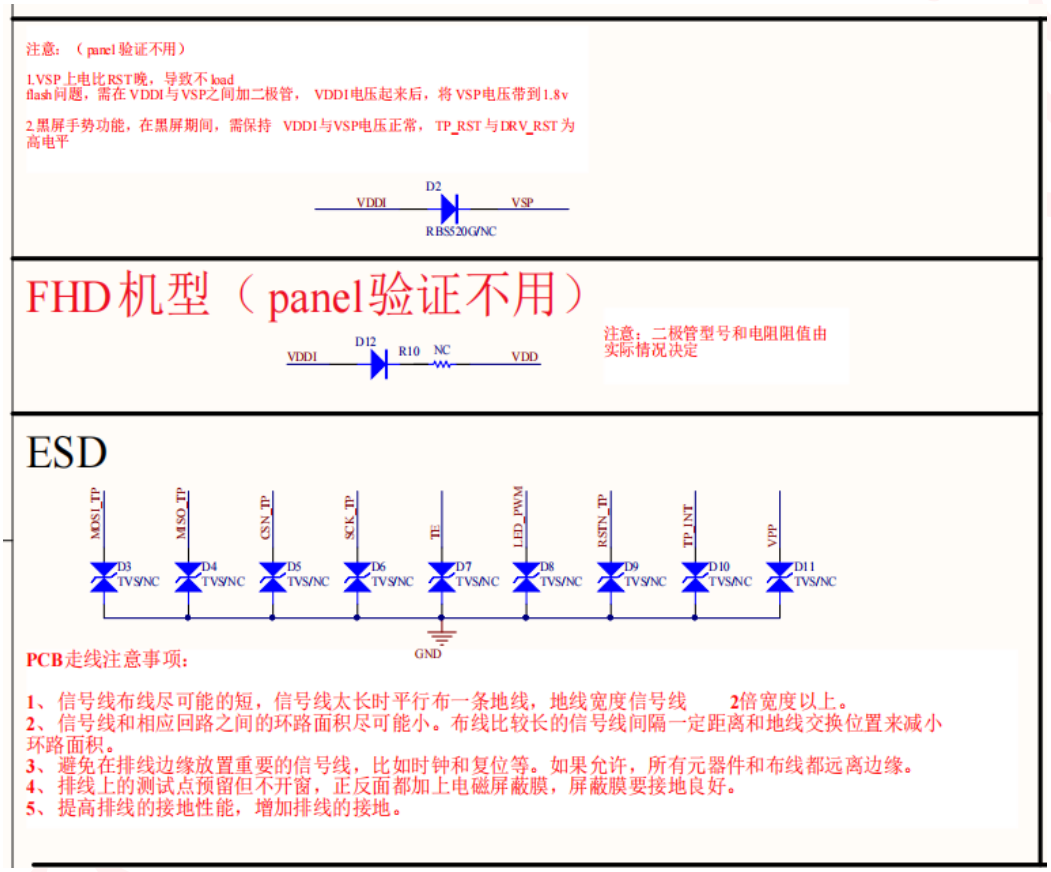
WORK_MODE is Selection of tp work interface ： 1:IIC， 0:SPI ,Default=0 补充SCK_TP 复用IIC的SCL ,MOSI_TP 复用IIC的SDA

PSW_PRIO_SEL ： 0:VDDI_TP 优先， 1:VDDI_DRV 优先 ,Default=1

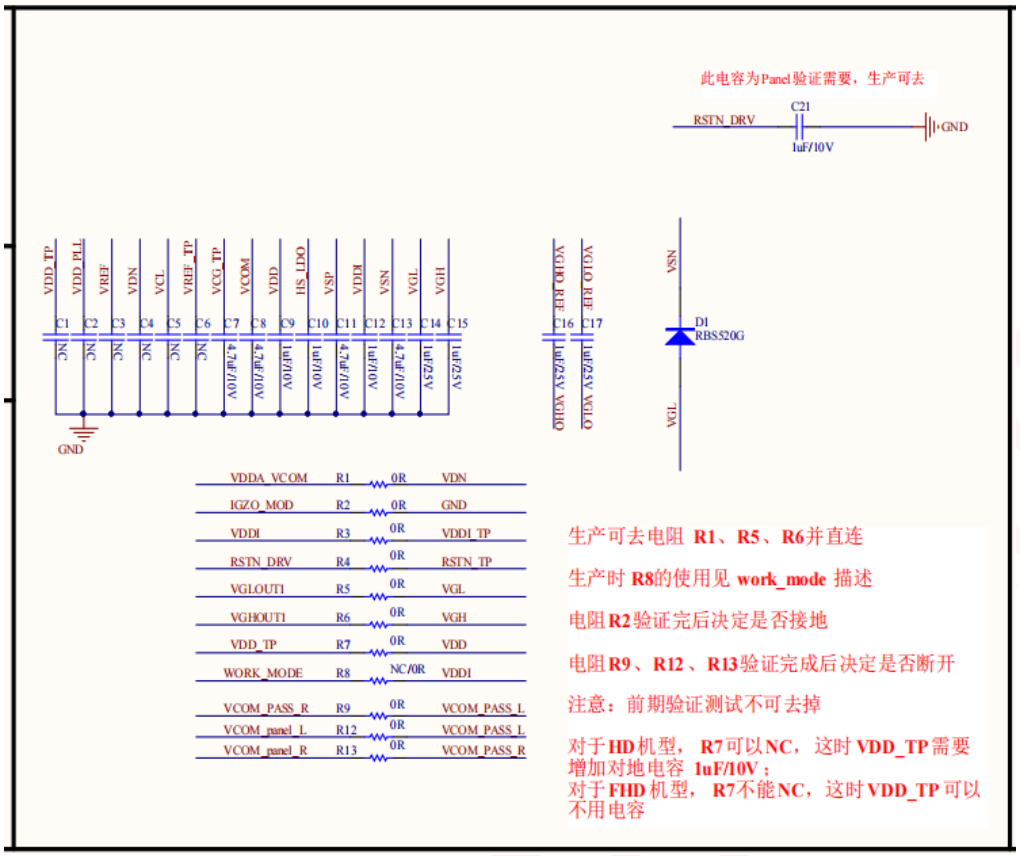
TS_SEL ： 测试pin

VPP： OTP电源， OTP时需外灌 8.5V

ATEST： 测试pin



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生产可去电阻 **R1**、**R5**、**R6**并直连

生产时 **R8**的使用见 **work_mode** 描述

电阻**R2**验证完后决定是否接地

电阻**R9**、**R12**、**R13**验证完成后决定是否断开

注意：前期验证测试不可去掉

对于**HD**机型，**R7**可以**NC**，这时**VDD_TP**需要增加对地电容 **1uF/10V**；
对于**FHD**机型，**R7**不能**NC**，这时**VDD_TP**可以不用电容

PCB走线注意事项：

1.MIPI走线需要按照标准差分走线要求，同时注意如果有过孔所有通道都要有相同的过孔，要么全过孔，要么全不过孔，不能有的通道过孔，有的通道不过孔；

2.TP_CLK 和**TP_CSN** 需要两边包地与其他线隔离开，防止干扰；如果条件允许 **SPI**的走线都建议包地

3.测试焊盘按参考原理图预留；

4.手机的 **TP_IOVCC** 和**TP_VCI** 需要引焊盘出来；

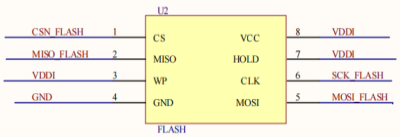


生产可去测试 **pad**：**PMOD_SEL**、**PSW_PRIO_SEL**、**PSW_ENN** 验证完后决定是否去掉

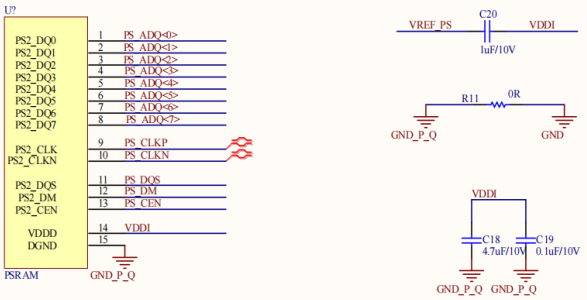
注意：前期验证测试不可去掉

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FLASH

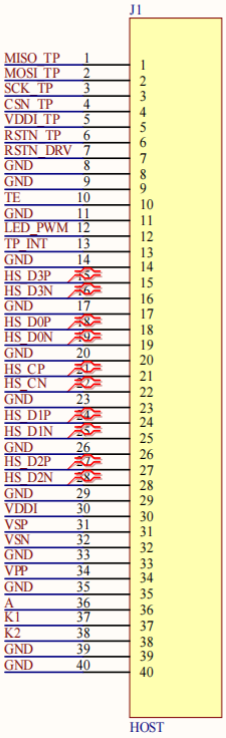


PSRAM （panel验证不用）

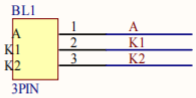


1.PSRAM 不用时，所有 pin 可以悬空
2.PSRAM 不用时，需要关掉使能（使能默认关）
3.psram数据线尽量短，等长，间隔不要太近

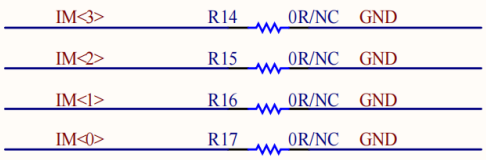
接口



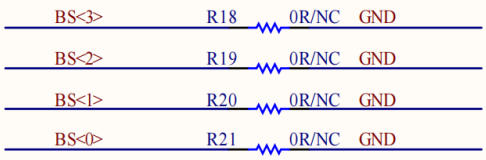
背光



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使用时请根据IM选择表拉出，内置上拉电阻，选 1时可直接悬空，选 0接GND



使用时请根据 BS选择表拉出，内置上拉电阻，选1时可直接悬空，选 0接GND

涉及15260 版本兼容，请优先选择左侧表格的红色字体顺序

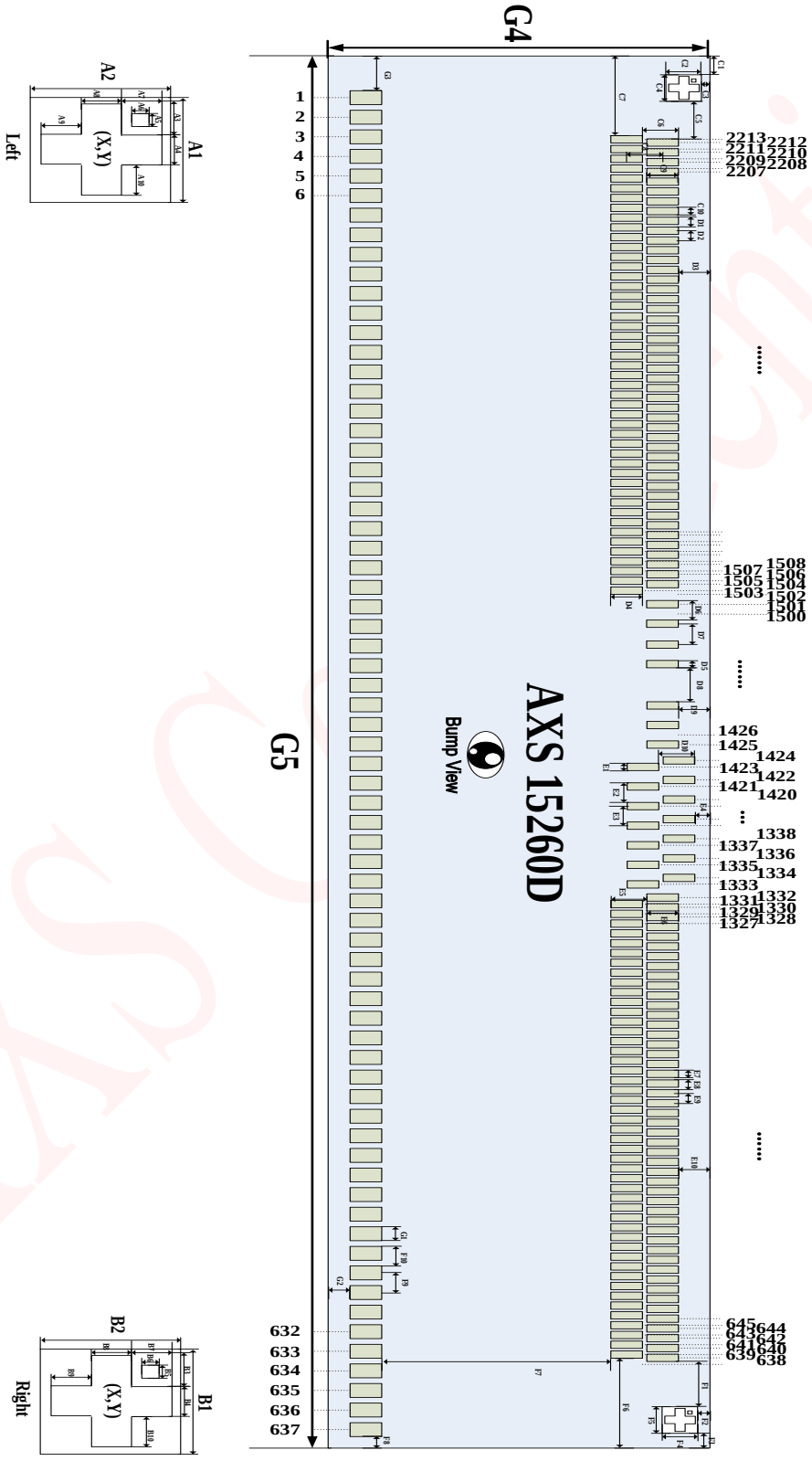
BS<3：0>内置上拉电阻，不需外部上拉

Note：pls contact AXS if you have any question.

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9 CHIP IMFORMATION

9.1 PAD Assignment



The information contained herein is the exclusive property of AXS Technology Corp. and shall not be distributed, reproduced, or disclosed in whole or in part without prior written permission of AXS Technology Corp.

Symbol	Size	Symbol	Size	Symbol	Size	Symbol	Size	Symbol	Size
A1	80	A2	80	A3	15	A4	20	A5	10
A6	10	A7	15	A8	20	A9	15	A10	15
B1	80	B2	80	B3	15	B4	20	B5	10
B6	10	B7	15	B8	20	B9	15	B10	15
C1	50	C2	80	C3	50	C4	80	C5	264/262
C6	100	C7	382/380	C8	100	C9	75	C10	12
D1	24/23.99	D2	24/23.99	D3	81.28	D4	75	D5	12
D6	47.98/47.99/48	D7	47.98/47.99/48	D8	335.92/336	D9	81.28	D10	100
E1	12	E2	71.99/71.98/72	E3	71.99/71.98/72	E4	35.28	E5	100
E6	75	E7	12	E8	24/23.99	E9	24/23.99	E10	81.28
F1	252/250	F2	50	F3	50	F4	80	F5	80
F6	394/392	F7	441.22	F8	88.76/86	F9	38.99/39	F10	38.99/39
G1	24	G2	47.5	G3	88.76/86	G4	820	G5	25000
								Unit=um	

Item	PAD No.	Size		Unit
		X	Y	
Chip Size	-	25000	820	um
Chip thickness	-	200		
Pad Size	1 - 637	24	75	
	638 - 1332	12	75	
	1333 - 1424	12	75	
	1425 - 1463	12	75	
	1464 - 1502	12	75	
	1503 - 2213	12	75	
Pad Pitch	1 - 637	38.99/39	-	
	638 - 1332	11.99/12	100	
	1333 - 1424	11.99/12	100	
	1425 - 1463	47.98/47.99/48	-	
	1464 - 1502	47.98/47.99/48	-	
	1503 - 2213	11.99/12	100	

Note1: Have Temperature compensation design.

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9.2 PAD CENTER COORDINATE

PAD NO	PAD name	X	Y
1	VGLOUT1	-12399.24	-325
2	VGLOUT1	-12360.25	-325
3	VGL	-12321.26	-325
4	VGL	-12282.27	-325
5	VGLOUT2	-12243.28	-325
6	VGLOUT2	-12204.29	-325
7	VPP	-12165.3	-325
8	VPP	-12126.3	-325
9	VDDA_OTP	-12087.31	-325
10	VDDA_OTP	-12048.32	-325
11	OTP_EXT_EN	-12009.33	-325
12	VCOM_PASS_L	-11970.34	-325
13	VSN	-11931.35	-325
14	VSN	-11892.36	-325
15	VSP	-11853.37	-325
16	VSP	-11814.37	-325
17	VSSA	-11775.38	-325
18	VSSA	-11736.39	-325
19	VSSA	-11697.4	-325
20	VGLO_L	-11658.41	-325
21	VGLO_L	-11619.42	-325
22	VGLO_L	-11580.43	-325
23	VGLO_L	-11541.43	-325

PAD NO	PAD name	X	Y
1108	VSSA	6472.56	291.22
1109	VSSA	6460.56	191.22
1110	VSSA	6448.57	291.22
1111	VSSA	6436.57	191.22
1112	VSSA	6424.57	291.22
1113	SX<320>	6412.57	191.22
1114	S<540>	6400.58	291.22
1115	S<539>	6388.58	191.22
1116	VSSA	6376.58	291.22
1117	S<538>	6364.59	191.22
1118	SX<319>	6352.59	291.22
1119	S<537>	6340.59	191.22
1120	S<536>	6328.59	291.22
1121	VSSA	6316.6	191.22
1122	S<535>	6304.6	291.22
1123	SX<318>	6292.6	191.22
1124	S<534>	6280.6	291.22
1125	S<533>	6268.61	191.22
1126	VSSA	6256.61	291.22
1127	S<532>	6244.61	191.22
1128	SX<317>	6232.61	291.22
1129	S<531>	6220.62	191.22
1130	S<530>	6208.62	291.22

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24	VGLO_L	-11502.44	-325	1131	VSSA	6196.62	191.22
25	VGH_L	-11463.45	-325	1132	S<529>	6184.63	291.22
26	VGH_L	-11424.46	-325	1133	SX<316>	6172.63	191.22
27	VGH_L	-11385.47	-325	1134	S<528>	6160.63	291.22
28	VGH_L	-11346.48	-325	1135	S<527>	6148.63	191.22
29	VGH_L	-11307.49	-325	1136	SX<315>	6136.64	291.22
30	VGHO_L	-11268.5	-325	1137	S<526>	6124.64	191.22
31	VGHO_L	-11229.5	-325	1138	SX<314>	6112.64	291.22
32	VGHO_L	-11190.51	-325	1139	S<525>	6100.64	191.22
33	VGHO_L	-11151.52	-325	1140	S<524>	6088.65	291.22
34	VGHO_L	-11112.53	-325	1141	SX<313>	6076.65	191.22
35	CGOUT_L<1>	-11073.54	-325	1142	S<523>	6064.65	291.22
36	CGOUT_L<2>	-11034.55	-325	1143	SX<312>	6052.65	191.22
37	CGOUT_L<3>	-10995.56	-325	1144	S<522>	6040.66	291.22
38	CGOUT_L<4>	-10956.56	-325	1145	S<521>	6028.66	191.22
39	CGOUT_L<5>	-10917.57	-325	1146	SX<311>	6016.66	291.22
40	CGOUT_L<6>	-10878.58	-325	1147	S<520>	6004.67	191.22
41	CGOUT_L<7>	-10839.59	-325	1148	SX<310>	5992.67	291.22
42	CGOUT_L<8>	-10800.6	-325	1149	S<519>	5980.67	191.22
43	CGOUT_L<9>	-10761.61	-325	1150	S<518>	5968.67	291.22
44	CGOUT_L<10>	-10722.62	-325	1151	SX<309>	5956.68	191.22
45	CGOUT_L<11>	-10683.63	-325	1152	S<517>	5944.68	291.22
46	CGOUT_L<12>	-10644.63	-325	1153	SX<308>	5932.68	191.22
47	CGOUT_L<13>	-10605.64	-325	1154	S<516>	5920.68	291.22
48	CGOUT_L<14>	-10566.65	-325	1155	S<515>	5908.69	191.22
49	CGOUT_L<15>	-10527.66	-325	1156	SX<307>	5896.69	291.22

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50	CGOUT_L<16>	-10488.67	-325	1157	S<514>	5884.69	191.22
51	CGOUT_L<17>	-10449.68	-325	1158	SX<306>	5872.69	291.22
52	VSSD	-10410.69	-325	1159	S<513>	5860.7	191.22
53	VSSD	-10371.69	-325	1160	S<512>	5848.7	291.22
54	VSSD	-10332.7	-325	1161	SX<305>	5836.7	191.22
55	VDD	-10293.71	-325	1162	S<511>	5824.71	291.22
56	VDD	-10254.72	-325	1163	SX<304>	5812.71	191.22
57	VDD	-10215.73	-325	1164	S<510>	5800.71	291.22
58	VDD_HS	-10176.74	-325	1165	S<509>	5788.71	191.22
59	VDD_HS	-10137.75	-325	1166	SX<303>	5776.72	291.22
60	VDD_HS	-10098.76	-325	1167	S<508>	5764.72	191.22
61	VDD_HS	-10059.76	-325	1168	SX<302>	5752.72	291.22
62	VDDI_HS	-10020.77	-325	1169	S<507>	5740.72	191.22
63	VDDI_HS	-9981.78	-325	1170	S<506>	5728.73	291.22
64	VDDI_HS	-9942.79	-325	1171	SX<301>	5716.73	191.22
65	VDDI_HS	-9903.8	-325	1172	S<505>	5704.73	291.22
66	HS_LDO	-9864.81	-325	1173	SX<300>	5692.73	191.22
67	HS_LDO	-9825.82	-325	1174	S<504>	5680.74	291.22
68	HS_LDO	-9786.82	-325	1175	S<503>	5668.74	191.22
69	HS_LDO	-9747.83	-325	1176	SX<299>	5656.74	291.22
70	VSS_HS	-9708.84	-325	1177	S<502>	5644.75	191.22
71	VSS_HS	-9669.85	-325	1178	SX<298>	5632.75	291.22
72	VSS_HS	-9630.86	-325	1179	S<501>	5620.75	191.22
73	VSS_HS	-9591.87	-325	1180	S<500>	5608.75	291.22
74	HS_D3P	-9552.88	-325	1181	SX<297>	5596.76	191.22
75	HS_D3P	-9513.89	-325	1182	S<499>	5584.76	291.22

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76	HS_D3P	-9474.89	-325	1183	SX<296>	5572.76	191.22
77	HS_D3P	-9435.9	-325	1184	S<498>	5560.76	291.22
78	HS_D3N	-9396.91	-325	1185	S<497>	5548.77	191.22
79	HS_D3N	-9357.92	-325	1186	SX<295>	5536.77	291.22
80	HS_D3N	-9318.93	-325	1187	S<496>	5524.77	191.22
81	HS_D3N	-9279.94	-325	1188	SX<294>	5512.77	291.22
82	VSS_HS	-9240.95	-325	1189	S<495>	5500.78	191.22
83	VSS_HS	-9201.95	-325	1190	S<494>	5488.78	291.22
84	HS_D0P	-9162.96	-325	1191	SX<293>	5476.78	191.22
85	HS_D0P	-9123.97	-325	1192	S<493>	5464.79	291.22
86	HS_D0P	-9084.98	-325	1193	SX<292>	5452.79	191.22
87	HS_D0P	-9045.99	-325	1194	S<492>	5440.79	291.22
88	HS_D0N	-9007	-325	1195	S<491>	5428.79	191.22
89	HS_D0N	-8968.01	-325	1196	SX<291>	5416.8	291.22
90	HS_D0N	-8929.02	-325	1197	S<490>	5404.8	191.22
91	HS_D0N	-8890.02	-325	1198	SX<290>	5392.8	291.22
92	VSS_HS	-8851.03	-325	1199	S<489>	5380.8	191.22
93	VSS_HS	-8812.04	-325	1200	S<488>	5368.81	291.22
94	HS_CP	-8773.05	-325	1201	SX<289>	5356.81	191.22
95	HS_CP	-8734.06	-325	1202	S<487>	5344.81	291.22
96	HS_CP	-8695.07	-325	1203	SX<288>	5332.81	191.22
97	HS_CP	-8656.08	-325	1204	S<486>	5320.82	291.22
98	HS_CN	-8617.08	-325	1205	S<485>	5308.82	191.22
99	HS_CN	-8578.09	-325	1206	VSSA	5296.63	291.22
100	HS_CN	-8539.1	-325	1207	S<484>	5284.64	191.22
101	HS_CN	-8500.11	-325	1208	SX<287>	5272.64	291.22

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102	VSS_HS	-8461.12	-325	1209	S<483>	5260.64	191.22
103	VSS_HS	-8422.13	-325	1210	S<482>	5248.64	291.22
104	HS_D1P	-8383.14	-325	1211	VSSA	5236.65	191.22
105	HS_D1P	-8344.15	-325	1212	S<481>	5224.65	291.22
106	HS_D1P	-8305.15	-325	1213	SX<286>	5212.65	191.22
107	HS_D1P	-8266.16	-325	1214	S<480>	5200.65	291.22
108	HS_D1N	-8227.17	-325	1215	S<479>	5188.66	191.22
109	HS_D1N	-8188.18	-325	1216	VSSA	5176.66	291.22
110	HS_D1N	-8149.19	-325	1217	S<478>	5164.66	191.22
111	HS_D1N	-8110.2	-325	1218	SX<285>	5152.66	291.22
112	VSS_HS	-8071.21	-325	1219	S<477>	5140.67	191.22
113	VSS_HS	-8032.21	-325	1220	S<476>	5128.67	291.22
114	HS_D2P	-7993.22	-325	1221	VSSA	5116.67	191.22
115	HS_D2P	-7954.23	-325	1222	S<475>	5104.68	291.22
116	HS_D2P	-7915.24	-325	1223	SX<284>	5092.68	191.22
117	HS_D2P	-7876.25	-325	1224	S<474>	5080.68	291.22
118	HS_D2N	-7837.26	-325	1225	S<473>	5068.68	191.22
119	HS_D2N	-7798.27	-325	1226	SX<283>	5056.69	291.22
120	HS_D2N	-7759.28	-325	1227	S<472>	5044.69	191.22
121	HS_D2N	-7720.28	-325	1228	SX<282>	5032.69	291.22
122	VSS_HS	-7681.29	-325	1229	S<471>	5020.69	191.22
123	VSS_HS	-7642.3	-325	1230	S<470>	5008.7	291.22
124	VSS_HS	-7603.31	-325	1231	SX<281>	4996.7	191.22
125	VSS_HS	-7564.32	-325	1232	S<469>	4984.7	291.22
126	HS_LDO	-7525.33	-325	1233	SX<280>	4972.7	191.22
127	HS_LDO	-7486.34	-325	1234	S<468>	4960.71	291.22

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128	HS_LDO	-7447.34	-325	1235	S<467>	4948.71	191.22
129	HS_LDO	-7408.35	-325	1236	SX<279>	4936.71	291.22
130	VCG_TP	-7369.36	-325	1237	S<466>	4924.72	191.22
131	VCG_TP	-7330.37	-325	1238	SX<278>	4912.72	291.22
132	VREF_TP	-7291.38	-325	1239	S<465>	4900.72	191.22
133	VREF_TP	-7252.39	-325	1240	S<464>	4888.72	291.22
134	VCOM	-7213.4	-325	1241	SX<277>	4876.73	191.22
135	VCOM	-7174.41	-325	1242	S<463>	4864.73	291.22
136	ATEST0_NV	-7135.41	-325	1243	SX<276>	4852.73	191.22
137	VCL	-7096.42	-325	1244	S<462>	4840.73	291.22
138	VCL	-7057.43	-325	1245	S<461>	4828.74	191.22
139	VSN	-7018.44	-325	1246	SX<275>	4816.74	291.22
140	VSN	-6979.45	-325	1247	S<460>	4804.74	191.22
141	VSN	-6940.46	-325	1248	SX<274>	4792.74	291.22
142	VDN	-6901.47	-325	1249	S<459>	4780.75	191.22
143	VDN	-6862.47	-325	1250	S<458>	4768.75	291.22
144	VDN	-6823.48	-325	1251	SX<273>	4756.75	191.22
145	VSSA	-6784.49	-325	1252	S<457>	4744.76	291.22
146	VSSA	-6745.5	-325	1253	SX<272>	4732.76	191.22
147	VSSA	-6706.51	-325	1254	S<456>	4720.76	291.22
148	VSP	-6667.52	-325	1255	S<455>	4708.76	191.22
149	VSP	-6628.53	-325	1256	SX<271>	4696.77	291.22
150	VSP	-6589.54	-325	1257	S<454>	4684.77	191.22
151	ATEST1_PV	-6550.54	-325	1258	SX<270>	4672.77	291.22
152	ATEST2_LV	-6511.55	-325	1259	S<453>	4660.77	191.22
153	VDD_TP	-6472.56	-325	1260	S<452>	4648.78	291.22

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154	VDD_TP	-6433.57	-325	1261	SX<269>	4636.78	191.22
155	VDD	-6394.58	-325	1262	S<451>	4624.78	291.22
156	VDD	-6355.59	-325	1263	SX<268>	4612.78	191.22
157	VDD	-6316.6	-325	1264	S<450>	4600.79	291.22
158	VDD	-6277.6	-325	1265	S<449>	4588.79	191.22
159	VSSD	-6238.61	-325	1266	SX<267>	4576.79	291.22
160	VSSD	-6199.62	-325	1267	S<448>	4564.8	191.22
161	VSSD	-6160.63	-325	1268	SX<266>	4552.8	291.22
162	VSSD	-6121.64	-325	1269	S<447>	4540.8	191.22
163	VDDI_DRV	-6082.65	-325	1270	S<446>	4528.8	291.22
164	VDDI_DRV	-6043.66	-325	1271	SX<265>	4516.81	191.22
165	DB<23>	-6004.67	-325	1272	S<445>	4504.81	291.22
166	DB<23>	-5965.67	-325	1273	SX<264>	4492.81	191.22
167	DB<22>	-5926.68	-325	1274	S<444>	4480.81	291.22
168	DB<22>	-5887.69	-325	1275	S<443>	4468.82	191.22
169	DB<21>	-5848.7	-325	1276	SX<263>	4456.82	291.22
170	DB<21>	-5809.71	-325	1277	S<442>	4444.82	191.22
171	DB<20>	-5770.72	-325	1278	SX<262>	4432.82	291.22
172	DB<20>	-5731.73	-325	1279	S<441>	4420.83	191.22
173	DB<19>	-5692.73	-325	1280	S<440>	4408.83	291.22
174	DB<19>	-5653.74	-325	1281	SX<261>	4396.83	191.22
175	DB<18>	-5614.75	-325	1282	S<439>	4384.84	291.22
176	DB<18>	-5575.76	-325	1283	SX<260>	4372.84	191.22
177	DB<17>	-5536.77	-325	1284	S<438>	4360.84	291.22
178	DB<17>	-5497.78	-325	1285	S<437>	4348.84	191.22
179	DB<16>	-5458.79	-325	1286	SX<259>	4336.85	291.22

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180	DB<16>	-5419.8	-325	1287	S<436>	4324.85	191.22
181	DB<15>	-5380.8	-325	1288	SX<258>	4312.85	291.22
182	DB<15>	-5341.81	-325	1289	S<435>	4300.85	191.22
183	DB<14>	-5302.82	-325	1290	S<434>	4288.86	291.22
184	DB<14>	-5263.83	-325	1291	SX<257>	4276.86	191.22
185	DB<13>	-5224.84	-325	1292	S<433>	4264.86	291.22
186	DB<13>	-5185.85	-325	1293	SX<256>	4252.86	191.22
187	DB<12>	-5146.86	-325	1294	S<432>	4240.87	291.22
188	DB<12>	-5107.86	-325	1295	S<431>	4228.87	191.22
189	HSYNC	-5068.87	-325	1296	SX<255>	4216.87	291.22
190	HSYNC	-5029.88	-325	1297	S<430>	4204.88	191.22
191	VSYNC	-4990.89	-325	1298	SX<254>	4192.88	291.22
192	VSYNC	-4951.9	-325	1299	S<429>	4180.88	191.22
193	DE	-4912.91	-325	1300	S<428>	4168.88	291.22
194	DE	-4873.92	-325	1301	SX<253>	4156.89	191.22
195	VDDI_DRV	-4834.93	-325	1302	S<427>	4144.89	291.22
196	VDDI_DRV	-4795.93	-325	1303	SX<252>	4132.89	191.22
197	VDD	-4756.94	-325	1304	S<426>	4120.89	291.22
198	VDD	-4717.95	-325	1305	S<425>	4108.9	191.22
199	VSSD	-4678.96	-325	1306	SX<251>	4096.9	291.22
200	VSSD	-4639.97	-325	1307	S<424>	4084.9	191.22
201	PCLK	-4600.98	-325	1308	SX<250>	4072.9	291.22
202	PCLK	-4561.99	-325	1309	S<423>	4060.91	191.22
203	DB<11>	-4522.99	-325	1310	S<422>	4048.91	291.22
204	DB<11>	-4484	-325	1311	SX<249>	4036.91	191.22
205	DB<10>	-4445.01	-325	1312	S<421>	4024.92	291.22

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206	DB<10>	-4406.02	-325	1313	SX<248>	4012.92	191.22
207	DB<9>	-4367.03	-325	1314	S<420>	4000.92	291.22
208	DB<9>	-4328.04	-325	1315	S<419>	3988.92	191.22
209	DB<8>	-4289.05	-325	1316	SX<247>	3976.93	291.22
210	DB<8>	-4250.06	-325	1317	S<418>	3964.93	191.22
211	DB<7>	-4211.06	-325	1318	SX<246>	3952.93	291.22
212	DB<7>	-4172.07	-325	1319	S<417>	3940.93	191.22
213	DB<6>	-4133.08	-325	1320	S<416>	3928.94	291.22
214	DB<6>	-4094.09	-325	1321	SX<245>	3916.94	191.22
215	DB<5>	-4055.1	-325	1322	S<415>	3904.94	291.22
216	DB<5>	-4016.11	-325	1323	SX<244>	3892.94	191.22
217	DB<4>	-3977.12	-325	1324	S<414>	3880.95	291.22
218	DB<4>	-3938.12	-325	1325	S<413>	3868.95	191.22
219	DB<3>	-3899.13	-325	1326	SX<243>	3856.95	291.22
220	DB<3>	-3860.14	-325	1327	S<412>	3844.96	191.22
221	DB<2>	-3821.15	-325	1328	SX<242>	3832.96	291.22
222	DB<2>	-3782.16	-325	1329	S<411>	3820.96	191.22
223	DB<1>	-3743.17	-325	1330	S<410>	3808.96	291.22
224	DB<1>	-3704.18	-325	1331	SX<241>	3796.97	191.22
225	DB<0>	-3665.19	-325	1332	S<409>	3784.97	291.22
226	DB<0>	-3626.19	-325	1333	DUMMY	3725.17	237.22
227	VSSD	-3587.2	-325	1334	DUMMY	3713.17	337.22
228	VSSD	-3548.21	-325	1335	DUMMY	3653.19	237.22
229	VDDI_DRV	-3509.22	-325	1336	DUMMY	3641.19	337.22
230	VDDI_DRV	-3470.23	-325	1337	DUMMY	3581.2	237.22
231	CSX	-3431.24	-325	1338	DUMMY	3569.21	337.22

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232	CSX	-3392.25	-325	1339	DUMMY	3509.22	237.22
233	SCL	-3353.25	-325	1340	DUMMY	3497.22	337.22
234	SCL	-3314.26	-325	1341	DUMMY	3437.24	237.22
235	RS	-3275.27	-325	1342	DUMMY	3425.24	337.22
236	RS	-3236.28	-325	1343	DUMMY	3365.25	237.22
237	DIN_SDA_DUAL	-3197.29	-325	1344	DUMMY	3353.25	337.22
238	DIN_SDA_DUAL	-3158.3	-325	1345	DUMMY	3293.27	237.22
239	DIN_SDA	-3119.31	-325	1346	DUMMY	3281.27	337.22
240	DIN_SDA	-3080.32	-325	1347	DUMMY	3221.28	237.22
241	TE	-3041.32	-325	1348	DUMMY	3209.29	337.22
242	TE	-3002.33	-325	1349	DUMMY	3149.3	237.22
243	LED_PWM	-2963.34	-325	1350	DUMMY	3137.3	337.22
244	LED_PWM	-2924.35	-325	1351	DUMMY	3077.32	237.22
245	VSSD	-2885.36	-325	1352	DUMMY	3065.32	337.22
246	VSSD	-2846.37	-325	1353	DUMMY	3005.33	237.22
247	VDDI_DRV	-2807.38	-325	1354	DUMMY	2993.33	337.22
248	VDDI_DRV	-2768.38	-325	1355	DUMMY	2933.35	237.22
249	SWIRE	-2729.39	-325	1356	DUMMY	2921.35	337.22
250	SWIRE	-2690.4	-325	1357	DUMMY	2861.36	237.22
251	RSTN_DRV	-2651.41	-325	1358	DUMMY	2849.37	337.22
252	RSTN_DRV	-2612.42	-325	1359	DUMMY	2789.38	237.22
253	VDD	-2573.43	-325	1360	DUMMY	2777.38	337.22
254	VDD	-2534.44	-325	1361	DUMMY	2717.4	237.22
255	VDD	-2495.45	-325	1362	DUMMY	2705.4	337.22
256	VDD	-2456.45	-325	1363	DUMMY	2645.41	237.22
257	VSSD	-2417.46	-325	1364	DUMMY	2633.41	337.22

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258	VSSD	-2378.47	-325	1365	DUMMY	2573.43	237.22
259	VSSD	-2339.48	-325	1366	DUMMY	2561.43	337.22
260	VSSD	-2300.49	-325	1367	DUMMY	2501.44	237.22
261	VSSD	-2261.5	-325	1368	DUMMY	2489.45	337.22
262	VDDI	-2222.51	-325	1369	DUMMY	2429.46	237.22
263	VDDI	-2183.51	-325	1370	DUMMY	2417.46	337.22
264	VDDI	-2144.52	-325	1371	DUMMY	2357.48	237.22
265	VDDI	-2105.53	-325	1372	DUMMY	2345.48	337.22
266	ATEST3_LV	-2066.54	-325	1373	DUMMY	2285.49	237.22
267	VDDI_PS	-2027.55	-325	1374	DUMMY	2273.49	337.22
268	VDDI_PS	-1988.56	-325	1375	DUMMY	2213.51	237.22
269	VDDI_PS	-1949.57	-325	1376	DUMMY	2201.51	337.22
270	VDDI_PS	-1910.58	-325	1377	DUMMY	2141.52	237.22
271	PS_ADQ<7>	-1871.58	-325	1378	DUMMY	2129.53	337.22
272	PS_ADQ<7>	-1832.59	-325	1379	DUMMY	2069.54	237.22
273	PS_ADQ<6>	-1793.6	-325	1380	DUMMY	2057.54	337.22
274	PS_ADQ<6>	-1754.61	-325	1381	DUMMY	1997.56	237.22
275	PS_ADQ<5>	-1715.62	-325	1382	DUMMY	1985.56	337.22
276	PS_ADQ<5>	-1676.63	-325	1383	DUMMY	1925.57	237.22
277	PS_ADQ<4>	-1637.64	-325	1384	DUMMY	1913.57	337.22
278	PS_ADQ<4>	-1598.64	-325	1385	DUMMY	1853.59	237.22
279	PS_DQS	-1559.65	-325	1386	DUMMY	1841.59	337.22
280	PS_DQS	-1520.66	-325	1387	DUMMY	1781.6	237.22
281	PS_CLKP	-1481.67	-325	1388	DUMMY	1769.61	337.22
282	PS_CLKP	-1442.68	-325	1389	DUMMY	1709.62	237.22
283	PS_CLKN	-1403.69	-325	1390	DUMMY	1697.62	337.22

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284	PS_CLKN	-1364.7	-325	1391	DUMMY	1637.64	237.22
285	VDDI_PS	-1325.71	-325	1392	DUMMY	1625.64	337.22
286	VDDI_PS	-1286.71	-325	1393	DUMMY	1565.65	237.22
287	VDDI_PS	-1247.72	-325	1394	DUMMY	1553.65	337.22
288	VDDI_PS	-1208.73	-325	1395	DUMMY	1493.67	237.22
289	VSSD	-1169.74	-325	1396	DUMMY	1481.67	337.22
290	VSSD	-1130.75	-325	1397	DUMMY	1421.68	237.22
291	VSSD	-1091.76	-325	1398	DUMMY	1409.69	337.22
292	VSSD	-1052.77	-325	1399	DUMMY	1349.7	237.22
293	VREF_PS	-1013.77	-325	1400	DUMMY	1337.7	337.22
294	VREF_PS	-974.78	-325	1401	DUMMY	1277.72	237.22
295	VREF_PS	-935.79	-325	1402	DUMMY	1265.72	337.22
296	VREF_PS	-896.8	-325	1403	DUMMY	1205.73	237.22
297	PS_CEN	-857.81	-325	1404	DUMMY	1193.73	337.22
298	PS_CEN	-818.82	-325	1405	DUMMY	1133.75	237.22
299	PS_DM	-779.83	-325	1406	DUMMY	1121.75	337.22
300	PS_DM	-740.84	-325	1407	DUMMY	1061.76	237.22
301	PS_ADQ<3>	-701.84	-325	1408	DUMMY	1049.77	337.22
302	PS_ADQ<3>	-662.85	-325	1409	DUMMY	989.78	237.22
303	PS_ADQ<2>	-623.86	-325	1410	DUMMY	977.78	337.22
304	PS_ADQ<2>	-584.87	-325	1411	DUMMY	917.8	237.22
305	PS_ADQ<1>	-545.88	-325	1412	DUMMY	905.8	337.22
306	PS_ADQ<1>	-506.89	-325	1413	DUMMY	845.81	237.22
307	PS_ADQ<0>	-467.9	-325	1414	DUMMY	833.81	337.22
308	PS_ADQ<0>	-428.9	-325	1415	DUMMY	773.83	237.22
309	VSSD	-389.91	-325	1416	DUMMY	761.83	337.22

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310	VSSD	-350.92	-325
311	VSSD	-311.93	-325
312	VSSD	-272.94	-325
313	VDDI_PS	-233.95	-325
314	VDDI_PS	-194.96	-325
315	VDDI_PS	-155.97	-325
316	VDDI_PS	-116.97	-325
317	VDD	-77.98	-325
318	VDD	-38.99	-325
319	VSSD	0	-325
320	VSSD	38.99	-325
321	VSSD	77.98	-325
322	VDDI	116.97	-325
323	VDDI	155.97	-325
324	VDDI	194.96	-325
325	VDDI	233.95	-325
326	VDD_PLL	272.94	-325
327	VDD_PLL	311.93	-325
328	VDD_OSC	350.92	-325
329	VDD_OSC	389.91	-325
330	VSS_OSC	428.9	-325
331	VSS_OSC	467.9	-325
332	OSC_IN	506.89	-325
333	OSC_TE	545.88	-325
334	VDDI_AO	584.87	-325
335	VDDI_AO	623.86	-325

1417	DUMMY	701.84	237.22
1418	DUMMY	689.85	337.22
1419	DUMMY	629.86	237.22
1420	DUMMY	617.86	337.22
1421	DUMMY	557.88	237.22
1422	DUMMY	545.88	337.22
1423	DUMMY	485.89	237.22
1424	DUMMY	473.89	337.22
1425	DUMMY	437.9	291.22
1426	DUMMY	389.91	291.22
1427	DUMMY	341.92	291.22
1428	DUMMY	293.93	291.22
1429	DUMMY	245.95	291.22
1430	DUMMY	197.96	291.22
1431	DUMMY	149.97	291.22
1432	DUMMY	101.98	291.22
1433	DUMMY	53.99	291.22
1434	DUMMY	6	291.22
1435	DUMMY	-41.99	291.22
1436	DUMMY	-89.98	291.22
1437	DUMMY	-137.97	291.22
1438	DUMMY	-185.96	291.22
1439	DUMMY	-233.95	291.22
1440	DUMMY	-281.94	291.22
1441	DUMMY	-329.93	291.22
1442	DUMMY	-377.92	291.22

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336	VSSA_REF	662.85	-325	1443	DUMMY	-425.91	291.22
337	VSSA_REF	701.84	-325	1444	DUMMY	-473.89	291.22
338	VREF	740.84	-325	1445	DUMMY	-521.88	291.22
339	VREF	779.83	-325	1446	DUMMY	-569.87	291.22
340	VREF	818.82	-325	1447	DUMMY	-617.86	291.22
341	ATEST5_PV	857.81	-325	1448	DUMMY	-665.85	291.22
342	VSP	896.8	-325	1449	DUMMY	-713.84	291.22
343	VSP	935.79	-325	1450	DUMMY	-761.83	291.22
344	VSP	974.78	-325	1451	DUMMY	-809.82	291.22
345	VSP	1013.77	-325	1452	DUMMY	-857.81	291.22
346	VSP	1052.77	-325	1453	DUMMY	-905.8	291.22
347	VSSA	1091.76	-325	1454	DUMMY	-953.79	291.22
348	VSSA	1130.75	-325	1455	DUMMY	-1001.78	291.22
349	VSSA	1169.74	-325	1456	DUMMY	-1049.77	291.22
350	VSSA	1208.73	-325	1457	DUMMY	-1097.76	291.22
351	VDN	1247.72	-325	1458	DUMMY	-1145.75	291.22
352	VDN	1286.71	-325	1459	DUMMY	-1193.73	291.22
353	VDN	1325.71	-325	1460	DUMMY	-1241.72	291.22
354	VDN	1364.7	-325	1461	DUMMY	-1289.71	291.22
355	VSN	1403.69	-325	1462	DUMMY	-1337.7	291.22
356	VSN	1442.68	-325	1463	DUMMY	-1385.69	291.22
357	VSN	1481.67	-325	1464	DUMMY	-1733.61	291.22
358	VSN	1520.66	-325	1465	DUMMY	-1781.6	291.22
359	VCL	1559.65	-325	1466	DUMMY	-1829.59	291.22
360	VCL	1598.64	-325	1467	DUMMY	-1877.58	291.22
361	ATEST4_NV	1637.64	-325	1468	DUMMY	-1925.57	291.22

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362	VDD	1676.63	-325	1469	DUMMY	-1973.56	291.22
363	VDD	1715.62	-325	1470	DUMMY	-2021.55	291.22
364	VSSD	1754.61	-325	1471	DUMMY	-2069.54	291.22
365	VSSD	1793.6	-325	1472	DUMMY	-2117.53	291.22
366	PSW_ENN	1832.59	-325	1473	DUMMY	-2165.52	291.22
367	PSW_PRIO_SEL	1871.58	-325	1474	DUMMY	-2213.51	291.22
368	VSSD	1910.58	-325	1475	DUMMY	-2261.5	291.22
369	VSSD	1949.57	-325	1476	DUMMY	-2309.49	291.22
370	VSSD	1988.56	-325	1477	DUMMY	-2357.48	291.22
371	VSSD	2027.55	-325	1478	DUMMY	-2405.47	291.22
372	PMOD_SEL	2066.54	-325	1479	DUMMY	-2453.45	291.22
373	IGZO_MOD	2105.53	-325	1480	DUMMY	-2501.44	291.22
374	VSSD	2144.52	-325	1481	DUMMY	-2549.43	291.22
375	VSSD	2183.51	-325	1482	DUMMY	-2597.42	291.22
376	BS<3>	2222.51	-325	1483	DUMMY	-2645.41	291.22
377	BS<2>	2261.5	-325	1484	DUMMY	-2693.4	291.22
378	BS<1>	2300.49	-325	1485	DUMMY	-2741.39	291.22
379	BS<0>	2339.48	-325	1486	DUMMY	-2789.38	291.22
380	IM<3>	2378.47	-325	1487	DUMMY	-2837.37	291.22
381	IM<2>	2417.46	-325	1488	DUMMY	-2885.36	291.22
382	IM<1>	2456.45	-325	1489	DUMMY	-2933.35	291.22
383	IM<0>	2495.45	-325	1490	DUMMY	-2981.34	291.22
384	VSSD	2534.44	-325	1491	DUMMY	-3029.33	291.22
385	VSSD	2573.43	-325	1492	DUMMY	-3077.32	291.22
386	VSSD	2612.42	-325	1493	DUMMY	-3125.31	291.22
387	VSSD	2651.41	-325	1494	DUMMY	-3173.29	291.22

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388	CSN_FLH	2690.4	-325	1495	DUMMY	-3221.28	291.22
389	CSN_FLH	2729.39	-325	1496	DUMMY	-3269.27	291.22
390	SCK_FLH	2768.38	-325	1497	DUMMY	-3317.26	291.22
391	SCK_FLH	2807.38	-325	1498	DUMMY	-3365.25	291.22
392	MISO_FLH	2846.37	-325	1499	DUMMY	-3413.24	291.22
393	MISO_FLH	2885.36	-325	1500	DUMMY	-3461.23	291.22
394	MOSI_FLH	2924.35	-325	1501	DUMMY	-3509.22	291.22
395	MOSI_FLH	2963.34	-325	1502	DUMMY	-3557.21	291.22
396	VDDI_FLH	3002.33	-325	1503	DUMMY	-3593.01	191.22
397	VDDI_FLH	3041.32	-325	1504	DUMMY	-3605.01	291.22
398	VDD_TP	3080.32	-325	1505	DUMMY	-3617.01	191.22
399	VDD_TP	3119.31	-325	1506	DUMMY	-3629	291.22
400	VSSD	3158.3	-325	1507	DUMMY	-3641	191.22
401	VSSD	3197.29	-325	1508	DUMMY	-3653	291.22
402	VSSD	3236.28	-325	1509	SX<240>	-3665	191.22
403	VSSD	3275.27	-325	1510	S<408>	-3676.99	291.22
404	VSSD	3314.26	-325	1511	S<407>	-3688.99	191.22
405	SCAN_MODE	3353.25	-325	1512	VSSA	-3700.99	291.22
406	SCAN_MODE	3392.25	-325	1513	S<406>	-3712.98	191.22
407	SCAN_EN	3431.24	-325	1514	SX<239>	-3724.98	291.22
408	SCAN_EN	3470.23	-325	1515	S<405>	-3736.98	191.22
409	SCAN_IN<2>	3509.22	-325	1516	S<404>	-3748.98	291.22
410	SCAN_IN<2>	3548.21	-325	1517	VSSA	-3760.97	191.22
411	SCAN_IN<1>	3587.2	-325	1518	S<403>	-3772.97	291.22
412	SCAN_IN<1>	3626.19	-325	1519	SX<238>	-3784.97	191.22
413	SCAN_IN<0>	3665.19	-325	1520	S<402>	-3796.97	291.22

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414	SCAN_IN<0>	3704.18	-325	1521	S<401>	-3808.96	191.22
415	SCAN_OUT<2>	3743.17	-325	1522	VSSA	-3820.96	291.22
416	SCAN_OUT<2>	3782.16	-325	1523	S<400>	-3832.96	191.22
417	SCAN_OUT<1>	3821.15	-325	1524	SX<237>	-3844.96	291.22
418	SCAN_OUT<1>	3860.14	-325	1525	S<399>	-3856.95	191.22
419	SCAN_OUT<0>	3899.13	-325	1526	S<398>	-3868.95	291.22
420	SCAN_OUT<0>	3938.12	-325	1527	VSSA	-3880.95	191.22
421	WORK_MODE	3977.12	-325	1528	S<397>	-3892.94	291.22
422	WORK_MODE	4016.11	-325	1529	SX<236>	-3904.94	191.22
423	TS_SEL	4055.1	-325	1530	S<396>	-3916.94	291.22
424	TS_SEL	4094.09	-325	1531	S<395>	-3928.94	191.22
425	RSTN_TP	4133.08	-325	1532	SX<235>	-3940.93	291.22
426	RSTN_TP	4172.07	-325	1533	S<394>	-3952.93	191.22
427	VDDI_TP	4211.06	-325	1534	SX<234>	-3964.93	291.22
428	VDDI_TP	4250.06	-325	1535	S<393>	-3976.93	191.22
429	VDDI_TP	4289.05	-325	1536	S<392>	-3988.92	291.22
430	VDDI_TP	4328.04	-325	1537	SX<233>	-4000.92	191.22
431	VDDI_TP	4367.03	-325	1538	S<391>	-4012.92	291.22
432	VSSD	4406.02	-325	1539	SX<232>	-4024.92	191.22
433	VSSD	4445.01	-325	1540	S<390>	-4036.91	291.22
434	CSN_TP	4484	-325	1541	S<389>	-4048.91	191.22
435	CSN_TP	4522.99	-325	1542	SX<231>	-4060.91	291.22
436	SCK_TP	4561.99	-325	1543	S<388>	-4072.9	191.22
437	SCK_TP	4600.98	-325	1544	SX<230>	-4084.9	291.22
438	MISO_TP	4639.97	-325	1545	S<387>	-4096.9	191.22
439	MISO_TP	4678.96	-325	1546	S<386>	-4108.9	291.22

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440	MOSI_TP	4717.95	-325	1547	SX<229>	-4120.89	191.22
441	MOSI_TP	4756.94	-325	1548	S<385>	-4132.89	291.22
442	GPIO<7>	4795.93	-325	1549	SX<228>	-4144.89	191.22
443	GPIO<7>	4834.93	-325	1550	S<384>	-4156.89	291.22
444	VDD_TP	4873.92	-325	1551	S<383>	-4168.88	191.22
445	VDD_TP	4912.91	-325	1552	SX<227>	-4180.88	291.22
446	VSSD	4951.9	-325	1553	S<382>	-4192.88	191.22
447	VSSD	4990.89	-325	1554	SX<226>	-4204.88	291.22
448	VSSD	5029.88	-325	1555	S<381>	-4216.87	191.22
449	GPIO<6>	5068.87	-325	1556	S<380>	-4228.87	291.22
450	GPIO<6>	5107.86	-325	1557	SX<225>	-4240.87	191.22
451	GPIO<5>	5146.86	-325	1558	S<379>	-4252.86	291.22
452	GPIO<5>	5185.85	-325	1559	SX<224>	-4264.86	191.22
453	GPIO<4>	5224.84	-325	1560	S<378>	-4276.86	291.22
454	GPIO<4>	5263.83	-325	1561	S<377>	-4288.86	191.22
455	GPIO<3>	5302.82	-325	1562	VSSA	-4300.85	291.22
456	GPIO<3>	5341.81	-325	1563	S<376>	-4312.85	191.22
457	GPIO<2>	5380.8	-325	1564	SX<223>	-4324.85	291.22
458	GPIO<2>	5419.8	-325	1565	S<375>	-4336.85	191.22
459	GPIO<1>	5458.79	-325	1566	S<374>	-4348.84	291.22
460	GPIO<1>	5497.78	-325	1567	VSSA	-4360.84	191.22
461	GPIO<0>	5536.77	-325	1568	S<373>	-4372.84	291.22
462	GPIO<0>	5575.76	-325	1569	SX<222>	-4384.84	191.22
463	VSSD	5614.75	-325	1570	S<372>	-4396.83	291.22
464	VSSD	5653.74	-325	1571	S<371>	-4408.83	191.22
465	VSSD	5692.73	-325	1572	VSSA	-4420.83	291.22

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466	VSSD	5731.73	-325	1573	S<370>	-4432.82	191.22
467	VSSD	5770.72	-325	1574	SX<221>	-4444.82	291.22
468	VSSD	5809.71	-325	1575	S<369>	-4456.82	191.22
469	VSSD	5848.7	-325	1576	S<368>	-4468.82	291.22
470	VSSD	5887.69	-325	1577	VSSA	-4480.81	191.22
471	VDDI_TP	5926.68	-325	1578	S<367>	-4492.81	291.22
472	VDDI_TP	5965.67	-325	1579	SX<220>	-4504.81	191.22
473	VDDI_TP	6004.67	-325	1580	S<366>	-4516.81	291.22
474	VDDI_TP	6043.66	-325	1581	S<365>	-4528.8	191.22
475	VDD_TP	6082.65	-325	1582	SX<219>	-4540.8	291.22
476	VDD_TP	6121.64	-325	1583	S<364>	-4552.8	191.22
477	VSSD	6160.63	-325	1584	SX<218>	-4564.8	291.22
478	VSSD	6199.62	-325	1585	S<363>	-4576.79	191.22
479	VSSD	6238.61	-325	1586	S<362>	-4588.79	291.22
480	VSSD	6277.6	-325	1587	SX<217>	-4600.79	191.22
481	VDD	6316.6	-325	1588	S<361>	-4612.78	291.22
482	VDD	6355.59	-325	1589	SX<216>	-4624.78	191.22
483	VDD	6394.58	-325	1590	S<360>	-4636.78	291.22
484	VDD	6433.57	-325	1591	S<359>	-4648.78	191.22
485	ATEST6_PV	6472.56	-325	1592	SX<215>	-4660.77	291.22
486	VSP	6511.55	-325	1593	S<358>	-4672.77	191.22
487	VSP	6550.54	-325	1594	SX<214>	-4684.77	291.22
488	VSP	6589.54	-325	1595	S<357>	-4696.77	191.22
489	VSSA	6628.53	-325	1596	S<356>	-4708.76	291.22
490	VSSA	6667.52	-325	1597	SX<213>	-4720.76	191.22
491	VSSA	6706.51	-325	1598	S<355>	-4732.76	291.22

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492	VDN	6745.5	-325	1599	SX<212>	-4744.76	191.22
493	VDN	6784.49	-325	1600	S<354>	-4756.75	291.22
494	VDN	6823.48	-325	1601	S<353>	-4768.75	191.22
495	VSN	6862.47	-325	1602	SX<211>	-4780.75	291.22
496	VSN	6901.47	-325	1603	S<352>	-4792.74	191.22
497	VSN	6940.46	-325	1604	SX<210>	-4804.74	291.22
498	VCL	6979.45	-325	1605	S<351>	-4816.74	191.22
499	VCL	7018.44	-325	1606	S<350>	-4828.74	291.22
500	VDDA_VCOM	7057.43	-325	1607	SX<209>	-4840.73	191.22
501	VDDA_VCOM	7096.42	-325	1608	S<349>	-4852.73	291.22
502	VCOM_OUT	7135.41	-325	1609	SX<208>	-4864.73	191.22
503	VCOM_OUT	7174.41	-325	1610	S<348>	-4876.73	291.22
504	VCOM_OUT	7213.4	-325	1611	S<347>	-4888.72	191.22
505	VCOM_OUT	7252.39	-325	1612	SX<207>	-4900.72	291.22
506	VCOM	7291.38	-325	1613	S<346>	-4912.72	191.22
507	VCOM	7330.37	-325	1614	SX<206>	-4924.72	291.22
508	VCOM	7369.36	-325	1615	S<345>	-4936.71	191.22
509	VCOM	7408.35	-325	1616	S<344>	-4948.71	291.22
510	ATEST7_NV	7447.34	-325	1617	SX<205>	-4960.71	191.22
511	VREF_TP	7486.34	-325	1618	S<343>	-4972.7	291.22
512	VREF_TP	7525.33	-325	1619	SX<204>	-4984.7	191.22
513	VCG_TP	7564.32	-325	1620	S<342>	-4996.7	291.22
514	VCG_TP	7603.31	-325	1621	S<341>	-5008.7	191.22
515	VSSA	7642.3	-325	1622	SX<203>	-5020.69	291.22
516	VSSA	7681.29	-325	1623	S<340>	-5032.69	191.22
517	VSSA	7720.28	-325	1624	SX<202>	-5044.69	291.22

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518	VSSA	7759.28	-325	1625	S<339>	-5056.69	191.22
519	VSSA	7798.27	-325	1626	S<338>	-5068.68	291.22
520	VSSA	7837.26	-325	1627	SX<201>	-5080.68	191.22
521	VSNOUT	7876.25	-325	1628	S<337>	-5092.68	291.22
522	VSNOUT	7915.24	-325	1629	SX<200>	-5104.68	191.22
523	VSNOUT	7954.23	-325	1630	S<336>	-5116.67	291.22
524	VSNOUT	7993.22	-325	1631	S<335>	-5128.67	191.22
525	VSNOUT	8032.21	-325	1632	SX<199>	-5140.67	291.22
526	C21N	8071.21	-325	1633	S<334>	-5152.66	191.22
527	C21N	8110.2	-325	1634	SX<198>	-5164.66	291.22
528	C21N	8149.19	-325	1635	S<333>	-5176.66	191.22
529	C21N	8188.18	-325	1636	S<332>	-5188.66	291.22
530	C21N	8227.17	-325	1637	SX<197>	-5200.65	191.22
531	C21P	8266.16	-325	1638	S<331>	-5212.65	291.22
532	C21P	8305.15	-325	1639	SX<196>	-5224.65	191.22
533	C21P	8344.15	-325	1640	S<330>	-5236.65	291.22
534	C21P	8383.14	-325	1641	S<329>	-5248.64	191.22
535	C21P	8422.13	-325	1642	SX<195>	-5260.64	291.22
536	VSPOUT	8461.12	-325	1643	S<328>	-5272.64	191.22
537	VSPOUT	8500.11	-325	1644	SX<194>	-5284.64	291.22
538	VSPOUT	8539.1	-325	1645	S<327>	-5296.63	191.22
539	VSPOUT	8578.09	-325	1646	S<326>	-5308.63	291.22
540	VSPOUT	8617.08	-325	1647	SX<193>	-5320.63	191.22
541	C12N	8656.08	-325	1648	S<325>	-5332.62	291.22
542	C12N	8695.07	-325	1649	SX<192>	-5344.62	191.22
543	C12N	8734.06	-325	1650	S<324>	-5356.62	291.22

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544	C12N	8773.05	-325	1651	S<323>	-5368.62	191.22
545	C12N	8812.04	-325	1652	VSSA	-5380.8	291.22
546	C12P	8851.03	-325	1653	S<322>	-5392.8	191.22
547	C12P	8890.02	-325	1654	SX<191>	-5404.8	291.22
548	C12P	8929.02	-325	1655	S<321>	-5416.8	191.22
549	C12P	8968.01	-325	1656	S<320>	-5428.79	291.22
550	C12P	9007	-325	1657	VSSA	-5440.79	191.22
551	VCI	9045.99	-325	1658	S<319>	-5452.79	291.22
552	VCI	9084.98	-325	1659	SX<190>	-5464.79	191.22
553	VCI	9123.97	-325	1660	S<318>	-5476.78	291.22
554	VCI	9162.96	-325	1661	S<317>	-5488.78	191.22
555	VCI	9201.95	-325	1662	VSSA	-5500.78	291.22
556	VSSA	9240.95	-325	1663	S<316>	-5512.77	191.22
557	VSSA	9279.94	-325	1664	SX<189>	-5524.77	291.22
558	VSSA	9318.93	-325	1665	S<315>	-5536.77	191.22
559	VSSA	9357.92	-325	1666	S<314>	-5548.77	291.22
560	VSSA	9396.91	-325	1667	VSSA	-5560.76	191.22
561	C11N	9435.9	-325	1668	S<313>	-5572.76	291.22
562	C11N	9474.89	-325	1669	SX<188>	-5584.76	191.22
563	C11N	9513.89	-325	1670	S<312>	-5596.76	291.22
564	C11N	9552.88	-325	1671	S<311>	-5608.75	191.22
565	C11N	9591.87	-325	1672	SX<187>	-5620.75	291.22
566	C11P	9630.86	-325	1673	S<310>	-5632.75	191.22
567	C11P	9669.85	-325	1674	SX<186>	-5644.75	291.22
568	C11P	9708.84	-325	1675	S<309>	-5656.74	191.22
569	C11P	9747.83	-325	1676	S<308>	-5668.74	291.22

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570	C11P	9786.82	-325	1677	SX<185>	-5680.74	191.22
571	VDD_TP	9825.82	-325	1678	S<307>	-5692.73	291.22
572	VDD_TP	9864.81	-325	1679	SX<184>	-5704.73	191.22
573	VDD_TP	9903.8	-325	1680	S<306>	-5716.73	291.22
574	VDD_TP	9942.79	-325	1681	S<305>	-5728.73	191.22
575	VDDI_TP	9981.78	-325	1682	SX<183>	-5740.72	291.22
576	VDDI_TP	10020.77	-325	1683	S<304>	-5752.72	191.22
577	VDDI_TP	10059.76	-325	1684	SX<182>	-5764.72	291.22
578	VDDI_TP	10098.76	-325	1685	S<303>	-5776.72	191.22
579	VSSD	10137.75	-325	1686	S<302>	-5788.71	291.22
580	VSSD	10176.74	-325	1687	SX<181>	-5800.71	191.22
581	VDD	10215.73	-325	1688	S<301>	-5812.71	291.22
582	VDD	10254.72	-325	1689	SX<180>	-5824.71	191.22
583	VGL	10293.71	-325	1690	S<300>	-5836.7	291.22
584	VGL	10332.7	-325	1691	S<299>	-5848.7	191.22
585	VGH	10371.69	-325	1692	SX<179>	-5860.7	291.22
586	VGH	10410.69	-325	1693	S<298>	-5872.69	191.22
587	CGOUT_R<17>	10449.68	-325	1694	SX<178>	-5884.69	291.22
588	CGOUT_R<16>	10488.67	-325	1695	S<297>	-5896.69	191.22
589	CGOUT_R<15>	10527.66	-325	1696	S<296>	-5908.69	291.22
590	CGOUT_R<14>	10566.65	-325	1697	SX<177>	-5920.68	191.22
591	CGOUT_R<13>	10605.64	-325	1698	S<295>	-5932.68	291.22
592	CGOUT_R<12>	10644.63	-325	1699	SX<176>	-5944.68	191.22
593	CGOUT_R<11>	10683.63	-325	1700	S<294>	-5956.68	291.22
594	CGOUT_R<10>	10722.62	-325	1701	S<293>	-5968.67	191.22
595	CGOUT_R<9>	10761.61	-325	1702	SX<175>	-5980.67	291.22

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596	CGOUT_R<8>	10800.6	-325	1703	S<292>	-5992.67	191.22
597	CGOUT_R<7>	10839.59	-325	1704	SX<174>	-6004.67	291.22
598	CGOUT_R<6>	10878.58	-325	1705	S<291>	-6016.66	191.22
599	CGOUT_R<5>	10917.57	-325	1706	S<290>	-6028.66	291.22
600	CGOUT_R<4>	10956.56	-325	1707	SX<173>	-6040.66	191.22
601	CGOUT_R<3>	10995.56	-325	1708	S<289>	-6052.65	291.22
602	CGOUT_R<2>	11034.55	-325	1709	SX<172>	-6064.65	191.22
603	CGOUT_R<1>	11073.54	-325	1710	S<288>	-6076.65	291.22
604	VGHO_REF	11112.53	-325	1711	S<287>	-6088.65	191.22
605	VGHO_REF	11151.52	-325	1712	SX<171>	-6100.64	291.22
606	VGHO_REF	11190.51	-325	1713	S<286>	-6112.64	191.22
607	VGHO_R	11229.5	-325	1714	SX<170>	-6124.64	291.22
608	VGHO_R	11268.5	-325	1715	S<285>	-6136.64	191.22
609	VGHO_R	11307.49	-325	1716	S<284>	-6148.63	291.22
610	VGHO_R	11346.48	-325	1717	SX<169>	-6160.63	191.22
611	VGHO_R	11385.47	-325	1718	S<283>	-6172.63	291.22
612	VGLO_R	11424.46	-325	1719	SX<168>	-6184.63	191.22
613	VGLO_R	11463.45	-325	1720	S<282>	-6196.62	291.22
614	VGLO_R	11502.44	-325	1721	S<281>	-6208.62	191.22
615	VGLO_R	11541.43	-325	1722	SX<167>	-6220.62	291.22
616	VGLO_R	11580.43	-325	1723	S<280>	-6232.61	191.22
617	VGLO_REF	11619.42	-325	1724	SX<166>	-6244.61	291.22
618	VGLO_REF	11658.41	-325	1725	S<279>	-6256.61	191.22
619	VCOM_PASS_R	11697.4	-325	1726	S<278>	-6268.61	291.22
620	VSN	11736.39	-325	1727	SX<165>	-6280.6	191.22
621	VSN	11775.38	-325	1728	S<277>	-6292.6	291.22

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622	VSP	11814.37	-325	1729	SX<164>	-6304.6	191.22
623	VSP	11853.37	-325	1730	S<276>	-6316.6	291.22
624	VSSA	11892.36	-325	1731	S<275>	-6328.59	191.22
625	VSSA	11931.35	-325	1732	SX<163>	-6340.59	291.22
626	VSSA	11970.34	-325	1733	S<274>	-6352.59	191.22
627	VSSA	12009.33	-325	1734	SX<162>	-6364.59	291.22
628	VSSA	12048.32	-325	1735	S<273>	-6376.58	191.22
629	VGHOUT2	12087.31	-325	1736	S<272>	-6388.58	291.22
630	VGHOUT2	12126.3	-325	1737	SX<161>	-6400.58	191.22
631	VGHOUT2	12165.3	-325	1738	S<271>	-6412.57	291.22
632	VGH	12204.29	-325	1739	VSSA	-6424.57	191.22
633	VGH	12243.28	-325	1740	VSSA	-6436.57	291.22
634	VGH	12282.27	-325	1741	VSSA	-6448.57	191.22
635	VGHOUT1	12321.26	-325	1742	VSSA	-6460.56	291.22
636	VGHOUT1	12360.25	-325	1743	VSSA	-6472.56	191.22
637	VGHOUT1	12399.24	-325	1744	VSSA	-6484.56	291.22
638	VSSA	12112	291.22	1745	VSSA	-6496.56	191.22
639	VSSA	12100	191.22	1746	VSSA	-6508.55	291.22
640	VSSA	12088	291.22	1747	VSSA	-6520.55	191.22
641	VSSA	12076.01	191.22	1748	VSSA	-6532.55	291.22
642	VSSA	12064.01	291.22	1749	VSSA	-6544.55	191.22
643	VSSA	12052.01	191.22	1750	VSSA	-6556.54	291.22
644	VSSA	12040.01	291.22	1751	VSSA	-6568.54	191.22
645	VCOM_PASS_R	12028.02	191.22	1752	VSSA	-6580.54	291.22
646	VCOM_PASS_R	12016.02	291.22	1753	VSSA	-6592.53	191.22
647	SDUM2	12004.02	191.22	1754	SX<160>	-6604.53	291.22

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650	S<809>	11968.03	291.22	1757	VSSA	-6640.52	191.22
651	VSSA	11956.03	191.22	1758	S<268>	-6652.52	291.22
652	S<808>	11944.04	291.22	1759	SX<159>	-6664.52	191.22
653	SX<479>	11932.04	191.22	1760	S<267>	-6676.52	291.22
654	S<807>	11920.04	291.22	1761	S<266>	-6688.51	191.22
655	S<806>	11908.04	191.22	1762	VSSA	-6700.51	291.22
656	VSSA	11896.05	291.22	1763	S<265>	-6712.51	191.22
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659	S<804>	11860.05	191.22	1766	S<263>	-6748.5	291.22
660	S<803>	11848.06	291.22	1767	VSSA	-6760.5	191.22
661	VSSA	11836.06	191.22	1768	S<262>	-6772.49	291.22
662	S<802>	11824.06	291.22	1769	SX<157>	-6784.49	191.22
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667	S<799>	11764.08	191.22	1774	SX<156>	-6844.48	291.22
668	SX<476>	11752.08	291.22	1775	S<258>	-6856.48	191.22
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670	S<797>	11728.08	291.22	1777	SX<155>	-6880.47	191.22
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672	S<796>	11704.09	291.22	1779	SX<154>	-6904.47	191.22
673	SX<474>	11692.09	191.22	1780	S<255>	-6916.46	291.22

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674	S<795>	11680.09	291.22	1781	S<254>	-6928.46	191.22
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682	S<790>	11584.12	291.22	1789	SX<150>	-7024.44	191.22
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686	SX<469>	11536.13	291.22	1793	S<247>	-7072.43	191.22
687	S<787>	11524.13	191.22	1794	SX<148>	-7084.43	291.22
688	SX<468>	11512.13	291.22	1795	S<246>	-7096.42	191.22
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692	S<784>	11464.14	291.22	1799	SX<146>	-7144.41	191.22
693	SX<466>	11452.14	191.22	1800	S<243>	-7156.41	291.22
694	S<783>	11440.15	291.22	1801	S<242>	-7168.41	191.22
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696	SX<465>	11416.15	291.22	1803	S<241>	-7192.4	191.22
697	S<781>	11404.16	191.22	1804	SX<144>	-7204.4	291.22
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699	S<780>	11380.16	191.22	1806	S<239>	-7228.39	291.22

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700	S<779>	11368.16	291.22	1807	SX<143>	-7240.39	191.22
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703	SX<462>	11332.17	191.22	1810	S<237>	-7276.38	291.22
704	S<777>	11320.17	291.22	1811	S<236>	-7288.38	191.22
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716	SX<457>	11176.21	291.22	1823	S<229>	-7432.35	191.22
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722	S<766>	11104.22	291.22	1829	SX<134>	-7504.33	191.22
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726	SX<453>	11056.23	291.22	1833	S<223>	-7552.32	191.22
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732	S<760>	10984.25	291.22	1839	SX<130>	-7624.31	191.22
733	SX<450>	10972.25	191.22	1840	S<219>	-7636.3	291.22
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736	SX<449>	10936.26	291.22	1843	S<217>	-7672.29	191.22
737	S<757>	10924.26	191.22	1844	SX<128>	-7684.29	291.22
738	SX<448>	10912.26	291.22	1845	S<216>	-7696.29	191.22
739	S<756>	10900.27	191.22	1846	S<215>	-7708.29	291.22
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743	SX<447>	10852.28	191.22	1850	S<213>	-7756.28	291.22
744	S<753>	10840.28	291.22	1851	S<212>	-7768.27	191.22
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746	VSSA	10816.29	291.22	1853	S<211>	-7792.27	191.22
747	S<751>	10804.29	191.22	1854	SX<126>	-7804.27	291.22
748	SX<446>	10792.29	291.22	1855	S<210>	-7816.26	191.22
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752	S<748>	10744.3	291.22	1859	SX<125>	-7864.25	191.22
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756	VSSA	10696.31	291.22	1863	S<205>	-7912.93	191.22
757	S<745>	10684.32	191.22	1864	SX<124>	-7924.93	291.22
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759	S<744>	10660.32	191.22	1866	S<203>	-7948.92	291.22
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761	SX<443>	10636.33	191.22	1868	S<202>	-7972.92	291.22
762	S<742>	10624.33	291.22	1869	SX<122>	-7984.92	191.22
763	SX<442>	10612.33	191.22	1870	S<201>	-7996.91	291.22
764	S<741>	10600.33	291.22	1871	S<200>	-8008.91	191.22
765	S<740>	10588.34	191.22	1872	SX<121>	-8020.91	291.22
766	SX<441>	10576.34	291.22	1873	S<199>	-8032.9	191.22
767	S<739>	10564.34	191.22	1874	SX<120>	-8044.9	291.22
768	SX<440>	10552.34	291.22	1875	S<198>	-8056.9	191.22
769	S<738>	10540.35	191.22	1876	S<197>	-8068.9	291.22
770	S<737>	10528.35	291.22	1877	SX<119>	-8080.89	191.22
771	SX<439>	10516.35	191.22	1878	S<196>	-8092.89	291.22
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774	S<735>	10480.36	291.22	1881	S<194>	-8128.88	191.22
775	S<734>	10468.36	191.22	1882	SX<117>	-8140.88	291.22
776	SX<437>	10456.37	291.22	1883	S<193>	-8152.88	191.22
777	S<733>	10444.37	191.22	1884	SX<116>	-8164.88	291.22

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778	SX<436>	10432.37	291.22	1885	S<192>	-8176.87	191.22
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781	SX<435>	10396.38	191.22	1888	S<190>	-8212.86	291.22
782	S<730>	10384.38	291.22	1889	SX<114>	-8224.86	191.22
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784	S<729>	10360.39	291.22	1891	S<188>	-8248.86	191.22
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787	S<727>	10324.4	191.22	1894	SX<112>	-8284.85	291.22
788	SX<432>	10312.4	291.22	1895	S<186>	-8296.85	191.22
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791	SX<431>	10276.41	191.22	1898	S<184>	-8332.84	291.22
792	S<724>	10264.41	291.22	1899	SX<110>	-8344.84	191.22
793	SX<430>	10252.41	191.22	1900	S<183>	-8356.83	291.22
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804	S<717>	10120.44	291.22	1911	S<176>	-8488.8	191.22
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807	S<715>	10084.45	191.22	1914	SX<104>	-8524.8	291.22
808	SX<424>	10072.45	291.22	1915	S<174>	-8536.79	191.22
809	S<714>	10060.45	191.22	1916	S<173>	-8548.79	291.22
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812	S<712>	10024.46	291.22	1919	SX<102>	-8584.78	191.22
813	SX<422>	10012.46	191.22	1920	S<171>	-8596.78	291.22
814	S<711>	10000.47	291.22	1921	S<170>	-8608.78	191.22
815	S<710>	9988.47	191.22	1922	SX<101>	-8620.77	291.22
816	SX<421>	9976.47	291.22	1923	S<169>	-8632.77	191.22
817	S<709>	9964.48	191.22	1924	SX<100>	-8644.77	291.22
818	SX<420>	9952.48	291.22	1925	S<168>	-8656.77	191.22
819	S<708>	9940.48	191.22	1926	S<167>	-8668.76	291.22
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823	SX<418>	9892.49	191.22	1930	S<165>	-8716.75	291.22
824	S<705>	9880.49	291.22	1931	S<164>	-8728.75	191.22
825	S<704>	9868.5	191.22	1932	SX<97>	-8740.75	291.22
826	SX<417>	9856.5	291.22	1933	S<163>	-8752.74	191.22
827	S<703>	9844.5	191.22	1934	SX<96>	-8764.74	291.22
828	SX<416>	9832.5	291.22	1935	S<162>	-8776.74	191.22
829	S<702>	9820.51	191.22	1936	S<161>	-8788.74	291.22

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830	S<701>	9808.51	291.22	1937	VSSA	-8800.73	191.22
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833	SX<415>	9772.52	191.22	1940	S<159>	-8836.73	291.22
834	S<699>	9760.52	291.22	1941	S<158>	-8848.72	191.22
835	S<698>	9748.52	191.22	1942	VSSA	-8860.72	291.22
836	VSSA	9736.53	291.22	1943	S<157>	-8872.72	191.22
837	S<697>	9724.53	191.22	1944	SX<94>	-8884.72	291.22
838	SX<414>	9712.53	291.22	1945	S<156>	-8896.71	191.22
839	S<696>	9700.53	191.22	1946	S<155>	-8908.71	291.22
840	S<695>	9688.54	291.22	1947	VSSA	-8920.71	191.22
841	VSSA	9676.54	191.22	1948	S<154>	-8932.7	291.22
842	S<694>	9664.54	291.22	1949	SX<93>	-8944.7	191.22
843	SX<413>	9652.54	191.22	1950	S<153>	-8956.7	291.22
844	S<693>	9640.55	291.22	1951	S<152>	-8968.7	191.22
845	S<692>	9628.55	191.22	1952	VSSA	-8980.69	291.22
846	VSSA	9616.55	291.22	1953	S<151>	-8992.69	191.22
847	S<691>	9604.56	191.22	1954	SX<92>	-9004.69	291.22
848	SX<412>	9592.56	291.22	1955	S<150>	-9016.69	191.22
849	S<690>	9580.56	191.22	1956	S<149>	-9028.68	291.22
850	S<689>	9568.56	291.22	1957	SX<91>	-9040.68	191.22
851	SX<411>	9556.57	191.22	1958	S<148>	-9052.68	291.22
852	S<688>	9544.57	291.22	1959	SX<90>	-9064.68	191.22
853	SX<410>	9532.57	191.22	1960	S<147>	-9076.67	291.22
854	S<687>	9520.57	291.22	1961	S<146>	-9088.67	191.22
855	S<686>	9508.58	191.22	1962	SX<89>	-9100.67	291.22

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856	SX<409>	9496.58	291.22	1963	S<145>	-9112.66	191.22
857	S<685>	9484.58	191.22	1964	SX<88>	-9124.66	291.22
858	SX<408>	9472.58	291.22	1965	S<144>	-9136.66	191.22
859	S<684>	9460.59	191.22	1966	S<143>	-9148.66	291.22
860	S<683>	9448.59	291.22	1967	SX<87>	-9160.65	191.22
861	SX<407>	9436.59	191.22	1968	S<142>	-9172.65	291.22
862	S<682>	9424.6	291.22	1969	SX<86>	-9184.65	191.22
863	SX<406>	9412.6	191.22	1970	S<141>	-9196.65	291.22
864	S<681>	9400.6	291.22	1971	S<140>	-9208.64	191.22
865	S<680>	9388.6	191.22	1972	SX<85>	-9220.64	291.22
866	SX<405>	9376.61	291.22	1973	S<139>	-9232.64	191.22
867	S<679>	9364.61	191.22	1974	SX<84>	-9244.64	291.22
868	SX<404>	9352.61	291.22	1975	S<138>	-9256.63	191.22
869	S<678>	9340.61	191.22	1976	S<137>	-9268.63	291.22
870	S<677>	9328.62	291.22	1977	SX<83>	-9280.63	191.22
871	SX<403>	9316.62	191.22	1978	S<136>	-9292.62	291.22
872	S<676>	9304.62	291.22	1979	SX<82>	-9304.62	191.22
873	SX<402>	9292.62	191.22	1980	S<135>	-9316.62	291.22
874	S<675>	9280.63	291.22	1981	S<134>	-9328.62	191.22
875	S<674>	9268.63	191.22	1982	SX<81>	-9340.61	291.22
876	SX<401>	9256.63	291.22	1983	S<133>	-9352.61	191.22
877	S<673>	9244.64	191.22	1984	SX<80>	-9364.61	291.22
878	SX<400>	9232.64	291.22	1985	S<132>	-9376.61	191.22
879	S<672>	9220.64	191.22	1986	S<131>	-9388.6	291.22
880	S<671>	9208.64	291.22	1987	SX<79>	-9400.6	191.22
881	SX<399>	9196.65	191.22	1988	S<130>	-9412.6	291.22

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882	S<670>	9184.65	291.22	1989	SX<78>	-9424.6	191.22
883	SX<398>	9172.65	191.22	1990	S<129>	-9436.59	291.22
884	S<669>	9160.65	291.22	1991	S<128>	-9448.59	191.22
885	S<668>	9148.66	191.22	1992	SX<77>	-9460.59	291.22
886	SX<397>	9136.66	291.22	1993	S<127>	-9472.58	191.22
887	S<667>	9124.66	191.22	1994	SX<76>	-9484.58	291.22
888	SX<396>	9112.66	291.22	1995	S<126>	-9496.58	191.22
889	S<666>	9100.67	191.22	1996	S<125>	-9508.58	291.22
890	S<665>	9088.67	291.22	1997	SX<75>	-9520.57	191.22
891	SX<395>	9076.67	191.22	1998	S<124>	-9532.57	291.22
892	S<664>	9064.68	291.22	1999	SX<74>	-9544.57	191.22
893	SX<394>	9052.68	191.22	2000	S<123>	-9556.57	291.22
894	S<663>	9040.68	291.22	2001	S<122>	-9568.56	191.22
895	S<662>	9028.68	191.22	2002	SX<73>	-9580.56	291.22
896	SX<393>	9016.69	291.22	2003	S<121>	-9592.56	191.22
897	S<661>	9004.69	191.22	2004	SX<72>	-9604.56	291.22
898	SX<392>	8992.69	291.22	2005	S<120>	-9616.55	191.22
899	S<660>	8980.69	191.22	2006	S<119>	-9628.55	291.22
900	S<659>	8968.7	291.22	2007	SX<71>	-9640.55	191.22
901	SX<391>	8956.7	191.22	2008	S<118>	-9652.54	291.22
902	S<658>	8944.7	291.22	2009	SX<70>	-9664.54	191.22
903	SX<390>	8932.7	191.22	2010	S<117>	-9676.54	291.22
904	S<657>	8920.71	291.22	2011	S<116>	-9688.54	191.22
905	S<656>	8908.71	191.22	2012	SX<69>	-9700.53	291.22
906	SX<389>	8896.71	291.22	2013	S<115>	-9712.53	191.22
907	S<655>	8884.72	191.22	2014	SX<68>	-9724.53	291.22

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908	SX<388>	8872.72	291.22	2015	S<114>	-9736.53	191.22
909	S<654>	8860.72	191.22	2016	S<113>	-9748.52	291.22
910	S<653>	8848.72	291.22	2017	SX<67>	-9760.52	191.22
911	SX<387>	8836.73	191.22	2018	S<112>	-9772.52	291.22
912	S<652>	8824.73	291.22	2019	SX<66>	-9784.52	191.22
913	SX<386>	8812.73	191.22	2020	S<111>	-9796.51	291.22
914	S<651>	8800.73	291.22	2021	S<110>	-9808.51	191.22
915	S<650>	8788.74	191.22	2022	SX<65>	-9820.51	291.22
916	SX<385>	8776.74	291.22	2023	S<109>	-9832.5	191.22
917	S<649>	8764.74	191.22	2024	SX<64>	-9844.5	291.22
918	SX<384>	8752.74	291.22	2025	S<108>	-9856.5	191.22
919	S<648>	8740.75	191.22	2026	S<107>	-9868.5	291.22
920	S<647>	8728.75	291.22	2027	VSSA	-9880.49	191.22
921	VSSA	8716.06	191.22	2028	S<106>	-9892.49	291.22
922	S<646>	8704.07	291.22	2029	SX<63>	-9904.49	191.22
923	SX<383>	8692.07	191.22	2030	S<105>	-9916.49	291.22
924	S<645>	8680.07	291.22	2031	S<104>	-9928.48	191.22
925	S<644>	8668.07	191.22	2032	VSSA	-9940.48	291.22
926	VSSA	8656.08	291.22	2033	S<103>	-9952.48	191.22
927	S<643>	8644.08	191.22	2034	SX<62>	-9964.48	291.22
928	SX<382>	8632.08	291.22	2035	S<102>	-9976.47	191.22
929	S<642>	8620.08	191.22	2036	S<101>	-9988.47	291.22
930	S<641>	8608.09	291.22	2037	VSSA	-10000.47	191.22
931	VSSA	8596.09	191.22	2038	S<100>	-10012.46	291.22
932	S<640>	8584.09	291.22	2039	SX<61>	-10024.46	191.22
933	SX<381>	8572.09	191.22	2040	S<99>	-10036.46	291.22

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934	S<639>	8560.1	291.22	2041	S<98>	-10048.46	191.22
935	S<638>	8548.1	191.22	2042	VSSA	-10060.45	291.22
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937	S<637>	8524.11	191.22	2044	SX<60>	-10084.45	291.22
938	SX<380>	8512.11	291.22	2045	S<96>	-10096.45	191.22
939	S<636>	8500.11	191.22	2046	S<95>	-10108.44	291.22
940	S<635>	8488.11	291.22	2047	SX<59>	-10120.44	191.22
941	SX<379>	8476.12	191.22	2048	S<94>	-10132.44	291.22
942	S<634>	8464.12	291.22	2049	SX<58>	-10144.44	191.22
943	SX<378>	8452.12	191.22	2050	S<93>	-10156.43	291.22
944	S<633>	8440.12	291.22	2051	S<92>	-10168.43	191.22
945	S<632>	8428.13	191.22	2052	SX<57>	-10180.43	291.22
946	SX<377>	8416.13	291.22	2053	S<91>	-10192.42	191.22
947	S<631>	8404.13	191.22	2054	SX<56>	-10204.42	291.22
948	SX<376>	8392.13	291.22	2055	S<90>	-10216.42	191.22
949	S<630>	8380.14	191.22	2056	S<89>	-10228.42	291.22
950	S<629>	8368.14	291.22	2057	SX<55>	-10240.41	191.22
951	SX<375>	8356.14	191.22	2058	S<88>	-10252.41	291.22
952	S<628>	8344.15	291.22	2059	SX<54>	-10264.41	191.22
953	SX<374>	8332.15	191.22	2060	S<87>	-10276.41	291.22
954	S<627>	8320.15	291.22	2061	S<86>	-10288.4	191.22
955	S<626>	8308.15	191.22	2062	SX<53>	-10300.4	291.22
956	SX<373>	8296.16	291.22	2063	S<85>	-10312.4	191.22
957	S<625>	8284.16	191.22	2064	SX<52>	-10324.4	291.22
958	SX<372>	8272.16	291.22	2065	S<84>	-10336.39	191.22
959	S<624>	8260.16	191.22	2066	S<83>	-10348.39	291.22

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960	S<623>	8248.17	291.22	2067	SX<51>	-10360.39	191.22
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962	S<622>	8224.17	291.22	2069	SX<50>	-10384.38	191.22
963	SX<370>	8212.17	191.22	2070	S<81>	-10396.38	291.22
964	S<621>	8200.18	291.22	2071	S<80>	-10408.38	191.22
965	S<620>	8188.18	191.22	2072	SX<49>	-10420.37	291.22
966	SX<369>	8176.18	291.22	2073	S<79>	-10432.37	191.22
967	S<619>	8164.19	191.22	2074	SX<48>	-10444.37	291.22
968	SX<368>	8152.19	291.22	2075	S<78>	-10456.37	191.22
969	S<618>	8140.19	191.22	2076	S<77>	-10468.36	291.22
970	S<617>	8128.19	291.22	2077	SX<47>	-10480.36	191.22
971	SX<367>	8116.2	191.22	2078	S<76>	-10492.36	291.22
972	S<616>	8104.2	291.22	2079	SX<46>	-10504.36	191.22
973	SX<366>	8092.2	191.22	2080	S<75>	-10516.35	291.22
974	S<615>	8080.2	291.22	2081	S<74>	-10528.35	191.22
975	S<614>	8068.21	191.22	2082	SX<45>	-10540.35	291.22
976	SX<365>	8056.21	291.22	2083	S<73>	-10552.34	191.22
977	S<613>	8044.21	191.22	2084	SX<44>	-10564.34	291.22
978	SX<364>	8032.21	291.22	2085	S<72>	-10576.34	191.22
979	S<612>	8020.22	191.22	2086	S<71>	-10588.34	291.22
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981	SX<363>	7996.22	191.22	2088	S<70>	-10612.33	291.22
982	S<610>	7984.23	291.22	2089	SX<42>	-10624.33	191.22
983	SX<362>	7972.23	191.22	2090	S<69>	-10636.33	291.22
984	S<609>	7960.23	291.22	2091	S<68>	-10648.32	191.22
985	S<608>	7948.23	191.22	2092	SX<41>	-10660.32	291.22

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986	SX<361>	7936.24	291.22	2093	S<67>	-10672.32	191.22
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988	SX<360>	7912.24	291.22	2095	S<66>	-10696.31	191.22
989	S<606>	7900.24	191.22	2096	S<65>	-10708.31	291.22
990	S<605>	7888.25	291.22	2097	SX<39>	-10720.31	191.22
991	SX<359>	7876.25	191.22	2098	S<64>	-10732.3	291.22
992	S<604>	7864.25	291.22	2099	SX<38>	-10744.3	191.22
993	SX<358>	7852.25	191.22	2100	S<63>	-10756.3	291.22
994	S<603>	7840.26	291.22	2101	S<62>	-10768.3	191.22
995	S<602>	7828.26	191.22	2102	SX<37>	-10780.29	291.22
996	SX<357>	7816.26	291.22	2103	S<61>	-10792.29	191.22
997	S<601>	7804.27	191.22	2104	SX<36>	-10804.29	291.22
998	SX<356>	7792.27	291.22	2105	S<60>	-10816.29	191.22
999	S<600>	7780.27	191.22	2106	S<59>	-10828.28	291.22
1000	S<599>	7768.27	291.22	2107	SX<35>	-10840.28	191.22
1001	SX<355>	7756.28	191.22	2108	S<58>	-10852.28	291.22
1002	S<598>	7744.28	291.22	2109	SX<34>	-10864.28	191.22
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1004	S<597>	7720.28	291.22	2111	S<56>	-10888.27	191.22
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1006	SX<353>	7696.29	291.22	2113	S<55>	-10912.26	191.22
1007	S<595>	7684.29	191.22	2114	SX<32>	-10924.26	291.22
1008	SX<352>	7672.29	291.22	2115	S<54>	-10936.26	191.22
1009	S<594>	7660.3	191.22	2116	S<53>	-10948.26	291.22
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1012	S<592>	7624.31	291.22	2119	SX<31>	-10984.25	191.22
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1014	S<591>	7600.31	291.22	2121	S<50>	-11008.24	191.22
1015	S<590>	7588.31	191.22	2122	VSSA	-11020.24	291.22
1016	VSSA	7576.32	291.22	2123	S<49>	-11032.24	191.22
1017	S<589>	7564.32	191.22	2124	SX<30>	-11044.24	291.22
1018	SX<350>	7552.32	291.22	2125	S<48>	-11056.23	191.22
1019	S<588>	7540.32	191.22	2126	S<47>	-11068.23	291.22
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1022	S<586>	7504.33	291.22	2129	SX<29>	-11104.22	191.22
1023	SX<349>	7492.33	191.22	2130	S<45>	-11116.22	291.22
1024	S<585>	7480.34	291.22	2131	S<44>	-11128.22	191.22
1025	S<584>	7468.34	191.22	2132	VSSA	-11140.21	291.22
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1027	S<583>	7444.35	191.22	2134	SX<28>	-11164.21	291.22
1028	SX<348>	7432.35	291.22	2135	S<42>	-11176.21	191.22
1029	S<582>	7420.35	191.22	2136	S<41>	-11188.2	291.22
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1031	SX<347>	7396.36	191.22	2138	S<40>	-11212.2	291.22
1032	S<580>	7384.36	291.22	2139	SX<26>	-11224.2	191.22
1033	SX<346>	7372.36	191.22	2140	S<39>	-11236.19	291.22
1034	S<579>	7360.36	291.22	2141	S<38>	-11248.19	191.22
1035	S<578>	7348.37	191.22	2142	SX<25>	-11260.19	291.22
1036	SX<345>	7336.37	291.22	2143	S<37>	-11272.18	191.22
1037	S<577>	7324.37	191.22	2144	SX<24>	-11284.18	291.22

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1038	SX<344>	7312.37	291.22	2145	S<36>	-11296.18	191.22
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1041	SX<343>	7276.38	191.22	2148	S<34>	-11332.17	291.22
1042	S<574>	7264.39	291.22	2149	SX<22>	-11344.17	191.22
1043	SX<342>	7252.39	191.22	2150	S<33>	-11356.17	291.22
1044	S<573>	7240.39	291.22	2151	S<32>	-11368.16	191.22
1045	S<572>	7228.39	191.22	2152	SX<21>	-11380.16	291.22
1046	SX<341>	7216.4	291.22	2153	S<31>	-11392.16	191.22
1047	S<571>	7204.4	191.22	2154	SX<20>	-11404.16	291.22
1048	SX<340>	7192.4	291.22	2155	S<30>	-11416.15	191.22
1049	S<570>	7180.4	191.22	2156	S<29>	-11428.15	291.22
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1051	SX<339>	7156.41	191.22	2158	S<28>	-11452.14	291.22
1052	S<568>	7144.41	291.22	2159	SX<18>	-11464.14	191.22
1053	SX<338>	7132.41	191.22	2160	S<27>	-11476.14	291.22
1054	S<567>	7120.42	291.22	2161	S<26>	-11488.14	191.22
1055	S<566>	7108.42	191.22	2162	SX<17>	-11500.13	291.22
1056	SX<337>	7096.42	291.22	2163	S<25>	-11512.13	191.22
1057	S<565>	7084.43	191.22	2164	SX<16>	-11524.13	291.22
1058	SX<336>	7072.43	291.22	2165	S<24>	-11536.13	191.22
1059	S<564>	7060.43	191.22	2166	S<23>	-11548.12	291.22
1060	S<563>	7048.43	291.22	2167	SX<15>	-11560.12	191.22
1061	SX<335>	7036.44	191.22	2168	S<22>	-11572.12	291.22
1062	S<562>	7024.44	291.22	2169	SX<14>	-11584.12	191.22
1063	SX<334>	7012.44	191.22	2170	S<21>	-11596.11	291.22

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1064	S<561>	7000.44	291.22	2171	S<20>	-11608.11	191.22
1065	S<560>	6988.45	191.22	2172	SX<13>	-11620.11	291.22
1066	SX<333>	6976.45	291.22	2173	S<19>	-11632.1	191.22
1067	S<559>	6964.45	191.22	2174	SX<12>	-11644.1	291.22
1068	SX<332>	6952.45	291.22	2175	S<18>	-11656.1	191.22
1069	S<558>	6940.46	191.22	2176	S<17>	-11668.1	291.22
1070	S<557>	6928.46	291.22	2177	SX<11>	-11680.09	191.22
1071	SX<331>	6916.46	191.22	2178	S<16>	-11692.09	291.22
1072	S<556>	6904.47	291.22	2179	SX<10>	-11704.09	191.22
1073	SX<330>	6892.47	191.22	2180	S<15>	-11716.09	291.22
1074	S<555>	6880.47	291.22	2181	S<14>	-11728.08	191.22
1075	S<554>	6868.47	191.22	2182	SX<9>	-11740.08	291.22
1076	SX<329>	6856.48	291.22	2183	S<13>	-11752.08	191.22
1077	S<553>	6844.48	191.22	2184	SX<8>	-11764.08	291.22
1078	SX<328>	6832.48	291.22	2185	S<12>	-11776.07	191.22
1079	S<552>	6820.48	191.22	2186	S<11>	-11788.07	291.22
1080	S<551>	6808.49	291.22	2187	SX<7>	-11800.07	191.22
1081	SX<327>	6796.49	191.22	2188	S<10>	-11812.06	291.22
1082	S<550>	6784.49	291.22	2189	SX<6>	-11824.06	191.22
1083	SX<326>	6772.49	191.22	2190	S<9>	-11836.06	291.22
1084	S<549>	6760.5	291.22	2191	S<8>	-11848.06	191.22
1085	S<548>	6748.5	191.22	2192	SX<5>	-11860.05	291.22
1086	SX<325>	6736.5	291.22	2193	S<7>	-11872.05	191.22
1087	S<547>	6724.51	191.22	2194	SX<4>	-11884.05	291.22
1088	SX<324>	6712.51	291.22	2195	S<6>	-11896.05	191.22
1089	S<546>	6700.51	191.22	2196	S<5>	-11908.04	291.22

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1090	S<545>	6688.51	291.22	2197	SX<3>	-11920.04	191.22
1091	SX<323>	6676.52	191.22	2198	S<4>	-11932.04	291.22
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1093	SX<322>	6652.52	191.22	2200	S<3>	-11956.03	291.22
1094	S<543>	6640.52	291.22	2201	S<2>	-11968.03	191.22
1095	S<542>	6628.53	191.22	2202	SX<1>	-11980.03	291.22
1096	SX<321>	6616.53	291.22	2203	S<1>	-11992.02	191.22
1097	S<541>	6604.53	191.22	2204	SDUM1	-12004.02	291.22
1098	VSSA	6592.53	291.22	2205	VCOM_PASS_L	-12016.02	191.22
1099	VSSA	6580.54	191.22	2206	VCOM_PASS_L	-12028.02	291.22
1100	VSSA	6568.54	291.22	2207	VSSA	-12040.01	191.22
1101	VSSA	6556.54	191.22	2208	VSSA	-12052.01	291.22
1102	VSSA	6544.55	291.22	2209	VSSA	-12064.01	191.22
1103	VSSA	6532.55	191.22	2210	VSSA	-12076.01	291.22
1104	VSSA	6520.55	291.22	2211	VSSA	-12088	191.22
1105	VSSA	6508.55	191.22	2212	VSSA	-12100	291.22
1106	VSSA	6496.56	291.22	2213	VSSA	-12112	191.22
1107	VSSA	6484.56	191.22				

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